Class 7
Biology
(Solutions)

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1. Nutrition in plants

Solutions

LEVEL – I

1. Correct option: (A) Roots

Roots are the plant organs that provide anchorage to the plants and are responsible for the uptake of water and nutrients from the soil & transporting it to the other plant parts. These are specialized and have tiny root hairs that help them to achieve their desired functions.

2. Correct option: (B) Stephen Hales

Stephen Hales was an English Botanist who first discovered the role of light in plant nutrition. Stephen Hales realized that air could be fixed by plants under the influence of solar light. Priestley showed that plants could regenerate the bad air making it breathable. Ingenhousz demonstrated that the green parts of plants performed the complete purification of air only under the influence of the light.

3. Correct option: (A) Iodine Solution

Starch is a carbohydrate which is synthesized by plant leaves and is stored as energy reserve by a process known as photosynthesis. To test the presence of starch, Iodine is frequently used. Iodine has the property to bind to Starch molecules and impart a blue-black color which is an indication of presence of starch.

4. Correct option: (A) Potassium, K

Potassium helps regulate the opening and closing of the stomata on leaves, which regulates the exchange of water vapor, oxygen, and carbon dioxide. Potassium also activates enzymes responsible for starch synthesis in plants.

5. Correct option: (A) Stomata

In plants, minute pores are present on the surface of leaves that serves as the center for gaseous exchange. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

6. Correct option: (B) Haustoria

Parasitic plants and fungi have a root-like extension known as Haustorium, that grows into or around another plant to absorb water or nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium.

7. Correct option: (D) Fungi

Moist Bread and pickles if left outside, grows a black and white spot on the surface due to spoilage by Fungus, generally by Rhizopus and Mucor species. They generally grow on moist and damp habitat and in warm temperatures.

8. Correct option: (A) Leaves

Transpiration is the loss of water in the form of water vapor through stomata, cuticles and lenticles. Most of the transpiration occurs through stomata in leaves i.e., 50-90% of total transpiration.

9. Correct option: (A) Urea

Urea serves as fertilizers and is a rich source of nitrogen. The urea reacts with the gases present in the soil and is firstly converted into ammonia and then into ammonium. This ammonium is then converted into nitrates and nitrites. Plant leaves turn yellow when there is a deficiency of nitrogen content. Farmers sprinkle urea so that they absorb nitrogen and fulfill its needs. The leaves soon become green again, therefore increasing the rate of photosynthesis.

10. Correct option: (D) All of above
Drocera, Utricularia and Venus fly trap are insectivorous plants that feed on insects as well as their larvae. They generally have modified shoots especially for the purpose of capturing and ingesting the insect.

11. Correct option: (B) Parasite
Cuscuta also known as Dodder or Amarbel is a genus of parasitic plants which has no chlorophyll, and it absorbs its food and nutrients from its host by wrapping its whole flexible stem around the host plant and absorbing nutrients and stays attached to it until the host dies.

12. Correct option: (A) Roots
Parasitic plants and fungi have root-like structures known as Haustorium, that grow into or around another plant and its tissues to absorb water and nutrients from that plant instead of directly taking it from the soil. Parasitic plants like Dodder and Mistletoe have a haustorium as specialized root like appendages projecting out from stem.

13. Correct option: (D) All of them
In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose and O_2 inside the chloroplast present in the leaves of plants. Given below is the reaction.
$6CO_2+6H_2O \rightarrow C_6H_{12}O_6+6O_2$

14. Correct option: (C) Nepenthes
Tropical Pitcher plants or Nepenthes is a genus of insectivorous plants that grows in Nitrogen deficient soil and feeds on insects. Nepenthes pitchers produce a lot of sweet nectar to attract insects.

15. Correct option: (A) Sundew
Drocera is commonly known as Sundew, that are “flypaper” plants that trap prey in sticky hairs on their leaves. They make up one of the largest groups of carnivorous plants. Long tentacles protrude from their leaves, each with a sticky gland at the tip. These droplets look like dew glistening in the sun, thus their name.

16. Correct option: (B) Stem
Lenticels are specialized pores present on the surface of stems/ barks and permit the exchange of gases between the environment and the internal tissue spaces of the organs. They permit the entrance of oxygen and simultaneously the output of carbon dioxide and water vapor.

17. Correct option: (A) Stomata
Stomata are the openings present on the surface leaves that function in exchange of gases in plants continuously. The gases diffuse into the intracellular spaces of the leaf through these pores. Gaseous exchange (Air) is necessary for plants for the purpose of photosynthesis and respiration.

18. Correct option: (D) Nitrogen
Carnivorous plants such as Picher plants and Sundews grow in soil which is deficient in Nitrogen. Nitrogen is an essential component for plants to produce proteins and vitamins. Hence, they rely on insects for their nitrogen need.

19. Correct option: (D) All of them
Micronutrients are those nutrients that is needed in a small quantity by a body for plant metabolism. There are 7 essential plant nutrient elements defined as micronutrients [boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl)]. They constitute in total less than 1% of the dry weight of most plants.

20. Correct option: (C) O_2

In the process of photosynthesis, Carbon dioxide, CO_2 along with Water, H_2O in presence of solar energy is converted to sugars, specifically Sucrose inside the chloroplast present in the leaves of plants and releases O_2 gas outside as by-product. Given below is the reaction.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

21. Correct Matching is: a-S; b-P; c-T; d-R; e-Q

a) Autotroph – (S) Organisms which can produce their food.
b) Stomata – (P) Tiny holes in leaves.
c) Carnivorous plants – (T) Plants which trap insects
d) Heterotroph – (R) Organism which cannot produce their own food.
e) Chlorophyll – (Q) Green pigment in the leaves

22. Correct option: a-S; b-R; c-Q; d-P; e-T

a) Parasite – (S) An organism that lives in or on another organism.
b) Host – (R) The animal or plant on which or in which another organism lives.
c) Saprophyte – (Q) Any organism that lives on dead and decaying organic matter.
d) Symbiotic – (P) An organism in association with other.
e) Nutrition – (T) Supplying food to living beings.

LEVEL-II

1. Correct option: (A) Science of nutrition

The study of different components of diet including carbohydrates, fats, and protein of living organisms is known as science of nutrition.

2. Correct option: (C) Leaves

Plant leaves have organelles called as chloroplast that synthesis their food requirements by a process known as photosynthesis, where light energy is converted to chemical energy,

3. Correct option: (B) 17

Plants require essential nutrients for proper growth and development. There are in total 17 nutrients that are required by plants. Out of which 6 are macronutrients and others are micronutrients. These are Carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), Phosphorus (P), Potassium (K), Sulfur (S), Calcium (Ca), Magnesium (Mg), Boron (B), Chlorine (Cl), Copper (Cu), Iron (Fe), Manganese (Mn), Molybdenum (Mo), Nickel (Ni), and Zinc (Zn).

4. Correct option: (A, C) Water and Carbon Dioxide

In the process of photosynthesis, water and carbon dioxide are the precursor material for sugar synthesis by plants and in the process, oxygen is liberated. So, water as well as carbon dioxide cannot be the end product of it.

5. Correct option: (D) All of these

Chlorosis is the loss of the normal green coloration of leaves of plants due to loss of chlorophyll. It leads to the yellowing of leaves. This is caused by iron deficiency in lime-rich soils, or due to diseases or lack of light. Other elements that cause chlorosis are Nitrogen, Potassium, Zinc, Manganese, Sulphur etc.

6. Correct option: (A) Nitrogen

Dry matter in leafy vegetables and proteins are rich in nitrogen and its molecules such as amines. Thus, Nitrogen is a very crucial mineral that helps in improving both the quantity and quality of that matter.

7. Correct option: (D) Bean-shaped

In monocots the Guard cell surrounding stomata are dumb-bell shaped, but in case of dicots they are kidney bean shaped.

8. Correct option: (A) Saprophytes are green
Saprophytes are plants/fungi that feed on dead and decaying matter and have heterotrophic mode of nutrition. They lack chloroplast in their cells and thus they aren't green.
9. Correct option: (C) Croton plants have chlorophyll, but it is hidden by dark red color pigments.
The leaves of the croton plant have chlorophyll, but they appear dark red in color. This is because the higher percentages of other pigments anthocyanins and carotenoid present in them, hides the green color of the chlorophyll. So, the leaves of the croton plants can also make food by the process of photosynthesis.
10. Correct option: (D) All of these
Fungi are ubiquitous that is are found everywhere in this earth, majority of them can grow on the surface of the soil, foods items, in liquid, cloths and leathers, decaying matter etc.
11. Correct option: (A) Autotrophic
Plants can synthesize their own food requirement by a process known as photosynthesis, and thus are autotrophic in nature.
12. Correct option: (B) Mushroom
Saprophytes are those organisms that feed and grow on dead and decaying matter. Fungi are such organisms that can grow and derive nutrients from dead leaves and humus in soil. Mushrooms are an example of fungus. Venus's flytrap and the Dodders are insectivorous, Lichen is a symbiotic association between algae and fungi.
13. Correct option: (D) Nourishment
Food that we consume has nutrients and minerals and serves as a mode of nourishment for the body organs for the purpose of energy to carry out metabolism.
14. Correct option: (B) Thylakoids
Plant cells have chloroplast, in which stacks of thylakoids form grana and are suspended in matrix known as stroma. Thylakoids have chlorophyll inside which absorbs light and carries out photosynthesis.
15. Correct option: (C) Green plants
Partial parasitic plants are green as they perform photosynthesis, but they are dependent on other plants for certain nutrients.
16. Correct option: (B) Crow
Crows draw their nutrition from both plant and animal sources. The organisms that draw their nutrition from both animals and plants are called omnivores. Hence, the crows are omnivorous organisms.
17. Correct option: (A) Blue-Black Color
Starch is a polymer of sugars present in plants, its presence is detected by iodine, iodine binds to starch turns it into blue-black color.
18. Correct option: (B) Plant hormones are nutrients
Plant hormones are not nutrients but chemicals that in small amounts promote and influence the growth, development, and differentiation of cells and tissues.
19. Correct option: (D) Chlorophyll
Plant leaves appear green due to chlorophyll present in their chloroplast. Chloroplasts are the sites of photosynthesis in plants and chlorophyll serves as a light absorbing pigment.
20. Correct option: (A) Solar energy is converted into chemical energy
During Photosynthesis, plants use solar energy to convert water and carbon dioxide into sugars and oxygen is released and this sugar is then used as chemical energy during respiration.

21. Correct option: (D) Glucose

The end product of photosynthesis is Glucose, $C_6H_{12}O_6$ which is formed from water and carbon dioxide in the presence of light.

22. Correct option: (B) Symbiosis

The associations of organism where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

23. Correct option: (A) Mango

Except for Mango rest all are heterotrophs, that is they cannot synthesize their own food and must derive their nutrition from other sources.

24. Correct option: (D) Iron

Iron and magnesium are essential for the synthesis of chlorophyll in plants. A deficiency of these elements causes chlorotic patches on leaves with the leaves turning yellow.

25. Correct option: (B) grows in soils which lacks nitrogen

They can trap insects because they do not grow in nitrogen-rich soils. Soil lacking nitrogen causes pitcher plants to get nitrogen from the trapped insects.

26. Correct option: (A) Marshy areas

The soil of marshy land is deficient in nitrogen. Plants living in marshy areas do not get nitrogen from the soil. To fulfil their nitrogenous need, they need to eat insects.

27. Correct option: (B) Stomata

These tiny pores present in the leaves are known as stomata. They help in the exchange of gases and give out water vapour through transpiration.

28. Correct option: (D) Symbiosis

The associations between organisms where they live together and share shelter and nutrients and are mutually beneficial for each other is known as Symbiosis. Lichen is an example of Symbiosis.

29. Correct option: (A) Rhizobium

Rhizobium is a nitrogen fixing bacteria present in the soil that converts atmospheric nitrogen into ammonia which can be taken up by plant roots.

30. Correct option: (A) i and iv

All green plants have chlorophyll in their cells to prepare their own food by photosynthesis. During the process of photosynthesis, water combines with carbon dioxide to form sugars and oxygen in liberated in the process.

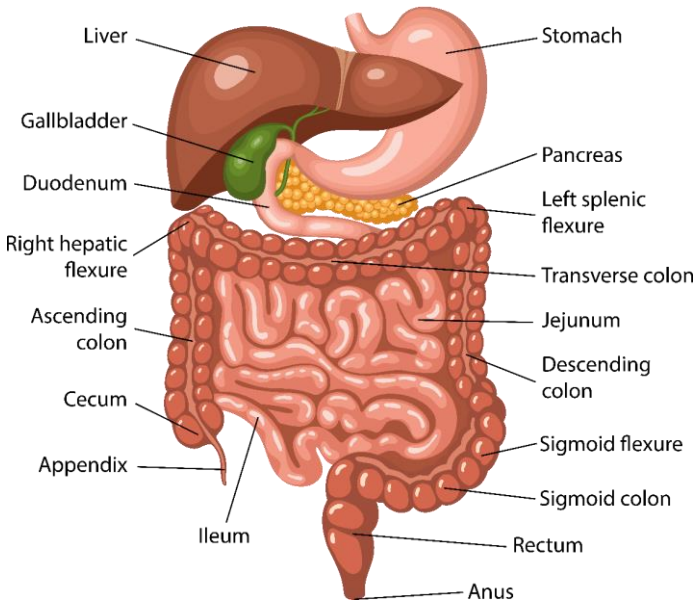
2. NUTRITION IN ANIMALS

Solutions

LEVEL I

1. Human beings ingest food through the mouth. Digestion begins in the mouth when we chew food. The salivary glands present in the mouth secrete their digestive juice called the saliva. Saliva breaks down the starch present in the food and converts it into sugar. The saliva also keeps the food moist making it easy to swallow. The food that we ingest goes down from the mouth through the oesophagus into the stomach. Stomach receives the partly digested food coming through the food pipe. Digestion of food in the stomach takes one to four hours, depending on the type of food eaten. Digestion of food happens with the help of various secretions such as hydrochloric acid and gastric juice secreted by the walls of the stomach. The semi-digested food, called chyme, leaves the stomach and enters the small intestine. When the food enters the small intestine, it mixes with the bile and the pancreatic juice, secreted into it by the liver and the pancreas, respectively. The small intestine also produces digestive juices, which help to complete the digestion of all the components of food. Here absorption of food takes place in the small intestine. The food that remains undigested and unabsorbed then enters the large intestine. The function of the large intestine is to absorb water and some salts from the undigested and unabsorbed food. The semi-solid food that is left behind is then stored as faeces in the rectum, the last part of the large intestine. The semi-solid faeces is removed from our body from anus.

Digestive system



2. Ruminants have a special stomach with four chambers, namely, rumen, reticulum, omasum and abomasum. The swallowed food goes straight into the rumen—the first and the largest of the four chambers in the stomach. The food is partially digested in the rumen. The partially digested food in the rumen is known as cud. The cud is pushed into the reticulum—the second chamber of the stomach and from here it is sent back into the mouth for thorough chewing. The chewed food is swallowed and sent into the omasum—the third and the smallest chamber of the stomach to absorb excess water from the food. In the omasum, the food is broken down to even smaller

pieces. The last chamber—abomasum is called the true stomach. Gastric juices are secreted in the abomasum to help in digestion. Enzymes act upon the food to complete the digestion. The digested food then goes into the small intestine, where absorption of nutrients takes place. A large sac-like structure called caecum present between the small intestine and large intestine also contain bacteria that help in cellulose digestion.

3. 1-e, 2-a, 3-d, 4-b, 5-c

4. False

5. True

6. True

7. True

8. False

9. Ingestion, digestion, absorption, assimilation, and egestion are the five stages of holozoic nutrition.

10. Assimilation

11. The salivary gland secretes digestive juice called saliva which breakdown starch into sugars. It also keeps the food moist and easy to swallow.

12. The antelope swallows grass as the first chamber of its stomach called rumen contains cellulose digesting bacteria which digests the cellulose present in the grass. This is sent to the second chamber and then regurgitated back to the mouth where it is thoroughly chewed.

13. Bile juice

14. I am amoeba. I form a pellet of the undigested food and throw it out of the food vacuole through a temporary opening of the cell membrane.

15. Enzyme

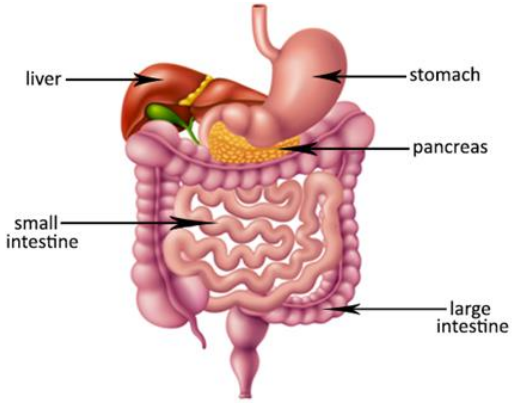
16. Digestion

17. Mouth/Buccal cavity

18. I am any cooked grain such as rice or wheat. I test positive the first time as there is starch present. The second time, I test negative as the saliva breaks down my starch into sugar when I am chewed.

19. The image is villi present on the inner walls of a part of the small intestine. There are millions of finger-like projections of villi which absorb food. They increase the surface area for absorption of food.

20. Pancreas and liver are the two digestive glands. Their secretions are found in the small intestine.



21. Correct option: (B) Digestion

The whole process starting from ingestion of food by mouth, its breakdown, absorption, and utilization by the body is known as digestion. It's a process by which complex food is taken up by the body.

22. Correct option: (B) Amoeba
Amoeba ingests its food with the help of its false feet or pseudopodia. The food is digested in the food vacuole.
23. Correct option: (B) Starch
Human saliva contains salivary amylase which breakdown starch into maltose and dextrin.
24. Correct option: (B) Stomach
The hydrochloric acid in the gastric juice breaks down the food and the digestive enzymes split up the proteins. The acidic gastric juice also kills bacteria. The mucus covers the stomach wall with a protective coating.
25. Correct option: (B) small intestine
Digestion and absorption of all biomolecules are completed in the intestine. The small intestine has 3 parts namely- the duodenum (c- shaped), jejunum, and ileum. Complete digestion of food takes place in the ileum. After being digested the food is absorbed by the blood vessels present in the small intestine.
26. Correct option: (C) Villi
The millions of finger-like outgrowths on the inner surface of the small intestine are called as Villi. They help in increasing the surface area for the absorption of the digested food.
27. Correct option: (C) 7.5 m
Although the small intestine is narrower than the large intestine, it is the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.
28. Correct option: (B) proteins
Your body uses amino acids to make proteins. The different types of amino acids and the way they're put together determine the function of each protein.
29. Correct option: (C) small intestine
Fats are present in the form of large globules in the small intestine. The small intestine receives the secretions from the liver and the pancreas. The bile salts (from the liver) break down the large fat globules into smaller globules so that the pancreatic enzyme lipase can easily act on them.
30. Correct option: (D) Large intestine
Water from the undigested food is absorbed in the large intestine to the body.
31. Correct option: (D) proteins
Protein after digestion by proteinases is converted into Amino acids which are absorbed into the bloodstream.
32. Correct option: (D) B and D
Digestion of groundnut oil produces fatty acids and glycerol. Groundnut oil is rich in fatty acids.
33. Correct option: (D) none of these
The boiled rice and saliva are added to the iodine solution then the colour of iodine will not change because of the digestion of starch present in rice into monomers of sugars.
34. Correct option: (B) Canines
Canines are used for ripping and tearing food such as meat.
35. Correct option: (D) Incisors
Incisors – These are the four teeth which are present in both the upper and lower jaw known as the incisors. These are having the one root. The major function of the incisors is to cut food because they have the sharp edge.

36. Correct option: (C) 4 incisors, 2 canines, 4 premolars and 6 molars
4 incisors, 2 canines, 4 premolars and 6 molars (Each half of the Jaw has 2,1,2,3 number of teeth's, 2 Incisors, 1 Canine, 2 premolars and 3 molars on each side of the Jaw)
37. Correct option: (B) Pulp cavity
The root contains blood vessels and nerves, which supply blood and feeling to the whole tooth. This area is known as the "pulp" of the tooth.
38. Correct option: (D) Tooth decay
Excessive intake of foods containing sugar causes tooth decay. The germs in the mouth form plaque on the surface of teeth
39. Correct option: (C) Pseudopodia
Amoebae use pseudopodia, which are arm-like projections on the cell surface, to engulf and ingest food.
40. Correct option: (D) It has digestive system
Amoeba is a protist that has pseudopodia or false feet which it uses for locomotion and capturing of food. It has no fixed shape, no fixed mouth and possess no digestive system.
41. Correct option: (C) Esophagus
Peristalsis is a series of wave-like muscle contractions that move food through the digestive tract. It starts in the esophagus where strong wave-like motions of the smooth muscle move balls of swallowed food to the stomach.
42. Correct option: (B) Intracellular
Amoeba is a unicellular organism. It undergoes intracellular digestion that is food is engulfed inside, fused with lysosome and broken down by enzymes inside the cells. The mode of nutrition in Amoeba is holozoic. The process of obtaining food by Amoeba is called phagocytosis.
43. Correct option: (B) Premolars and Molars
Premolars and Molars are the teeth that do most of the chewing because those flat surfaces are perfect for grinding and crushing food until it's ready to be swallowed.
44. Correct option: (A) five
Holozoic Nutrition is a process in which complex food is taken into a specialist digestive system and broken down into small pieces to be absorbed. This consists of 5 stages, ingestion, digestion, absorption, assimilation, and egestion.
45. Correct Answer: liver.
Bile is a fluid that is made and released by the liver and stored in the gallbladder.
46. Correct Answer: Small intestine.
Stomach is connected to Duodenum of small intestine at its lower portion.

LEVEL-II

1. Correct option: (C) Hookworm
Hookworms are parasites that live inside the intestines. There, they feed on blood from the intestinal wall, mature and lay eggs.
2. Correct option: (C) Sucking
The bees climb onto or into the flower and suck up the nectar with their straw-like mouth and collect it in a little sac called a crop.
3. Correct option: (D) Snakes
Snakes do not have the right kind of teeth to chew their food so they must eat their catch whole. Their jaw is structured in such a way that it allows the mouth to open wider than their own body to swallow their prey whole.

4. Correct option: (A) salivary gland
Saliva is an extracellular fluid produced and secreted by salivary glands present in the mouth.
5. Correct option: (D) both A and B
The food vacuole containing the food molecule fuses with lysosome which contains amylase and proteinase enzymes. These enzymes digests carbohydrates and proteins present in the food.
6. Correct option: (B) Parietal cells
Hydrochloric acid is the main component of gastric juice and is secreted by the parietal cells or oxyntic cells of the stomach walls in humans.
7. Correct option: (D) All of the above
The vacuole is pushed deeper into the amoeba by cytoplasmic movements where it fuses with lysosomes - organelles containing digestive enzymes like amylase, cellulase and proteinases which can digest carbohydrates such as Sugars and cellulose and proteins.
8. Correct option: (C) Absorb digested food
The finger-like outgrowths (Pseudopodia) of Amoeba help ingest food. However, the finger-like outgrowths or villi of human intestines increases the surface area of the walls and thus help to absorb the digested food.
9. Correct option: (B) Swallowed and partially digested
Cud is the name given to the food of ruminants which is swallowed and partially digested. Cud is a food component that returns from the stomach to the mouth of an animal for a second chew. Rumination is the process by which cud is generated during digestion. Ruminants are other names for cud-chewing animals.
10. Correct option: (A) Water
The large intestine has 3 primary functions: absorbing water and electrolytes, producing and absorbing vitamins, and forming and propelling faces toward the rectum for elimination.
11. Correct option: (C) Digestion
Digestion is the mechanical and chemical break down of food into small organic fragments.
12. Correct option: (D) all the of above
In a holozoic mode of nutrition, the organism feeds on complex organic matter. They ingest the complex organic matter, digest, and absorb the nutrients. In animals' mode of nutrition is holozoic because they depend on plants for food, cannot synthesize their own food, thus are heterotrophs, and they swallow their food.
13. Correct option: (D) fatty acid and glycerol
Lacteals are lymph vessels responsible for absorbing fat in the small intestine. Occurring in the finger-like villi of the ileum, lacteals have a milky appearance and drain into the lymphatic system. Before fat can pass into the lacteal, bile from the liver causes its emulsification into small droplets for action of the enzyme lipase.
14. Correct option: (B) Milk teeth
Deciduous teeth are the official term for baby teeth, milk teeth, or primary teeth. Deciduous teeth start developing during the embryonic stage and then commonly begin to come in about 6 months after birth.
15. Correct option: (C) Capturing food and movement
The false feet of Amoeba or the pseudopodia are used to engulf and capture food and also in movement and locomotion.

16. Correct option: (B) 6-8
The first set of teeth which are lost between the ages of **6 -12 years**, are termed as the milk teeth.
17. Correct option: (D) 4
There are four types of teeth Incisors, Canines, Premolars and Molars.
18. Correct option: (A) Acidic medium
In stomach, the medium is acidic with a pH of about 1-2 due to presence of Hydrochloric acid in the Gastric juice. It serves as suitable medium for digestion of proteins in the stomach.
19. Correct option: (D) Egestion
The removal of faeces through anus is called excretion.
20. Correct option: (B) Molars and Premolars
Premolars and molars pair of teeth differ in structure but are similar in function that is grinding of food.
21. Correct option: (A) Saliva
Salivary Amylases digest starch into smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules by maltase.
22. Correct option: (C) Chyme
Chyme is a churned semi-solid food that is formed in the stomach of the digestive tract.
23. Correct option: (B) U Shaped bag
The stomach is a thick-walled muscular bag. Its shape is like a flattened U and it is the widest part of the alimentary canal.
24. Correct option: (D) 7.5 m long
Although the small intestine is narrower than the large intestine, it is actually the longest section of your digestive tube, measuring about 7.5m on average, or three-and-a-half times the length of your body.
25. Correct option: (C) Amino acid
Proteins are chains of amino acids linked together, once they are broken by proteinases, they yield small units of amino acids.
26. Correct option: (D) Oral Rehydration Solution
Plenty of boiled and cooled water with a pinch of salt and sugar dissolved in it. This is called Oral Rehydration Solution (ORS).
27. Correct option: (B) Euglena
Euglena has both an Autotropic as well as heterotrophic mode of nutrition. It's a protozoon that can synthesize its own food in presence of sunlight and can capture and consume its prey when deprived of sunlight.
28. Correct option: (D) bitter
The tip of the tongue is most sensitive to sweet and salty substances, the side to sour, while the back is most sensitive to bitter taste.
29. Correct option: (A) i, ii and iv
Villi are finger-like projections in the small intestines that increases surface area for absorption of digested food. They have very thin walls and have network of thin and small blood vessels close to the surface.
30. Correct option: (B) The contraction of the muscle of the food pipe
The swallowed food moves downwards in the alimentary canal because of the contraction of muscles in the wall of the food pipe. This movement is under the control of nervous system and is known as peristalsis.

3. Adaptation of Animals to Climate

Solutions

LEVEL-I

1. Correct option: (A) Very hot and wet

The climate found in Kerala is categorized as "very hot and wet". Accordingly, Kerala has high temperatures and humidity all year round. The area has high rainfall throughout the monsoon season, which adds to its profusion of wildlife and beautiful foliage. The distinctive natural beauty and picturesque landscapes of Kerala are mostly due to its warm and humid climate, which also makes it a great location for a variety of plant and animal species to thrive.

2. Correct option: (A) In the afternoon

When the sun rises, the temperature reaches its peak for the day. The Earth's surface starts to warm when the sun rises, which raises temperatures throughout the morning and early afternoon.

Although the maximum temperature is normally measured around midday when the sun is at its highest point in the sky, this heating impact peaks in the late afternoon. After then, when the sun sets and the Earth radiates heat away, the temperatures begin to steadily fall. Thus, the hottest time of the day often occurs in the morning, around noon, before it begins to drop off in the evening.

Early in the day, when the sun is low in the sky and its rays are less direct, the temperature is at its lowest of the day. The air above the Earth's surface has cooled down throughout the course of the previous night.

3. Correct option: (B) Sand cools down faster

Nights in deserts are much cooler primarily because of the low heat-retaining capacity of sand. During the day, the desert sand absorbs a significant amount of heat from the sun and becomes very hot. However, as the sun sets and the desert loses its direct exposure to sunlight, the sand cools down rapidly due to its low heat retention properties.

This leads to a substantial drop in temperature, making desert nights considerably cooler compared to the scorching daytime temperatures. Option (b) correctly identifies this phenomenon as the reason for cooler nights in deserts.

The other choices are incorrect. In deserts, the humidity does not rise significantly at night. Although wind does blow in deserts, it does not blow hard enough to considerably chill the air. Deserts have no nighttime precipitation.

4. Correct option: (A) Climate

Climate is the annual average of a location's long-term average temperature and precipitation. A region's climate is made up of the regular weather factors that prevail there for a prolonged period of time, often 30 years or more, including temperature, precipitation, humidity, and other atmospheric components.

It gives us insight into the typical weather patterns and trends in a region, enabling us to distinguish between various climatic zones like tropical, temperate, dry, etc. Climate is the proper identification of this phrase in Option (a).

The other choices are incorrect. Climate is the long-term average of a region's weather conditions. The state of the atmosphere at a specific location and time is called the weather. The technique of anticipating future weather conditions is called weather forecasting.

5. Correct option: (D) All of these

Temperature, humidity, and rainfall are a few examples of weather components. The term "weather" describes the transient atmospheric conditions and changes that take place from day to day or even hour to hour. These factors are very important in influencing the

local weather at any given time. Other weather factors may include cloud cover, visibility, wind speed, air pressure, cloud cover, and atmospheric conditions like fog or frost. Option (d) is correct as it includes all the mentioned elements of weather.

6. Correct option: (D) All of these

Forecasting the weather has many uses and benefits in daily life. By giving people early warnings and enabling them to take the required steps to preserve lives and property, it assists in preparing for adverse weather conditions, such as storms, hurricanes, or excessive rains.

It helps with daily activity planning since it allows people to plan outside events, travel, and other activities in accordance with the weather prediction. Weather forecasting is also essential for organizing agricultural activities since it allows farmers to arrange irrigation, planting, and harvesting operations depending on the expected weather. The right response is option (d), which contains all of the advantages of weather forecasting that have been highlighted.

7. Correct option: (D) All the above

The humidity of a place refers to the amount of water vapor present in the air. It is an indicator of the amount of moisture in the air and has a big impact on how comfortable or uncomfortable people feel in different weather situations. Low humidity can result in dryness, which can have an adverse effect on the health of the skin, the respiratory system, and some materials. High humidity can make the air seem heavy and stuffy.

Precipitation, or falling water from the sky, includes both rainfall and snowfall. Humidity and precipitation are not the same. Precipitation is the amount of water that falls from the sky, whereas humidity is the amount of water vapor in the air.

8. Correct option: (C) Polar region

Penguins have evolved to exist in the polar regions, especially in Antarctica and its surroundings. They are equipped to endure in severely cold and icy conditions. Penguins have various adaptations that allow them to survive in these extreme temperatures, including a coat of insulating blubber, thick feathers that trap air for further insulation, and unique glands to remove extra salt from their bodies. Their distinctive body composition and streamlined form also help them swim and dive more effectively in frigid waters. The polar area is where penguins are evolved to reside, hence option (c) is the right response.

Equatorial areas are hot and humid, whereas rainforests are warm and moist. Deserts are hot and dry. Because they are not suited to the heat or the humidity, penguins would not be able to live in these environments.

9. Correct option: (B) Rajasthan

The state of Rajasthan, which lies in the northwest of India, is renowned for its hot and dry weather. It frequently encounters summertime temperatures far beyond 40^0 Celsius (104 degrees Fahrenheit), which are considered to be severe. The area is known for its extremely low rainfall and dry and semi-arid climate. Rajasthan is among the hottest and driest places in India due to its climate. The Thar Desert, the largest desert in India, is located in Rajasthan.

The other choices are incorrect. The climates of Kashmir are chilly and mountainous, Kerala is hot and humid, and the North East has a humid, subtropical environment.

10. Correct option: (B) They cannot swim

Penguins are well-suited to live in the water and have excellent swimming abilities. Being excellent swimmers, they move through the water with tremendous agility using their wings, which have changed into flippers. Penguins spend a large amount of their lives in

the water where they forage for fish and other marine life. They are widely recognized for their swimming prowess.
The other statements are all correct. Penguins have webbed feet, streamlined bodies, and do snuggle together. Penguins keep warm by huddling together, and their webbed feet and sleek bodies enable them to swim and dive.

11. Correct option: (C) Desert regions

The adaptive traits listed in the query, such as being nocturnal, having big, cushioned feet, sweating less, and having water-saving systems, are traits frequently observed in animals living in desert environments. Animals living in deserts may survive in their environment despite the intense heat and lack of water because to these characteristics.
Numerous desert animals are nocturnal, meaning they are most active at night when it is cooler and less water evaporates. Desert animals are able to avoid sinking into the sand because to their wide, cushioned feet, which help them disperse their weight over a broader area. Because sweating causes water loss and desert animals must preserve water, they do not sweat as much. Some desert animals have unique water-saving strategies to aid them. For instance, camels can create water when necessary thanks to the fat that is stored in their hump.

12. Correct option: (A) Climate

The average weather pattern taken over a long period, such as 25 years or more, is known as "climate." The term "climate" refers to the normal atmospheric conditions, such as temperature, precipitation, humidity, wind patterns, and other meteorological factors, that are long-term fixtures in a certain area. It helps us comprehend the typical weather patterns encountered in many parts of the world and gives us a larger perspective on the long-term weather trends in a certain place.
Weather refers to the atmosphere's present condition at a specific location and time. The process through which a species or creature adapts to its environment over time is change. Humidity refers to how much water vapor is present in the atmosphere.

13. Correct option: (A) Polar region

The coldest region on Earth is the polar region, specifically the areas near the North Pole and the South Pole. Due to their proximity to the Earth's poles and the prolonged darkness of the winter, these areas suffer very frigid temperatures all year long. The main polar regions recognized for their arctic climates, ice, and snow are Antarctica and the Arctic, both close to the poles.
On the other hand, tropical locations are close to the equator and receive the most direct sunshine, which causes the temperature to rise as well. The temperate zones, which lie in between the tropical and polar regions, are more temperate than any of them. Near the oceans are the coastal zones. Coastal places often have milder temperatures than inland areas because the seas serve to regulate local temperatures there.

14. Correct option: (A) Fatty tissue

The hump of a camel is a reservoir of fatty tissue, not water or milk. The capacity of camels to store fat in their humps, which they may use as an energy reserve during periods of food scarcity, is widely documented. Contrary to common opinion, camels are extremely adept at conserving water and may go for extended periods without drinking. Instead of acting as a water storage organ, the hump. During its travels through dry and desert areas, the camel receives vital nutrition and energy from the fat tissue in the hump. The camel's bodily fluids, not its hump, are where it stores water. The kidneys of the camel are excellent at retaining water. The camel's mammary glands, not its hump, are where milk is produced.

15. Correct option: (A) Pressure

The appropriate response is (a) Pressure. Usually, a daily weather report does not include pressure. In order to convey information about the present or predicted weather conditions, the daily weather report often concentrates on factors like Temperature (b), Rainfall (c), and Humidity (d). Understanding the current atmospheric conditions and making appropriate plans for your everyday activities depends on these factors.

Although crucial for analyzing weather patterns and formulating forecasts, pressure is not often covered in a brief and uncomplicated daily weather report intended for the general audience. Instead, to evaluate and forecast more complicated weather systems, meteorologists frequently study and analyses pressure values and trends.

16. Correct option: (C) Meteorological department

The department that prepares the weather report is the meteorological department. It is in charge of tracking, evaluating, and predicting weather patterns. To provide precise and fast weather reports and predictions, the meteorological department gathers data from numerous weather stations, weather satellites, and other sources. These reports are essential for many industries that rely on weather data, including public safety, aviation, agriculture, and many others.

A nation's telecommunications infrastructure, including phone lines, cell towers, and internet cables, is the responsibility of the telecom department. A nation's electrical system, which distributes electricity to residences and businesses, is under the control of the electrical department. Buildings, bridges, and other public works projects are within the purview of the civil department.

17. Correct option: (D) Moisture

The amount of humidity in the atmosphere is a measure of the moisture content in the air. In order to evaluate weather conditions and determine how comfortable it is for people to be outside, it is crucial to know the air's humidity level. Our sense of temperature can be affected by the air feeling damp and high humidity levels. On the other side, low humidity can result in dry air, which can have an adverse effect on the skin, respiratory health, and some materials.

To determine how hot or cold something is, use the term "heat." A substance's temperature is a measure for the amount of heat it contains. Materials comprised of atoms or molecules are known as chemicals.

18. Correct option: (B) Rain gauge

The instrument used to measure rainfall is called a rain gauge. It is a simple meteorological instrument made to gather and measure the total amount of precipitation—mostly rainfall—over a predetermined time frame. Although there are many other types of rain gauges, the most popular one is a cylindrical container with markings showing the quantity of rainfall in inches or millimeters. Rain comes into the gauge, gathers in the container, and the height of the water gathered reflects the quantity of precipitation that fell during the specified time period.

An instrument used to gauge atmospheric pressure is a barometer. An instrument used to measure temperature is called a thermometer. A maximum-minimum thermometer is a device that measures the temperatures at their highest and lowest points throughout a specific time period.

19. Correct option: (C) Maximum – minimum thermometer

The maximum and minimum temperatures of a day are recorded using a maximum-minimum thermometer. A 24-hour day is typically the time span for which this sort of thermometer is intended to monitor and record the peak and lowest temperatures. Two distinct scales—one for monitoring the highest temperature and the other for the

minimum temperature—and a U-shaped glass tube make up the device. The liquid expands as the temperature rises and travels up the tube, revealing the highest temperature achieved. The liquid, on the other hand, contracts as the temperature decreases and flows down the tube, indicating the minimum temperature.
Both a Celsius and a Fahrenheit thermometer are used to monitor temperature, however neither one records the peak and lowest temperatures for a certain time period. Although a hydrogen thermometer may measure extremely low temperatures, it is uncommon to use one to note the high and low temperatures for a given day.

20. Correct option: (B) Sun

All the changes taking place in the atmosphere are primarily caused by the (b) Sun. The fundamental energy source for the Earth's weather and climate systems is the Sun. The uneven heating of the Earth's surface caused by solar radiation leads to the creation of air masses, wind patterns, and weather events. The water cycle, which produces clouds and precipitation through the evaporation and condensation of water vapor, is likewise fueled by the energy of the Sun.
The atmosphere's changes are not significantly influenced by the Moon, Earth, or stars. The tides are impacted by the Moon's gravity, although the atmosphere is not significantly affected. There are minor changes in the atmosphere brought on by the Earth's rotation, although these changes are minimal. The atmosphere is not much impacted by the stars at all.

21. Correct option: (A) Polar regions

The places where the sun does not rise for six months and does not set for the other six months are the (a) polar regions. These areas are close to the poles of the Earth, notably the North and South Poles. The tilt of the Earth's axis causes continuous darkness for roughly six months during the polar winter and continuous sunshine for about six months during the polar summer. The Earth's axis is tilted in relation to its orbit around the Sun, which affects the length of the day at various latitudes throughout the year.
This phenomenon is not present in the equatorial, northern, or southern areas. Even though the amount of sunshine varies throughout the year, the sun rises and sets in these areas every day.

22. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.
Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

23. Correct option: (D) Weather

he short-term variation that occurs in the atmosphere is called "weather." The daily variations in temperature, humidity, precipitation, wind speed, and cloud cover that can occur over short periods of time are referred to as weather. It is the state of the atmosphere at a specific location and time, offering details about the current circumstances.
Contrarily, climate refers to the typical weather patterns for a certain place over an extended period of time. Usually, the length of a climate measurement is 30 years. The quantity of water vapor in the air is known as humidity. What determines how hot or cold anything is its temperature.

24. Correct option: (B) altitude or height above sea-level

The climate at a place depends on various factors, but one of the most significant factors is altitude or height above sea-level. The atmospheric pressure drops as altitude rises, resulting in thinner air. Weather patterns and temperature are directly impacted by this alteration in atmospheric circumstances. In general, temperatures are cooler at greater altitudes, which explains why mountains and high plateaus typically have colder climates than low-lying locations.

While elements such as air pressure, wind speed, and temperature affect climate in crucial ways, the height or elevation above sea level frequently has a decisive impact on the general climatic features of a given region.

25. Correct option: (D) Polar bear

The animal that possesses blubber under its skin is the (d) Polar bear. Blubber is a thick layer of fat found under the skin of certain marine mammals, such as whales, seals, and polar bears. It serves as an insulating layer, helping these animals to stay warm in cold environments, particularly in the frigid waters of the polar regions. Polar bears, in particular, rely on their blubber to maintain body heat and survive in the Arctic's harsh and icy conditions.

There is no blubber under the skin of red-eyed frogs, elephants, or birds. In tropical jungles, there exist tiny amphibians called red-eyed frogs. In savannas and woodlands, elephants are huge creatures. Feathered creatures with wings are birds. Due to their climatic adaptations, none of these creatures require blubber to keep warm.

LEVEL-II

1. Correct option: (C) Toucan

A big, long beak is one of the distinguishing characteristics of toucans, which is known about. In addition to its amazing size, its beak has rich coloring that is frequently decorated with vivid colors. In addition to other functions, a toucan's beak enables it to reach fruits and seeds that are higher up on branches than it is physically able to reach. It also participates in communication, territorial displays, and assists in mate attraction.

Pigeons, penguins, and kiwis don't have very lengthy beaks. In order to dig for food, kiwis have long, narrow beaks. The small, stubby beaks of penguins are employed for capturing fish. Pigeons utilize their short, pointed beaks to devour seeds and other small foods.

2. Correct option: (D) Catch its prey

A sharp beak is a specialized weapon used by eagles to effectively grab and eat their prey. Eagles are raptors with exceptional hunting skills, and their beaks are built to shred and tear apart their prey's flesh. They can grip and hold onto their catch tightly thanks to their sharp, hooked beak.

Additionally, it helps the eagle swiftly break off bite-sized chunks of meat, making it simpler for it to devour. While eagles occasionally migrate, they are not known for hiding from or fleeing from predators, and their beaks are mostly specialized for their carnivorous diet and hunting ability.

They are apex predators, meaning they are at the top of the food chain and lack natural predators, therefore option A is to flee from them. Because they are huge birds, they are not particularly adept at hiding from predators, thus choose option B. Eagles migrate, option C, although their beak does not aid them in this endeavor.

3. Correct option: (D) All of these

Numerous elements, such as average temperature, soil type, and yearly precipitation, have an impact on the variety of species in varied ecosystems. Each of these elements has a significant impact on the habitat's environmental characteristics, which in turn impacts the kinds of species that may survive there. Different species have distinct adaptations

that enable them to endure and procreate in certain environmental circumstances. For instance, certain species could do better in hot, dry habitats with particular soil types, whereas others might do better in cold, damp conditions with various soil traits. The sum of these elements leads to the total biodiversity seen in different environments across the world.

4. Correct option: (B) Orchids

The term "epiphyte" refers to plants that grow on top of other plants (often trees) without taking nutrients from those plants. The most popular epiphytes are orchids. With the aid of specialized adaptations, they may adhere to tree trunks or branches and draw water and nutrients from the surrounding trash, the rain, and the air.

The popularity of orchids in horticulture and decorative plant collections can be attributed to their well-known for having an array of exquisite blossoms. The well-known tree species fir and pine do not develop as epiphytes on other plants, despite the fact that they are not epiphytes.

Both fir and pine are tree species, not epiphytes. They are rooted to the ground and develop roots as they grow in the soil.

5. Correct option: (B) Hibernation

Hibernation is the term for when animals sleep or go into a state of reduced activity in the winter to preserve energy and endure severe weather. Many animals, especially tiny mammals like bears, groundhogs, and bats, hibernate as a seasonal adaptation to withstand lower temperatures and scarcer food sources. The animal may use its stored energy stores to survive until the warmer months return since during hibernation its metabolic rate and body temperature drop dramatically to fit the environment. Torpor enables animals to endure the harsh winter months when food is in short supply.

Different methods that animals adapt to their surroundings include camouflage, movement, and mimicry. With the use of camouflage, an animal can conceal itself from predators by blending in with its environment. An animal can move about its surroundings through locomotion. Animals can mimic other organisms to look like them, frequently for trickery.

6. Correct option: (D) All of these

In addition to India, Malaysia is located in Southeast Asia, Brazil is in South America, Kenya is in East Africa, the Republic of Congo is in Central Africa, Indonesia is in Southeast Asia, Nigeria is in West Africa, and Uganda is in East Africa. Warm temperatures, high humidity levels, and year-round rainfall in these areas often provide the perfect environment for the luxuriant and diversified ecosystems seen in tropical rainforests. Numerous plant and animal species found nowhere else on Earth may be found in these jungles, which are renowned for their extraordinary biodiversity

7. Correct option: (D) - 37 °C

The polar regions are the coldest regions on Earth. The average temperature in the polar regions in winter can be as low as -37°C. Due to the tilt of the Earth's axis, the sun's rays during the winter months strike the polar areas at a relatively oblique angle. This implies that the Earth's surface is not warmed by the sun's beams for as long.

Snow and ice in the polar regions reflect the sun's rays back into space. The surface of the Earth gets further cooled as a result. Since land surrounds the polar regions, it does not have the same cooling impact as oceans. Oceans absorb heat, preventing extremes in temperature.

8. Correct option: (B) Camouflage

A turtle may camouflage itself by blending in with its surroundings using the colors and patterns on its shell. The ability to blend in with their environment allows animals of

many different kinds to use camouflage, which makes it more difficult for predators to identify them and for prey to detect their presence.
In the case of turtles, the patterns and hues of their shells frequently match the forest floor or other aspects of their natural habitat, giving them some kind of defense against possible dangers. By lowering their probability of being seen by predators or prey, this adaptation boosts their odds of surviving.
Some creatures go into hibernation during the winter months, which is a condition of inactivity and low metabolic rate. Animals travel from one location to another during migration, frequently in pursuit of food or a better environment. Behavior is how an animal acts or responds to its surroundings.

9. Correct option: (C) Average in weather pattern in a few years

Climate is the long-term average of the weather in a certain area over a lengthy period of time, usually spanning decades or even centuries. It considers variables including temperature, humidity, precipitation, wind patterns, and other long-term atmospheric conditions. For a variety of uses, including agriculture, urban planning, and environmental research, climate provides important information that helps us understand the normal or predicted weather patterns in a particular place.
On the other hand, weather is the momentary condition of the atmosphere. From day to day, or even from hour to hour, it might alter. Many of the same elements that affect climate also affect weather, although it is also impacted by more regional causes like thunderstorms and hurricanes.

10. Correct option: (A) A white fur, fat below skin, keen sense of smell

Polar bears can survive in extremely cold settings because to a number of adaptations, including:
Their fur is really translucent, giving it the appearance of being white, which helps them conceal themselves when hunting and blend in with their snowy surroundings.
Polar bears have a thick layer of blubber under their skin that serves as insulation and aids in the preservation of body heat in the chilly Arctic conditions.
Polar bears have a keen sense of smell that enables them to locate seals and other prey at a distance. This skill, which enables them to find food even when it is concealed beneath the ice, is essential to their survival in their frigid habitat.
The alternative choices are incorrect. Large eyes, thin skin, and a long tail would all be problems for bears in the cold. Large eyes would make it difficult for bears to see in the snow. The bears' huge, white paws would aid in swimming but not in keeping them warm.

11. Correct option: (B) Swimming in water

Frogs and ducks have webbed feet as a form of adaptation to help them swim effectively in water. Their feet have larger surface area thanks to the webbing between their toes, enabling them to push against the water with more power. This helps them swim and move through watery surroundings. For ducks and aquatic frogs who spend a lot of time in the water, this adaption is extremely advantageous.
The other options are not correct. Frogs and ducks do not need their webbed feet to hold branches tightly, fly in the air, or grab and eat insects.

12. Correct option: (C) Adaptation

An organism's ability to grow and live in a given climate or habitat is referred to as adaptation. These characteristics are the product of natural selection and evolutionary processes, in which qualities that offer an organism a competitive edge in a particular environment is favored and handed down through generations. In order to properly obtain food, shelter, and reproduce within their ecological niche, organisms undergo adaptations to deal with elements including temperature, precipitation, humidity, and other

environmental circumstances. Therefore, an organism's fitness and success in a given environment are greatly influenced by its adaptations.
The alternative choices are inaccurate. Climate variables including temperature, humidity, and wind are referred to as the short-term weather. Climate is the collection of long-term, typical weather patterns. None of these words discuss the specific traits that an organism possesses that make it suitable to a certain environment.

13. Correct option: (A) 15 º – 40 º C

The temperature in the tropics ranges from 59 º to 104 º Fahrenheit, or 15 º to 40 ºC. Near the equator, tropical zones receive warm to hot temperatures all year long. Due to these areas' close closeness to the equator, where they receive constant, bright sunshine, the temperature there stays relatively high. Depending on the location, altitude, and other local conditions, the precise temperature within this range might change, but tropical locations often have warm climates with high humidity levels that sustain a wide variety of ecosystems and lush flora.

14. Correct option: (B) Arboreal fi burrow

Burrows and arboreal habitats are two separate categories of animal habitats. Burrowing animals dwell underground in tunnels or shelters, whereas arboreal creatures live in trees. Some creatures, like squirrels, may dwell in both burrows and trees, therefore these two habitat types are not mutually exclusive. The pair Arboreal fi Burrow is incorrect since the two phrases are not equivalent.
The other choices are the right ones. Animals classified as terrestrial or aquatic or amphibian live on land, in water, or both.

15. Correct option: (C) Long grasping tail

All the other choices (A, B, and D) are traits that are typically associated with creatures that are evolved to surviving in cold conditions, such as the Arctic or polar areas, which is why C) is the odd one out.
A) Thick layer of fat under the skin: Polar bears and seals use this adaption, known as blubber, to help them insulate themselves.
B) White fur: As a kind of camouflage, several animals have evolved white fur to assist them blend in with their environment.
D) Animals with broad, big feet and keen claws, such as polar bears, frequently exhibit this adaptation, which makes it easier for them to move around and grasp frozen surfaces and to successfully acquire prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

16. Correct option: (B) They cannot swim

Penguins have exceptional swimming skills, making them well-suited to life in the water. They are adept swimmers and use their wings, which have become flippers, to navigate through the water with great dexterity. Penguins spend a significant portion of their lives foraging for fish and other aquatic creatures in the ocean. They are also known for having excellent swimming skills.
All of the other claims are true. Penguins do cuddle up together and have webbed feet and streamlined bodies. Penguins can swim and dive thanks to their webbed feet and streamlined bodies, and they keep warm by huddling together.

17. Correct option: (C) To blend in and hide

The ability to change color allows chameleons to blend in with their environment. This enables them to ambush prey and hide from predators. They can adjust their skin color to

match the colors of nearby objects, such as leaves or branches, which helps them remain unnoticed. Chromatophores are peculiar skin cells that are present in chameleons. These cells carry pigment in a variety of colors. These cells are really growing or shrinking as a chameleon changes color. This alters how much pigment is exposed, which in turn alters the skin color of the chameleon.
The other options are not correct. Chameleons do not change color to use less energy, to change their temperature, or to help them hibernate.

18. Correct option: (A) Highest day temperature and lowest night temperature of the day

The highest recorded temperature during the day, often in the afternoon, is referenced by the maximum temperature that is shown in the weather broadcast. The highest temperature that day was represented by this.
The minimum temperature listed in the weather broadcast is the lowest reading made overnight or early in the morning, usually just before daybreak. The lowest temperature that day was represented by this.
The alternate choices are incorrect. Usually, the weather bulletin does not include the month's highest daytime or nighttime temperatures. Additionally, it is not customary to include the temperature recorded at noon and at midnight because these are not necessarily the highest and lowest temperatures of the day. The average highest and lowest temperatures of the day and night are also not often shown since they are less useful than the actual highest and lowest temperatures.

19. Correct option: (C) Red-eyed frog and beard ape

Red-eyed frogs and beard apes are both found in tropical rainforests, which are hot and humid environments. The rainforests provide these animals with the moisture and shade they need to survive.
All around the world, tropical rainforests are home to red-eyed frogs. They can see in the dark thanks to their vivid red iris. Additionally, sticky pads on the foot of red-eyed frogs enable them to climb trees.
The Western Ghats of India are the home of the bearded ape. They go by the name "lion-tailed macaques." A long, thick mane of hair in bearded apes serves as sun protection. Additionally, they have a long tail that aids in balance while climbing trees.
Elephants in Africa and polar bears can only be found in cold regions; hence they cannot live in hot and humid settings. The tundra and desert, which are both cold and dry habitats, are where you may find camels and reindeer.

20. Correct option: (C) Average weather pattern of many years

Climate is the long-term, typically spanning decades or even centuries, average of the weather in a particular place. Temperature, humidity, precipitation, wind patterns, and other long-term climatic conditions are all factors that are taken into account. Climate offers crucial information that helps us comprehend the typical or anticipated weather patterns in a certain location for a range of tasks, including agriculture, urban planning, and environmental research.
Contrarily, weather refers to the current state of the atmosphere. It might change from day to day, or even from hour to hour. The same factors that influence climate also have an impact on weather, albeit it is also influenced by more localized factors like thunderstorms and hurricanes.

21. Correct option: (D) All of these

Tropical rainforests are characterized by hot and humid climates, enormous numbers and varieties of animals, and competition for food and shelter.

The environment in tropical rainforests is often hot and humid, with average temperatures of 15 ° – 40 ° C. Additionally, there is a lot of humidity, with typical percentages ranging from 70 to 90% Huge variety and quantity of creatures: The tropical rainforests are home to a huge diversity and number of animals. Tropical rainforests are home to about half of all species on Earth, or an estimated 10 million different kinds of plants and animals.
Competition for food and shelter is a result of the high levels of biodiversity seen in tropical rainforests. These creatures may evolve specialized adaptations as a result of this intensive competition.

22. Correct option: (A) They cannot swim

Penguins can swim quite effectively, which makes them perfect for a life in the sea. They are skilled swimmers and can navigate through the water with incredible agility using their wings, which have evolved into flippers. Penguins forage on fish and other aquatic animals in the ocean for a sizable portion of their life. They are also renowned for their prodigious swimming abilities.
The other assertions are all accurate. Penguins have webbed feet, streamlined bodies, and they do snuggle up together. Penguins' webbed feet and streamlined bodies enable them to swim and dive, and they keep warm by huddling together.

23. Correct option: (C) Minimum and maximum thermometer

A unique kind of thermometer known as a minimum-maximum thermometer is used to track the daily minimum and maximum temperatures. There are two bulbs in it, one loaded with mercury and the other with alcohol. The minimum temperature is recorded using an alcohol bulb, while the highest temperature is recorded using a mercury bulb.
The temperature of a person or a material is measured using clinical thermometers or laboratory thermometers, respectively. They are not intended to keep track of the daily minimum and maximum temperatures.

24. Correct option: (D) Tropical rainforest

A tiger is a carnivorous animal that fits the definition; it is distinguished by its striking stripes and extraordinary speed and agility when pursuing prey. In Asia, grasslands and tropical rainforests are the main habitats of tigers. They are strong predators that have evolved to live in these luxuriant, thick forest habitats. These animals have stripes on their body that help them to camouflage in the dense vegetation of the rainforest. They also have sharp claws and teeth that help them to catch their prey.
The Arctic, which is significantly dissimilar to the tropical jungle in terms of habitat, is where polar bears may be found. Tropical rainforests are often very hot and humid, while deserts are typically very hot and dry. Ocean-dwelling animals have gills or a thick covering of blubber as adaptations to help them survive in the saline water.

25. Correct option: (C) Long grasping tail

The qualities of all the other options (A, B, and D) are often associated with animals that have evolved to survive in frigid climates, such the Arctic or polar regions, which makes C) the odd one out.
Polar bears and seals employ this adaptation, known as blubber, to help them insulate themselves. A) Thick layer of fat under the skin.
B) White fur: Several animals have developed white fur as a form of camouflage to help them blend in with their surroundings.
D) This adaptation is typically seen in animals with large, broad feet and sharp claws, such as polar bears, which makes it simpler for them to walk about, grab icy surfaces, and effectively catch prey.
Animals living in cold climates are not often thought to have long, gripping tails. For animals that dwell in trees or deep woods and depend on holding on to branches for

support or movement, a long gripping tail may be more appropriate. It is not a trait frequently observed in creatures that have evolved to cold regions.

26. Correct option: (C) Mix with the surroundings

Tigers and leopards have evolved to have stripes on their body, which gives them the ability to blend in with their environment and serve as an excellent kind of camouflage. Their ability to blend in and approach their target covertly is especially helpful for their hunting tactics.

In the dappled light and shadows of their native environments, such as grasslands and woodlands, the stripes blur the shape of their bodies, making it more difficult for their prey to identify them. Their ability to blend in with their surroundings and approach their victim undetected until they are prepared to strike is crucial to their effectiveness as predators.

27. Correct option: (A)

28. Correct option: (D) a – iv; b – ii; c – i; d – iii

a. Red-eyed frog has iv. Sticky pads on feet. They can climb trees because to this adaptability. Arboreal creatures, red-eyed frogs spend the most of their time in trees. Their feet include sticky pads that allow them to effortlessly move around in trees and hold onto branches.

b. Penguin has ii. Streamline body. This modification enhances their capacity for swift and effective swimming. Penguins can hunt fish and other prey using their streamlined bodies since they are skilled divers and swimmers.

c. Tiger has i. Very sensitive hearing. They have the capacity to hear noises that are imperceptible to humans. This modification makes it easier for them to discover prey and recognize predators.

d. Lion-tailed macaque has iii. Silver-white mane. They can attract mates thanks to this modification. Additionally, the main makes lions appear bigger and more frightening, which can aid in fending off predators.

So, the answer is D. a – iv; b – ii; c – i; d – iii.

29. Correct option: (A) a – iv; b – iii; c – ii; d – i

a) Elements of weather - iv) Temperature, humidity: The term "elements of weather" refers to a variety of elements, such as temperature and humidity, that make up the atmospheric conditions.

b) Cause of all changes - iii) Sun: The sun is the main influence on Earth's weather and climate. All alterations in weather patterns and atmospheric conditions are influenced by its energy.

c) Tropical region - ii) Hot and wet: High temperatures and high humidity combine to create a hot and humid environment in tropical locations.

d) Desert climate - i) Hot and dry: Low humidity and high temperatures create a dry and warm atmosphere in desert settings.

30. Correct option: (B) (i), (iii)

The conditions from the above list that are responsible for the presence of a large number of plants and animals in tropical rainforests are:

(i) Humid and hot climate: Tropical rainforest’s humid and hot climates are ideal for the development and spread of a wide range of plant species. High temperatures and humidity encourage quick plant development, which results in a varied and thick forest canopy.

(iii) Abundant rainfall: Tropical rainforests get an abundance of rainfall, which guarantees a steady supply of water, which is essential for the survival and development of both plants and animals. It supports a wide range of plant growth and offers home to a large number of animal species.

4. Respiration in Organisms

Solutions

LEVEL-I

1. Correct option: (A) Trachea

The "trachea," or network of breathing tubes, is seen in cockroaches. In insects, a system of tubes called a trachea permits the exchange of gases such as oxygen and carbon dioxide. Without the need of lungs or other specialized respiratory organs, these tubes efficiently supply oxygen to the insect's cells and tissues.

In cockroaches and other insects, the tracheal system consists of a network of tiny branching tubes that extend throughout their bodies. These tubes transport oxygen and remove carbon dioxide through microscopic apertures called "spiracles" that are situated along the sides of their bodies.

2. Correct option: (A) Expiration

The act of exhaling involves taking air from the lungs. The diaphragm and intercostal muscles relax during expiration, which reduces the capacity of the chest cavity. The body exhales carbon dioxide-rich air as a result of the pressure rise forcing air out of the lungs.

The other choices are false. The act of being inspired involves breathing in air, not out. The act of breathing in air (option C) involves the diaphragm contracting and the chest cavity expanding, which allows air to enter the lungs. None is not a correct response.

3. Correct option: (D) Dry leaves

Respiration is a metabolic process that involves the exchange of gases, particularly the uptake of oxygen and the release of carbon dioxide. Dry leaves, being dead and devoid of living cells, do not undergo respiration as there are no cellular activities or metabolic processes occurring within them.

Living things use respiration to break down food molecules and release energy. There are several uses for this energy, including movement, reproduction, and growth. Because they lack the water and enzymes required to break down food molecules, dry leaves cannot respire.

In contrast, sprouted seeds, flowers and buds, and fruits are living or metabolically active tissues, and they require respiration to carry out their cellular functions and maintain their life processes.

4. Correct option: (C) Stem

On the outside of plant stems, there are tiny holes or apertures called lenticels. The stem's interior and the environment can interchange gases since they are porous structures.

For gas exchange in woody plants, lenticels are crucial. The stem's bark becomes impervious to gases as it thickens. Lenticels allow for the interchange of gases such as oxygen, carbon dioxide, and water vapour, all of which are necessary for the plant to survive. On leaves or flowers, lenticels are absent.

5. Correct option: (C) Both A and B

The cockroach's body has tiny holes called spiracles that allow air to enter the tracheae. A system of tubes called tracheae is responsible for delivering air to the body's cells.

Because they lack lungs, cockroaches must breathe through their spiracles and tracheae. The tracheae can receive airflow when the spiracles are open. Air flow from the tracheae is halted when the spiracles are closed.

The tracheae are a network of tubes that branch out and deliver air to the body's cells. A thin layer of cells lines the tracheae, allowing carbon dioxide to seep out and oxygen to diffuse into the cells.

6. Correct option: (B) Two

Aestivation in the summer and hibernation in the winter are the two phases of dormancy that frogs experience each year. Frogs experience aestivation, a period of inactivity, when the weather is hot and dry. Frogs endure a period of inactivity known as hibernation as the temperature drops.

Frogs hibernate underground during aestivation to avoid the heat. Until the weather begins to drop off, they could remain underground for several months. Frogs go underground during hibernation to avoid the cold. Until it warms up, they could spend many months underground.

So, a frog will spend two nights sleeping below the ground in a year: once during aestivation and once during hibernation. The solution is B.

7. Correct option: (C) Skin

Frogs can live both in water and on land since they are amphibians. In addition to having lungs for inhaling air, they also have thin, wet skin that enables them to take in oxygen from the water. Frogs hibernate, which causes a decrease in lung capacity, in the winter and summer. During these phases, they must remain in wet settings because they breathe via their skin. Although their metabolic rate drastically lowers when they are inactive, some gas exchange still happens via their skin, which allows them to continue to live without using their gills or lungs for active respiration.

8. Correct option: (A) Gills

Fish are water-based creatures that can breathe underwater. They do not have lungs; thus, they must get oxygen from the water via their gills. On the sides of the fish's head are gills, which are feathery structures. They have a thin layer of highly vascularized skin covering them, which means it is packed with blood vessels. The oxygen in the water diffuses into the blood vessels as the fish moves, causing water to flow over its gills. The oxygen is subsequently delivered to the body's other tissues by the blood.

The respiratory system of terrestrial animals, such as mammals, birds, and certain reptiles, is comprised of the lung. Although fish do have skin and a trachea, those are not their main breathing organs.

9. Correct option: (B) Skin

In order to breathe, earthworms have skin. Earthworms lack lungs or gills, in contrast to humans and many other creatures that rely on these organs for breathing. Instead, they have a skin that is delicate, wet and porous and that permits gas exchange. Through their skin, oxygen from the air around them diffuses into their circulation, while carbon dioxide is released in the opposite way. Because their epidermis has to be moist in order to promote this exchange, earthworms are frequently seen in moist areas like soil or compost piles. Earthworms are able to live and thrive in their underground environments because to their distinctive mode of breathing.

Other options ar4e incorrect. Since earthworms lack lungs, they are unable to breathe air. Segments are the physical components of an earthworm, whereas gills are the breathing portions of fish and other aquatic creatures.

10. Correct option: (C) Both A and B

On the surface of leaves and other plant components are stomata, which are tiny holes or openings. They are essential for transpiration and gas exchange. Although they may be found on the upper side of leaves as well, they are more common on the lower surface. Stomata helps to decrease water loss through transpiration, the bottom surface of leaves often has more stomata than the top surface.

11. Correct option: (A) Priestly

The discovery of oxygen is in fact attributed to English scientist Joseph Priestley. While performing studies on the "dephlogisticated air" (what he then termed oxygen), he discovered the gas in 1774 and noted that it was necessary for sustaining combustion and animal respiration. In terms of our understanding of gases, Priestley's discovery of oxygen was a key turning point. It also served as the foundation for further studies on the function of oxygen in chemical processes and respiration.

The other answer choices are incorrect. The discovery of carbon dioxide is attributed to Von Helmont, the name of oxygen is attributed to Lavoisier, and the Swedish botanist Linneaus is renowned for his method of classifying plants and animals.

12. Correct option: (B) Tracheae

The network of respiratory tubes that carries air directly to body tissues in cockroaches and many other insects is known as the tracheae. The respiratory system of the insect includes these tracheae, which are essential for allowing gas exchange. As air enters and exits the tracheal system through holes on an insect's body surface called spiracles, carbon dioxide and oxygen can be exchanged with the environment.

The other answer choices are incorrect. The apertures that link the tracheae to the insect's body's sides are called spiracles. Lenticels are tiny pores that allow for gas exchange in the bark of plants. On the surface of leaves, stomata are microscopic holes that permit gas exchange.

13. Correct option: (C) Gills

Gills serve as the tadpole larva's respiratory system for breathing. The aquatic larval stage of frogs and several other amphibians is known as a tadpole. They are aquatic creatures that breathe air from the water and expel carbon dioxide via gills. They go through considerable physiological changes throughout metamorphosis and maturation into adult frogs, including the creation of lungs, which become their main breathing system as they switch from an aquatic to a terrestrial habitat.

The other answer choices are incorrect. Mammals and other air-breathing creatures have lungs, which are respiratory organs. Insects can breathe thanks to their network of air passages called a trachea. Although skin is not a respiratory organ, it can assist in gas exchange in some species.

14. Correct option: (B) CO_2

Lime water (calcium hydroxide solution) becomes milky white when exposed to carbon dioxide (CO2). The presence of carbon dioxide gas is easily detected using this reaction as a test. Calcium carbonate, which is insoluble and manifests as a milky white precipitate, is created when carbon dioxide is added to lime water and interacts with the calcium hydroxide. The following is a representation of the reaction:

$CO_{2(g)} + Ca(OH)_{2(aq)} \longrightarrow CaCO_{3\,(s)} + H2O_{\,(l)}$

The alternative response options are wrong. Nitrogen dioxide (NO_2), sulphur dioxide (SO_2), and oxygen (O_2) do not convert lime water milky white.

15. Correct option: (A) Colorless

Like the blood of many other insects, cockroach blood is typically colorless or light yellow. Blood in insects is referred to as hemolymph. The insect's body is supplied with nutrition, hormones, and waste products through hemolymph, among other things. Hemocytes, which are involved in the insect's immunological response, and other types of cells, such as plasma, make up the majority of the hemolymph of insects.

The other answer choices are incorrect. Some insects, like the horseshoe crab, have blue blood. Some amphibians, like the fire salamander, have green blood. Most vertebrates, including humans, have red blood.

16. Correct option: (B) Inspiration

Inspiration or inhalation refers to the action of drawing air into the lungs while breathing. The muscles between the ribs and the diaphragm tighten during inspiration, enlarging the chest cavity. A rush of outside air enters the lungs to restore pressure after this expansion reduces the pressure there. After being breathed, oxygen from the air diffuses into the circulation, where it is delivered to cells for a variety of physiological functions.

The other answer choices are incorrect. Inspiration is the polar opposite of exhalation. Small pores called lenticels allow for gas exchange in the bark of plants.

Exhale is a colloquial word for exhalation.

17. Correct option: (D) Both A and B

Adult frogs breathe both through their skin and lungs. Frogs are amphibians that transform from tadpole larvae to adult frogs. They mainly breathe through their gills in their watery environment as tadpoles. But as they change into adult frogs and grow lungs, they can breathe air when they are on land. Additionally, their skin is semi-permeable and able to take in oxygen from the environment, which helps with breathing, particularly when they are in water or in wet situations. Adult frogs can adapt to a variety of environmental situations thanks to their dual mechanism of breathing (lungs and skin).

Gills are only found in tadpoles and a few other amphibians.

18. Correct option: (C) Cockroach

Insects and several other arthropods use a breathing technique called tracheal respiration. Without the need of lungs or gills, insects can directly transfer oxygen to their tissues thanks to a network of tubes called tracheae. Through tiny holes on the body's surface known as spiracles, these tracheae are linked to the outside world. The body parts of cockroaches have spiracles, which let air to enter and exit the tracheal system and facilitate gas exchange for breathing. Cockroaches utilize their tracheal system to breathe and satisfy their oxygen requirements, unlike animals like fish or dogs who use gills or lungs for breathing.

19. Correct Matching is: a-S; b-T; c-Q; d-P; e-R

a) Tadpole - S) Gills

Amphibians called tadpoles are aquatic creatures. They take oxygen from the water by breathing through gills, which are respiratory organs.

b) Cockroach - T) Tracheae

Using a network of tubes called tracheae to carry oxygen throughout their bodies, cockroaches are insects that breathe through these tubes.

c) Earthworm - Q) Skin

Having no lungs, earthworms are an invertebrate. Their skin, which is both thin and wet, serves as their respiratory system, allowing oxygen to pass through and into the circulation.

d) Frog - P) Lungs

Amphibians like frogs may live both on land and in water. When they are on land, they breathe via their lungs, and when they are in water, they breathe through their gills.

e) Plants - R) Lenticels

Plants lack lungs and gills. Stomata, which are microscopic holes on their leaves, are how they breathe. Stomata enable oxygen to enter the plant and carbon dioxide to leave.

20. Correct Matching is: a-R; b-T; c-P; d-Q; e-S

a) Respiration - R) Exchange of gases

The process of breathing involves exchanging gases between the body and the outside air. Gas exchange enables the organism to take in oxygen and expel carbon dioxide.

b) Lime water turns milky - T) Carbon dioxide

Calcium carbonate, a white precipitate, is created when carbon dioxide is bubbled through lime water and combines with the calcium hydroxide present in the lime water.
c) Cockroach - P) Network of respiratory tubes
A system of tubes called tracheae is used by cockroaches to breathe. Openings known as spiracles link the tracheae to the exterior of the body.
d) Fish - Q) Absorbs oxygen dissolved in water
Fish breathe by consuming oxygen that has been dissolved in water. They accomplish this by using their gills, which are feathery organs situated on the sides of their heads.
e) Roots - S) Respire through airspaces
Roots breathe through air voids since they lack lungs and gills. In the soil, there exist microscopic air pockets called airspaces.

LEVEL-II

1. Correct option: (A) Mitochondria
The mitochondria are where food is oxidized inside of cells. Eukaryotic cells include organelles called mitochondria that are in charge of producing energy during the process of cellular respiration. During cellular respiration, glucose and other nutrients are oxidized, and adenosine triphosphate (ATP) is created as the energy source. This activity, which takes place in the mitochondria's inner membrane, is crucial for supplying energy for many cellular functions.
The other answer choices are incorrect. The genetic makeup of the cell is housed in chromosomes. Protein synthesis takes place in ribosomes, which are organelles. The packaging and distribution of proteins is carried out by Golgi bodies, organelles.
2. Correct option: (C) Lungs
The lungs are the parts of the body in charge of giving the blood oxygen. Inhaling air through the mouth and nose, it is then passed through the alveoli, which are tiny air sacs in the lungs, to do this. Small blood veins called capillaries round the alveoli. Both carbon dioxide and oxygen from the air permeate into the blood in the capillaries and alveoli, respectively. After returning to the heart, the blood is subsequently circulated throughout the body by the heart.
Blood oxygenation does not occur at the heart. It is in charge of moving blood to all parts of the body, including the lungs. It is a gland called the pancreas that produces hormones and digestive enzymes. The liver is a sizable organ that performs several different tasks, including as detoxification, metabolism, and nutrition storage. These organs are not in charge of blood oxygenation.
3. Correct option: (A) 15 – 18 times
An adult individual breathing normally at rest takes 15 to 18 breaths per minute. Age, degree of exercise, and health state are a few variables that might affect this. Children often breathe more quickly than adults do, while persons who are exercising or suffering from a respiratory disease may breathe more slowly.
A medical issue may be indicated by breathing rates that are outside of the usual range. For instance, bradypnea or tachypnea, which are abnormally slow or rapid breathing rates, might indicate a cardiac issue, respiratory sickness, or brain disorder.
4. Correct option: (C) 12 pairs of ribs
Twelve pairs of ribs round the chest cavity. The sternum, or breastbone, is directly connected to the first seven pairs of ribs. By use of cartilage, the following three pairs of ribs are indirectly connected to the sternum. Because they are completely detached from the sternum, the final two pairs of ribs are known as floating ribs.

The heart, lungs, and other internal organs in the chest cavity are shielded by the ribs. The diaphragm, the muscle that divides the chest cavity from the abdominal cavity, is also assisted by them.

5. Correct option: (B) Diaphragm

The diaphragm is the term for the muscle that lines the chest cavity. It is a dome-shaped muscle that divides the abdominal cavity from the thoracic cavity in the chest. A key part of respiration is played by the diaphragm. It flattens and contracts, which causes the chest cavity to open up and enable air to enter the lungs during inhalation. The diaphragm rises when it relaxes, squeezing the lungs and releasing air during exhale. One of the main muscles involved in breathing, this muscle is crucial for the respiratory system's proper operation.

The other options are incorrect. The bony structure that encloses the chest cavity is known as the rib cage. The windpipe, or trachea, is where air travels from the larynx to the lungs. The trachea's branches that enter the lungs are known as bronchi.

6. Correct option: (D) Lactic acid only

The demand for energy in muscle cells is great during intense activity, but the oxygen supply is constrained. The muscle cells switch to anaerobic respiration, which doesn't need oxygen, under this scenario. Lactic acid is a waste product of anaerobic respiration.

Lactic acid has been linked to muscular discomfort and weariness. Additionally, it explains why our muscles feel exhausted after working out. Following exercise, the liver finally breaks down the lactic acid and transforms it into carbon dioxide and water.

Anaerobic respiration does not result in the production of alcohol in muscle cells. It is created by the anaerobic respiration process known as fermentation, which is carried out by yeast and bacteria.

7. Correct option: (C) Epiglottis

The epiglottis protects the entrance of the pharynx, which serves as a shared conduit for both food and air. The epiglottis, a flap-like structure at the base of the tongue, serves as a barrier to keep food and liquids out of the lungs and trachea while being swallowed. In order to prevent food or liquid from entering the airway when you swallow, the epiglottis covers the trachea's aperture. By doing this, the respiratory system is shielded against aspirational and choking hazards.

The other options are incorrect. The protrusion of the thyroid cartilage in the neck is known as the Adam's apple. It is not responsible for protecting the pharynx's entrance. The tiniest air sacs in the lung’s called alveoli are where gas exchange occurs. They are not engaged in securing the pharynx's opening. The windpipe that delivers air from the larynx to the lungs is called the trachea. It is not responsible for protecting the pharynx's entrance.

8. Correct option: (A) Spiracles

In insects, spiracles, or tiny apertures, allow air to enter the body. The insect's body has small holes called spiracles that are often found on the thorax and abdomen. The insect's respiratory system, which consists of a web of tubes called tracheae, can breathe thanks to these apertures. Without the need of lungs or gills, respiration is made possible via the tracheae, which supply oxygen straight to the insect's tissues. By efficiently exchanging air through spiracles and tracheae, insects are able to satisfy their oxygen needs and endure in a variety of situations.

The other options are incorrect. For gas exchange in water, gills are employed. Since insects cannot survive in water, they lack gills. For air gas exchange, lungs are employed. Because they lack lungs, insects do not breathe air the same way that humans do.

9. Correct option: (A) Food is broken down in the absence of oxygen to release energy.
A type of respiration that doesn't need oxygen is called anaerobic respiration. When oxygen is scarce or nonexistent, organisms use it. Anaerobic respiration is the breakdown of food to produce energy, although it is less effective than aerobic respiration. Less ATP is produced by anaerobic respiration than by aerobic respiration.
The other options are incorrect. The statement "Breakdown of food occurs in the presence of oxygen" is false since anaerobic respiration doesn't require oxygen. The statement "it is a slow process compared to aerobic respiration" is untrue since anaerobic respiration may occur quickly, especially in creatures that have evolved to it. It produces a significant quantity of energy compared to aerobic respiration, yet anaerobic respiration actually generates less ATP than aerobic respiration.
10. Correct option: (D) Tracheae
Insect tracheae are air passages. These are a system of tiny tubes that run the length of the insect's body and allow oxygen to reach the tissues directly. Air enters the insect's body through spiracles, or tiny apertures, and flows through the tracheae to all of the cells, facilitating gas exchange. Insects are highly adapted to their varied habitats because to their special respiratory mechanism that allows them to easily satisfy their oxygen requirements without the need for lungs or gills.
The other options are incorrect. For gas exchange in water, gills are employed. Since insects cannot survive in water, they lack gills. The word "air tubes" does not specifically refer to an insect's air breathing mechanism. Fish employ air bladders for buoyancy. Since insects do not inhabit aquatic environments, they lack air bladders.
11. Correct option: (B) Anaerobically producing alcohol
Yeast is used in alcohol industries because it undergoes anaerobic respiration, producing alcohol as a byproduct. Yeast cells ferment carbohydrates in the absence of oxygen to create ethanol (alcohol) and carbon dioxide. Fermentation of alcohol is the name given to this process. To generate alcoholic drinks like wine, beer, and other spirits, yeast is added to fermentable materials like fruit juices or grains in the alcohol industry. The generation of carbonation in beverages like beer and sparkling wines is one procedure that makes use of the carbon dioxide produced during fermentation.
12. Correct option: (C) Crocodile, dolphin, fish
The odd one out is crocodile, dolphin, fish because dolphins and crocodiles are both mammals and reptiles, respectively, and both have lungs as their respiratory organs. In contrast, fish have gills, which are very different from lungs in terms of how they breathe.
Cockroaches, grasshoppers, and ants breathe through tracheae, which are a network of tubes that carry air directly to the cells. Lizards, cows, and snakes breathe through lungs, which are sacs that expand and contract to bring air into and out of the body. Crows and goats also breathe through lungs.
13. Correct option: (D) All of these
An organism is considered to be aerobic if it can create energy through aerobic respiration and needs oxygen to survive. The three options—cat, dog, and person—all represent different types of aerobes. These creatures can breathe in oxygen from the air thanks to their specialized respiratory systems, which include lungs. The aerobic respiration mechanism, which produces energy in the form of ATP (adenosine triphosphate) by breaking down glucose and other nutrients in the presence of oxygen, depends on oxygen. The most effective method of creating energy is aerobic respiration, which is used by a variety of creatures, including humans, cats, dogs, and many other animals.
14. Correct option: (D) Fish
Fish have a substantially faster respiratory rate than the other alternatives, on average. Fish have a unique respiratory mechanism that enables them to take in oxygen from the

water. Their gills, which are situated on the sides of their heads, are in charge of absorbing oxygen dissolved in water through a process known as gill respiration. The gill filaments, which are covered with water, offer a substantial surface area for gas exchange. This enables fish to acquire enough oxygen for their metabolic requirements in water, where oxygen concentration is often lower than in the air. Fish thus breathe more often to make up for the reduced oxygen levels in their aquatic habitat.

In contrast, mammals like humans and dogs (options A and B) have lungs and breathe air, while birds like sparrows (option C) have efficient respiratory systems but still not as high as that of fish.

15. Correct option: (B) Lactic acid

Lactic acid is the byproduct of the metabolism of glucose in anaerobic circumstances (without oxygen) in muscle cells. Muscle cells use anaerobic respiration, which uses less oxygen than aerobic respiration, to create energy. Lactic acid and a tiny quantity of energy (ATP) are produced during anaerobic respiration from glucose's incomplete breakdown. As a result, less ATP and carbon dioxide are produced as a result of this process than through aerobic respiration, which is more effective. Muscle tiredness and pain after strenuous physical activity can be caused by the buildup of lactic acid in muscle cells.

The other answer choices are incorrect. Aerobic respiration, which needs oxygen to function, creates water, ethanol, and carbon dioxide.

16. Correct option: (D) Alveoli

Alveoli are the tiny, balloon-like structures with thin walls that aid in gas exchange. The lungs contain microscopic air sacs called alveoli, which serve as the main points of gas exchange between the respiratory system and the bloodstream. The oxygen from the air that is inhaled diffuses into the capillaries that surround the alveoli and is subsequently delivered to the cells of the body for a variety of physiological activities. As the body prepares to exhale during expiration, carbon dioxide, a waste product of cellular respiration, diffuses from the blood into the alveoli. The alveoli are crucial for the lungs' respiratory system because of their huge surface area and thin walls, which allow for efficient gas exchange.

The other answer choices are incorrect. Stomata, which are tiny apertures in plant leaves that permit gas exchange, are similar to veins in that they convey blood away from and towards the heart, respectively.

17. Correct option: (B) Tracheae

Tracheae, an organ, are used by cockroaches to breathe. Insects have a system of air passages called tracheae that allow oxygen to reach the tissues directly and enable breathing. Insects like cockroaches have tiny holes on their bodies called spiracles that connect these tubes to the outside world. Air enters the tracheal system through the spiracles and travels to various bodily tissues, supplying the oxygen needed for metabolic activities.

The other answer choices are incorrect. Humans and other mammals utilize their noses as a sense of smell. Since they lack a nose, cockroaches are unable to smell.

18. Correct option: (C) Ribs move upward and outward

The ribs expand and rise during inspiration (inhalation). The external intercostal muscles and the diaphragm both contract during inspiration. The external intercostal muscles between the ribs contract as the diaphragm goes lower, forcing the ribs higher and outward. The capacity of the chest cavity is increased by this expansion, which lowers the pressure inside the lungs. In order to equalize the pressure, air is pulled into the lungs from the outside environment, and oxygen enters the circulation through the respiratory system.

During expiration, the diaphragm relaxes and the ribs move downward and inward.

19. Correct option: (B) Yeast

An organism that has anaerobic respiration is yeast. The process of producing energy from glucose without the need of oxygen is known as anaerobic respiration, commonly referred to as fermentation. A form of fungus called yeast is capable of both anaerobic (without oxygen) and aerobic respiration. Anaerobic fermentation is the process by which yeast transforms glucose into ethanol (alcohol) and carbon dioxide when oxygen is not present. In baking, where the carbon dioxide created during fermentation makes bread rise, this method is frequently employed to make alcoholic drinks like beer and wine.

The other answer choices are incorrect. Amoeba, Paramecium, and Euglena are all single-celled organisms that require oxygen to respire.

20. Correct option: (D) All the above

A) Children breathe 20–30 times per minute on average. Children often have a greater respiratory rate than adults.

B) The body's need for oxygen rises when a person engages in physical activities or workouts that demand greater energy. The person breathes more quickly and deeply to satisfy the increased demand and provide the cells more oxygen for generating energy.

C) Heavy physical activity or exercise may result in a higher energy expenditure, which in turn may cause hunger pangs. In order to recuperate and restore the energy stores depleted during exercise, the body needs more nutrients and energy, which increases feelings of hunger.

Hence, All the statements are correct.

21. Correct option: (B) Lime water

The chemical that is used to check for CO2 gas in exhaled air is lime water. Calcium hydroxide (Ca (OH)2) in lime water combines with carbon dioxide (CO2) when it is bubbled through it to generate calcium carbonate (CaCO3), an insoluble deposit that is milky white in color. A quick test to determine whether carbon dioxide is present in a gaseous mixture, such as exhaled air, is this reaction. The presence of CO2 is shown by the milky color of lime water following exposure to exhaled air.

The other answer choices are incorrect. Limestone, lime juice, and quick lime are all forms of calcium oxide. They do not react with carbon dioxide to form a white precipitate.

22. Correct option: (D) Larynx

The vocal cords, commonly referred to as the voice box, are located in the larynx. A component of the respiratory system in the throat is called the larynx. It houses the vocal cords, which are two tissue bands that span the larynx. Inhaling air causes the vocal cords to vibrate, which in turn generates the sound waves that make up our voice. We may talk and sing by producing various pitches and tones by manipulating the tension and location of the vocal cords.

The other answer choices are incorrect. The area of the throat that joins the mouth and oesophagus is known as the pharynx. The mouth is the facial aperture used for speaking and eating. When swallowing, a flap of cartilage called the epiglottis covers the larynx to stop food from entering the trachea.

23. Correct option: (A) Cockroaches breathe with the help of air tubes called spiracles

On the exterior of insects, including cockroaches, are microscopic apertures or pores called spiracles. These spiracles allow air to enter the respiratory system and are linked to a system of tubes known as tracheae that directly delivers oxygen to the tissues without the need for lungs. Because cockroaches do not actually breathe through spiracles, option A is false.

The other answer choices are correct. Exchange of gases in the leaves take place with the help of stomata. The diaphragm forms the floor of the chest cavity. The roots of a plant take up oxygen from the air trapped between the soil particles

24. Correct option: (B) Carbon dioxide

Carbon dioxide (CO_2) is the primary component of the raw materials used in photosynthesis in the atmosphere. Photosynthesis is the process through which plants turn carbon dioxide and water into glucose, a kind of sugar, and oxygen. Through little holes known as stomata, plants' leaves absorb carbon dioxide from the atmosphere. The plant's tissues may absorb carbon dioxide through these stomata, where it forms one of the key ingredients in the process of photosynthesis, which creates food.

The other answer choices are incorrect. Nitrogen and hydrogen are not raw materials for photosynthesis, and oxygen is not present in the air in its constituent forms.

25. Correct option: (B) Oxyhemoglobin

Oxyhemoglobin is the type of haemoglobin that transports oxygen molecules. Haemoglobin and oxygen are combined to form oxyhemoglobin, a protein found in red blood cells. Oxyhemoglobin is created in the red blood cells when oxygen attaches to the iron atoms in the haemoglobin molecules during inhalation into the lungs. Then, when this oxyhemoglobin travels through the circulation to various tissues and organs, it releases oxygen for cellular use in various metabolic activities. Oxyhemoglobin is converted to deoxyhemoglobin during this release of oxygen, making it ready to absorb additional oxygen during the subsequent cycle of respiration.

The other answer choices are incorrect. Haemoglobin and carbon monoxide combine to generate the substance known as carboxyhemoglobin. Plant cells include organelles called chloroplasts. They carry out the process of photosynthesis.

26. Correct option: (B) Capture of food and movement

Amoebae have what are referred to as pseudopodia, or fake feet, which are employed for both mobility and food capture. Amoeba is a single-celled creature that moves and collects food via pseudopodia, which are transient extensions of its cell membrane. The amoeba may move ahead by expanding pseudopodia in the direction of movement. Additionally, it stretches pseudopodia around any food particles, such as bacteria or algae, and engulfs them, creating a food vacuole. Phagocytosis, the process of engulfing food, enables the amoeba to seize and ingest its meal.

The false feet of amoeba are not used for exchange of gases. Amoeba respires through its entire body surface.

27. Correct option: (A) Oesphagus, large intestine, rectum

Even though the oesophagus, large intestine, and rectum are all components of the digestive system, they are not involved in digestion. Food must go via the oesophagus from the mouth to the stomach. From the food that has been digested, the large intestine absorbs water and electrolytes. The digested food is kept in the rectum until it is time for elimination.

The digestive processes are performed by the stomach, small intestine, and buccal cavity. With the aid of the tongue and teeth, the buccal cavity mechanically breaks down food. Gastric juices are released by the stomach and aid in further food digestion. Nutrients from the broken-down meal are absorbed in the small intestine.

28. Correct option: (B) To trap dust particles

The purpose of the nasal hair, sometimes referred to as the cilia, is to collect airborne dust, pollen, and other contaminants that are breathed via the nose. These hairs function as a physical barrier to stop these particles from entering the lungs or other parts of the respiratory system. The cilia normally eject or propel the trapped particles backward in the direction of the throat, where they are either ingested or coughed up. This system aids in shielding the sensitive respiratory tissues from possible irritants and airborne contaminants.

The other answer choices are incorrect. Nose hair does not protect mucus glands, secrete mucus, or keep germs from getting into the lungs.

29. Correct option: (D) The contraction of muscles in the wall of food pipe

The peristaltic motions of the muscles in the wall of the food pipe (oesophagus) are principally responsible for the ingested food moving downhill through the alimentary canal. Peristalsis, a synchronised muscle contraction and relaxation, aids in moving food through the digestive system in a manner akin to a wave. The circular and longitudinal muscles in the esophageal wall cooperate to produce regular contractions when food is ingested and enters the oesophagus. The food is propelled downhill by these contractions as it moves from the mouth to the stomach and onward through the digestive tract for absorption and processing.

The other answer choices are incorrect. However, it is not the primary power that propels the food down the canal; rather, the muscular tongue assists in pushing the food down the food pipe. Although it is not the dominant force, gravitational pull helps the food go down the canal. Food does not go more quickly down the canal while water is flowing alongside it.

30. Correct option: (D) (i) and (iii)

The act of breathing involves taking in oxygen-rich air and expelling air that is carbon dioxide-rich. While breathing in, oxygen from the air enters the lungs and travels via the circulation to the body's tissues, giving cells the oxygen, they require for numerous metabolic activities. On the other hand, when you exhale, your body flushes out carbon dioxide, a waste product of cellular metabolism. The ingestion of oxygen and expulsion of carbon dioxide during this gas exchange are essential for the body's efficient operation and the preservation of a stable internal environment.

Food does not digest to release energy during breathing. Cellular respiration, a mechanism that occurs in the mitochondria of the cells, carries out this function. The cells do not create water when breathing. Although it is not created by breathing, water is a consequence of cellular respiration.

5. Transportation in Organism
Solutions

LEVEL-I

1. Correct option: (C) Diffusion

The picture provided exemplifies the process of diffusion. Due to the inherent tendency of particles to spread out and reach equilibrium, diffusion is the passive movement of particles or molecules from an area of higher concentration to an area of lower concentration. The presented graphic shows the flow of particles from a left-side area with a higher concentration to a right-side area with a lower concentration. This spontaneous movement without external energy is a distinguishing feature of passive transport mechanisms.

Osmosis, isotonic, and active transport are not viable possibilities in this situation since they relate to particular cellular transport processes that entail different mechanisms and factors.

2. Correct option: (A) water moves from high to low concentration.

Water does definitely migrate during osmosis from a region with a high concentration of water and a lower concentration of solutes (low water concentration). The objective of this movement is to balance the solute content on each side of a selectively permeable membrane. It is a passive process that doesn't need energy to be added. The core idea behind osmosis is correctly stated in this response.

Osmosis, unlike active transport, doesn't require energy expenditure and only works with small, simple molecules like water. This response is consistent with the basic idea that osmosis is a passive process fueled by concentration gradients.

3. Correct option: (B) Xylem

The plant tissue known as xylem is designed specifically for transporting water and minerals from the roots to other areas of the plant. It is made up of different cell types, such as tracheids and vessel components, which are made to transport water and nutrients through the plant effectively. This tissue is the proper response to the question since it supports the plant and keeps the water balance in check. The xylem predominantly conducts water and minerals, whereas parenchyma, phloem, and collenchyma serve other purposes like storage, nutrient delivery, and structural support.

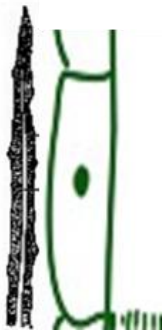
4. Correct option: (A) Phloem

The correct answer is A) Phloem. The plant tissue known as phloem is in charge of transporting material both upward (from leaves to other parts of the plant) and downward (from sources to sinks, such roots), chiefly sugars and other organic molecules created during photosynthesis. Phloem is a good option for the question since it transports nutrients and substances that are high in energy in both directions. In a plant, parenchyma, xylem, and collenchyma each have a different function, but only phloem is designed specifically for bidirectional material transfer.

5. Correct option: (B) Phloem fibers

The correct answer is B) Phloem fibers. In the phloem tissue of plants, sclerenchyma cells of the phloem fiber type are present. By giving the plant support and stiffness, they play a structural role. Phloem fibers are non-living at maturity because they are made of thick walls that are strengthened with lignin, unlike sieve tubes, companion cells, and phloem parenchyma. Their main job is to give the plant mechanical strength, which helps maintain its overall structural integrity. Unlike other phloem components like sieve tubes, companion cells, and phloem parenchyma, which are living cells active in a variety of physiological processes like nutrient transfer and support, these are not.

6. Correct option: (D)



Companion cells are specialised plant cells that collaborate with sieve tube components in the phloem tissue to ensure the effective delivery of nutrients, mostly sugars generated during photosynthesis, to different regions of the plant. These cells support the sieve tube elements' operations and are essential to the process of long-distance nutrition distribution.

Companion cells stand out from other cell types due to the following characteristics:

Sieve tube components have a close relationship with companion cells.

Companion cells have a noticeable nucleus that controls their metabolic processes and enables them to supply the essential materials to filter tube components.

7. Correct option: (A) Sieve elements

The lengthy, tube-like cells known as sieve elements are responsible for moving food and other substances through the phloem. Sieve plates, which have perforated areas that let chemicals move from one cell to the next, connect them end to end.

Companion cells, which are tiny cells situated next to the sieve elements, sustain the sieve elements. Companion cells assist in supplying the sieve elements with nutrients and energy as well as regulating the flow of materials through the sieve elements.

The phloem is made up of tough, fibrous cells called bast fibers. They aid in the transportation of water and nutrients and assist the companion cells and sieve components. The ground tissue of the phloem is called phloem parenchyma. It is made up of cells that assist carry water and nutrients as well as store food and other chemicals.

8. Correct option: (B) Arteries

Blood channels called arteries transport oxygenated blood from the heart and distribute it to various tissues and organs throughout the body. They have layers of muscle and elastic tissue and thicker walls than other blood veins, which enable them to withstand the strain brought on by the heart's pumping motion. Because of their muscular makeup, arteries are able to control blood flow and keep their internal pressure relatively constant. Contrarily, veins transport deoxygenated blood back to the heart while capillaries are microscopic, thin-walled blood arteries that enable the exchange of nutrients, oxygen, and waste products between the blood and surrounding tissues.

9. Correct option: (A) Veins

The blood vessels that transport deoxygenated blood are called veins. Veins return blood to the heart, as opposed to arteries, which carry oxygenated blood elsewhere. Blood that has already carried oxygen to the body's tissues and organs and is now returning to the heart to be reoxygenated is referred to as deoxygenated blood. Veins can accept the lower pressure and slower flow of deoxygenated blood because they have thinner walls and greater lumens than arteries. Additionally, veins have valves that help maintain the direction of blood flow toward the heart and prevent blood from flowing backward. Overall, veins are essential to the circulatory system because they return blood that has lost oxygen to the heart, where it is reoxygenated.

The body receives oxygenated blood from the heart through arteries. The body's tiniest blood vessels are called capillaries. They link veins and arteries. In the capillaries, carbon dioxide and oxygen are exchanged.

10. Correct option: (B) A and C

In the following picture, the cells next to the guard cells can be used to identify the subsidiary cells. A and C are the subordinate cells in this picture. In the epidermis of plants, particularly in the stomata, subsidiary cells are specialized cells. They help the guard cells' operation and are situated on either side of them. By supporting and directing the guard

cells' movement, the subsidiary cells aid in managing the stomata's opening and closing. Consequently, the subsidiary cells in this image are A and C.

11. Correct option: (C) 1.0 – 2.0

The answer to the question is C. 1.0 – 2.0 liters. This range represents the average amount of urine excreted by a human in a day. The kidneys play a crucial role in maintaining the body's water balance and removing waste products. They filter the blood and produce urine, which is then transported to the bladder for storage and eventual elimination. The amount of urine produced can vary depending on factors such as fluid intake, physical activity, and overall health. On average, a healthy adult excretes about 1.0 – 2.0 liters of urine per day. However, it is important to note that individual variations can occur, and certain medical conditions or medications may affect urine production.

12. Correct option: (A) 95

Water makes up the majority of the contents of urine—roughly 95% of it. Around 2.5 percent of urine is normally made up of urea, while the remaining 2.5 percent is made up of various waste products, electrolytes, minerals, and substances that the body has to expel. These substances can be found in different concentrations in urine, depending on things including hydration levels, food, and general health. The body's attempts to balance its fluid and electrolyte levels while removing waste items are reflected in the composition of urine, which is a reflection of proper hydration.

13. Correct option: (B) Urinary bladder

The urinary bladder is the organ that holds the urine. Behind the pubic bone in the pelvis, the urine bladder is a muscular sac. Its main job is to keep urine in reserve until the body is ready to expel it. The bladder extends and swells as it fills with urine, signifying the urge to urinate. The bladder may extend without leaking because it is coated by a specific tissue called transitional epithelium. Urine is released through the urethra when the bladder contracts. Overall, by temporarily holding urine until it is expelled, the urinary bladder contributes significantly to the excretory system.

14. Correct option: (B) Urethra

The duct that expels urine from the body is called the urethra. It is a tube that connects the urinary bladder to an aperture on the outside, enabling the body to expel pee. Urine passes through the urethra and leaves the body when the urinary bladder is sufficiently full and the person voluntarily relaxes the muscles around the bladder. The male urethra eliminates pee in addition to acting as a pathway for semen during ejaculation. Although the urethra can vary in length and structure between males and females, its main purpose is to provide a pathway for the body to expel urine.

15. Correct option: (D) Capillaries

Capillaries are largely responsible for the exchange of breathing gases between the blood and tissues as well as the absorption of nutrients. The tiniest blood veins in the body, capillaries have thin walls that enable chemical exchanges between the blood and tissues nearby. The capillaries in the small intestine absorb nutrients from digested food into the bloodstream, whereas the capillaries in the lungs absorb oxygen and distribute it to the body's cells. Also extracted from the tissues and carried through the capillaries to be expelled from the body are waste materials like carbon dioxide. Capillaries serve a critical function in allowing the flow of chemicals between the blood and tissues because of their large network and near closeness to cells.

16. Correct option: (C) Pulmonary vein carries blood to the heart.

The right answer is C. Blood is carried to the heart by the pulmonary vein. The pulmonary vein is in charge of returning oxygenated blood to the heart from the lungs. The pulmonary

veins gather the blood after it has been oxygenated in the lungs and return it to the left atrium of the heart. It is subsequently pushed into the left ventricle and distributed throughout the body via the systemic circulation from there. The pulmonary artery, in contrast, transports deoxygenated blood from the heart to the lungs where it is oxygenated. As a result, choice C perfectly sums up the role of the pulmonary vein in the cardiovascular system.

17. Correct option: (D) Upper part of the body to the heart.

A sizable vein called the superior vena cava transports blood depleted of oxygen from the upper body to the heart. It starts at the top of the body, close to the head and neck, and draws blood from various areas including the upper chest, upper arms, and head. The right atrium of the heart is where the deoxygenated blood is then moved before being pushed to the lungs for oxygenation. Because it appropriately defines the blood flow path through the superior vena cava, choice D, "upper part of the body to the heart," is the appropriate response.

18. Correct option: (A) Lower parts of the body to heart

Blood travels through the inferior vena cava from the lower body to the heart. The inferior vena cava is a sizable vein that returns blood that has lost oxygen to the right atrium of the heart from the lower body, including the abdomen, pelvis, and legs. After being pumped into the right ventricle, this blood is finally delivered to the lungs to receive oxygen. Blood circulation depends on the inferior vena cava, which makes sure that deoxygenated blood returns to the heart to be reoxygenated and circulated once more throughout the body.

19. Correct option: (B) Flow of blood

The regular expansion and contraction of an artery brought on by blood flow constitute the pulse. The wrist, neck, and temples are only a few places on the body where the pulse can be felt. The heart's contraction is what causes the pulse to beat. Blood is pumped into the arteries while the heart beats. The arteries widen as a result of this. The blood returns to the heart when it is at ease. The arteries constrict as a result. The heart rate determines the pulse's frequency. The number of times the heart beats each minute is known as the heart rate. The average heart rate for an adult is about 72 beats per minute. The strength of the pulse is determined by the amount of blood being pumped by the heart. The stronger the heart, the stronger the pulse.

20. Correct option: (C) Vessels

In general, vessels are tubes in a human or animal's body that transport red liquid, typically blood. Veins and arteries are examples of vessels, which are in charge of carrying blood around the body. While arteries carry oxygenated blood out from the heart, veins convey deoxygenated blood back to the heart. The exchange of nutrients, oxygen, and waste products between the blood and neighboring tissues is made possible by capillaries, which are microscopic blood channels that connect arteries and veins. Because of this, the word "vessels" is a more broad and appropriate way to refer to the numerous kinds of tubes that the red liquid in the body travels through.

Arteries carry blood away from the heart. Veins carry blood back to the heart. Capillaries are the smallest vessels and connect arteries and veins.

21. Correct option: (C) Blood from the wound.

Since blood is a necessary fluid that transports nutrients and oxygen to our cells and removes waste, blood from wounds is not seen as waste from our body. Blood clots and stops excessive bleeding when it is discharged from a wound in order to respond to harm. Blood from a wound can be considered a waste product in the context of the wound, but it is not a waste product from our body as a whole.

The other choices—carbon dioxide, perspiration, and urine—are all bodily wastes. During respiration, cells produce carbon dioxide, which is then evacuated from the body through the lungs. The sweat glands create perspiration to assist control body temperature. The kidneys produce urine to eliminate waste from the circulation.

22. Correct option: (D) It accumulates poisons in blood and various tissues

Poisons build up in the blood and other tissues as a result. The metabolic activities of the body produce waste products, which, if they are not effectively cleared, can build up in the circulation and different tissues and pose health risks. These waste products may include excess minerals, metabolic wastes, and other chemicals that must be eliminated to preserve a wholesome interior environment. Inadequate waste removal can lead to the accumulation of toxins and hazardous substances, which can impair biological processes and perhaps cause a number of health problems over time. Maintaining general health and limiting the buildup of toxic substances within the body depend on proper waste removal.

23. Correct option: (A) Excretion

Excretion is a biological process that helps living things get rid of trash that is produced inside their cells as a result of metabolic processes. This comprises waste materials that, if left to build up, could become dangerous, such as carbon dioxide, urea, too much salt, and other chemicals. By eliminating these waste products through excretion, organisms maintain a healthy internal environment and enable optimal operation of their biological systems. According to their unique physiological requirements and environmental circumstances, different creatures use a variety of excretory mechanisms, such as respiratory systems, kidneys, and specialized excretory structures.

Either liquids or gases are the byproducts of metabolism. Carbon dioxide and water vapor are the waste gases. Through the respiratory system, the body expels them. Urine, perspiration, and faeces are the liquid waste products. These are eliminated from the body through the skin, the digestive system, and the excretory system.

24. Correct option: (C) In transpiration, liquids are expelled out

The incorrect statement is C) "In transpiration, liquids are expelled out." Water vapor is expelled from the stomata (tiny openings) on the surfaces of leaves and other plant parts through the process of transpiration. Instead of liquid being ejected from the plant, the main source of the problem is the loss of water vapor in the form of gas. Water leaves the plant through the stomata after evaporating from the moist cell surfaces into the surrounding air spaces. This process aids plants in maintaining turgidity, transporting nutrients, and controlling temperature. The other options are correct: A) Guard cells do surround the stomatal pore, B) subsidiary cells do encircle the guard cells, and D) the transpiration pull does play a role in bringing water up through the roots due to the cohesion and adhesion properties of water molecules.

25. Correct option: (B) Transpiration pull

The transfer of water from the roots to the leaves during transpiration pull, a crucial process in plants, is caused by the evaporation of water vapor from the stomata in the leaves. Water molecules that evaporate from leaf surfaces exert a negative pressure or tension inside the xylem vessels of the plant. This negative pressure, which is sometimes referred to as tension or suction, creates a pulling force that causes water to be drawn upward through the vascular system of the plant. Water may be dragged up against gravity because of the cohesive and adhesive qualities of its molecules, which enable them to adhere to one another and to the walls of the xylem vessels. This process is often referred to as the "transpiration pull" and plays a crucial role in the transport of water and nutrients from the roots to the rest of the plant.

26. Correct option: (C) RBC

The correct answer is C) RBC, which stands for red blood cells. Hemoglobin, a protein found in high concentration in red blood cells, gives them their red hue. In order to facilitate oxygen exchange for metabolic activities, hemoglobin binds with oxygen in the lungs and transports it to numerous tissues and organs throughout the body. When oxygen is present, the hemoglobin's iron-containing heme group gives the cells their distinctive red hue. Due to this red pigment, oxygen-poor blood that is returning to the lungs appears a darker shade of red whereas blood that is oxygen-rich appears bright red.

27. Correct option: (D) Capillaries

The smallest blood veins in the human body are called capillaries, and they create a network throughout the tissues. They provide for the exchange of oxygen, nutrients, and waste products between the blood and the surrounding tissues. They join arteries to veins. Capillaries are a crucial part of the circulatory system since they are in charge of bringing oxygen-rich blood to the cells and eliminating waste.

Large blood channels called arteries are used to transport blood away from the heart. Large blood channels called veins transport blood to the heart. All of the blood vessels in the body are collectively referred to as "blood vessels" in this context.

LEVEL-II

1. Correct option: (B) Opening and closing its stomata

The correct answer is B) Opening and closing its stomata. Plant stomata are controlled in their opening and shutting by potassium ions (K+). On the surfaces of leaves and stems, there are tiny pores called stomata that permit gas exchange, including the uptake of carbon dioxide and the release of oxygen. Changes in the concentration of potassium ions govern how the guard cells, which control the opening and closing of the stomatal pore and surround it, move. Potassium ions cause osmotic water uptake when they enter guard cells, causing the cells to expand and the stomata to open. The guard cells contract and the stomata close as a result of the loss of potassium ions, which causes water to leave the cells. This mechanism helps plants regulate transpiration, water loss, and gas exchange in response to changing environmental conditions.

2. Correct option: (A) Less than 1%

The leaves only utilize a very little portion of the water that plants absorb for photosynthesis. Although the conversion of light energy into chemical energy during photosynthesis is an essential process for plants, only a small number of water molecules are really actively involved in the chemical reactions. The majority of the water taken by the plant is used for other crucial procedures such aiding numerous metabolic functions, sustaining cell turgidity, and moving nutrients and minerals throughout the plant. The majority of the water absorbed by the plant eventually evaporates as a result of transpiration, which occurs when the stomata in the leaves release water vapor.

3. Correct option: (C) RBC

The correct answer is C) RBC (Red Blood Cells). Red blood cells, also known as erythrocytes, are distinct from the other items on the list in that they lack a nucleus. Mature red blood cells in animals shed their nucleus as they grow, making room for the iron-containing hemoglobin molecules that carry oxygen. The red blood cells' biconcave form, which maximizes their surface area to volume ratio for effective oxygen and carbon dioxide exchange in the bloodstream, is made possible by their lack of a nucleus.

4. Correct option: (B) Adhesion

The correct answer is B) Adhesion. Water molecules being drawn to and adhering to polar surfaces is referred to as adhesion. Due to differences in electronegativity, water is a polar

molecule with a slightly positive end (hydrogen) and a slightly negative end (oxygen). The attraction interactions between the positive and negative ends of the water molecules and the surface molecules cause adhesion when water comes into contact with polar surfaces like the walls of a container or other substances containing polar molecules. This characteristic of water is crucial for processes like capillary action, in which liquid moves upward in small tubes defying gravity, and it also contributes to the cohesion-adhesion theory of water transport in plants, which allows liquid to move uphill through vascular systems.

5. Correct option: (D) Both blood platelets and damaged tissues

The protein thromboplastin, commonly referred to as tissue factor, is essential for the coagulation or blood clotting process. Blood platelets and injured tissues both produce it. A blood vessel injury or tissue damage causes platelets to become activated and release thromboplastin. Damaged tissues at the site of injury also release thromboplastin simultaneously. By starting a series of events that result in the creation of a blood clot, thromboplastin aids in limiting excessive bleeding. This mechanism is crucial for keeping the circulatory system functioning properly and ensuring that wounds are closed off to stop blood loss.

6. Correct option: (D) It's the only means for gaseous movement within the plant body

The correct answer is D) It's the only means for gaseous movement within the plant body. The major way for gases like oxygen, carbon dioxide, and water vapor to travel within the body of a plant is through a process called diffusion, which is a crucial function for plants. Plants lack specialized respiratory organs like those seen in mammals, which are necessary for active gas exchange. Instead, they use the passive process of diffusion to help transfer gases between different plant components. Gases diffuse between cells and via the holes in leaves known as stomata as they migrate from places of higher concentration to areas of lower concentration. The distribution of essential gases throughout the plant for functions like photosynthesis and respiration is ensured by this type of gaseous exchange.

7. Correct option: (B) Liberation of energy

The biological process of respiration is how living things' cells turn organic molecules like glucose into energy, usually in the form of adenosine triphosphate (ATP). Although breathing in oxygen (option A) is a necessary component of aerobic respiration, the main objective of respiration is the release of energy held in the bonds of organic molecules. Cells use this energy for a variety of purposes, such as growth, mobility, maintenance, and metabolic processes. Carbon dioxide, water, and energy are produced during respiration when glucose and oxygen are combined. The survival and operation of cells, and subsequently the health of the entire body, depend on this energy being released.

8. Correct option: (C) Synthesis of prothrombin

One of the proteins involved in blood clotting is prothrombin, and vitamin K is necessary for its formation. The precursor protein thrombin is transformed into thrombin, a crucial enzyme in the blood clotting cascade. Prothrombin's post-translational carboxylation, a process that modifies specific amino acid residues, cannot occur without vitamin K. This alteration enables thrombin to be activated from prothrombin and play a crucial part in converting fibrinogen into fibrin, resulting in the formation of a stable blood clot. Without vitamin K, prothrombin synthesis and subsequent thrombin activation cannot take place as intended, impairing blood clotting and raising the risk of bleeding problems. The other steps in blood clotting can still occur in the absence of vitamin K. Fibrinogen is a protein that is converted to fibrin, which forms the clot. Thromboplastin is a substance that activates prothrombin to thrombin.

9. Correct option: (B) Sinu-auricular node that provides impulse for heartbeat

The sinoatrial (SA) node is referred to as the pacemaker in terms of the physiology of the heart. The right atrium of the heart has a tiny group of specialized cells known as the SA node. It produces electrical impulses that start and control the heartbeat by establishing the heart's overall rhythm. These electrical impulses force the ventricles to contract and pump blood after reaching the atrioventricular (AV) node, which in turn transfers the impulses to the atria. The SA node serves as the body's internal pacemaker, generating regular electrical signals to synchronize the contraction and relaxation of the heart. An artificial pacemaker may be implanted to control the heartbeat in cases of irregular cardiac rhythms or disorders like bradycardia.

10. Correct option: (C) White in color

Like many reptiles, lizards mostly produce uric acid as a nitrogenous waste product. Comparatively insoluble in water to ammonia or urea, the main nitrogenous waste products in other animals like mammals and birds, uric acid takes less water to be removed from the body. When uric acid is expelled, it frequently takes less water to remove because it produces a white, pasty substance. Since it allows them to efficiently eliminate waste, this adaptation is especially helpful to reptiles like lizards that live in environments with limited access to water.

11. Correct option: (D) Transfusion

Transfusion is not a process used by human beings for the excretion of waste materials. Blood or blood products are transfused from one person (the donor) to another (the receiver), usually to replace lost blood or to treat certain medical disorders. It is not a biological process for excretion. The other choices—exhalation (A), faeces (B), and sweat (C), on the other hand, are all ways that people get rid of waste from their bodies. Defecation removes solid waste from the digestive tract, exhalation removes carbon dioxide from the atmosphere, and perspiration eliminates waste materials and aids in controlling body temperature by evaporating sweat.

12. Correct option: (B) Tidal volume

The amount of air that is inhaled and exhaled during regular, easy breathing is referred to as tidal volume. It reflects the volume of air that is normally inhaled and exhaled with each breath. Tidal volume is an essential element of lung ventilation and a significant metric used to evaluate lung function. Along with other lung volumes including inspiratory reserve volume and expiratory reserve volume, it is one of the elements that affects the total respiratory cycle. Tidal volume is essential for keeping the body's supply of oxygen stable and for eliminating carbon dioxide.

13. Correct option: (A) The root dies first

When the entire ring of bark covering the stem or trunk of a ring-girdled plant is removed, the flow of carbohydrates and nutrients between the leaves (the source) and the roots (the sink) through the phloem is disrupted. The movement of these nutrients occurs through the phloem. The nutrients produced in the leaves cannot reach the roots when the phloem is girdled, which causes the roots to progressively wither away from malnutrition. The xylem, which moves water and minerals from the roots to the rest of the plant, is what the shoot takes nourishment from, so it may initially survive. However, the long-term survival of both shoots and root is compromised, as the disruption in nutrient flow ultimately leads to the plant's decline.

14. Correct option: (B) Alkaline

Phloem sap tested positive and found to be alkaline. Sugars, amino acids, hormones, and other organic compounds are transported from the leaves, where they are made by photosynthesis, to other sections of the plant by the nutrient-rich sap known as phloem

sap. Due to the presence of organic acids like citric acid and basic substances like potassium and magnesium ions, phloem sap is frequently mildly alkaline. The phloem sap's ability to transfer nutrients and sustain the physiological functions of the plant is tied to its alkaline composition. Thus, if the sap is examined and confirmed to be alkaline, it is likely phloem sap.

15. Correct option: (C) Electrocardiograph

ECG stands for electrocardiograph in its full form. A medical instrument called an electrocardiograph is used to continuously record the electrical activity of the heart. An electrocardiogram is the name given to the resulting recording (ECG or EKG). The electrical impulses that cause each heartbeat can be measured using an ECG by applying electrodes to the skin in specified areas. This aids in the diagnosis of a number of heart diseases, including arrhythmias, aberrant conduction, and other cardiac disorders. The ECG helps medical professionals make appropriate diagnoses and treatment choices for their patients by providing important data regarding the heart's rhythm, beat rate, and general electrical health.

16. Correct option: (C) Antigen – antibody reaction

Red blood cells (RBCs) clump together when blood from one person is combined with serum or blood from another person due to an antigen-antibody response. Antigens are distinctive molecules that are present on the surface of red blood cells. An individual's immune system will develop antibodies to fight and neutralize foreign or blood type-incompatible antigens when they are encountered. The antibodies in the recipient's blood react with the antigens in the donor's blood when blood or serum from people with various blood types—who each contain different antigens—is combined with blood from people with the same blood type. This antigen-antibody reaction can cause the RBCs to clump together or agglutinate, which can lead to serious health complications such as hemolysis (destruction of RBCs) and other immune responses. This is why blood compatibility and blood typing are critical considerations in blood transfusions.

17. Correct option: (C) both (A) & (B)

In vertebrates, oxygen is transported by both its dissolution in the plasma and its binding to hemoglobin. About 1.5 percent of the oxygen in the blood is dissolved in the plasma. Although the bloodstream carries this oxygen to the tissues, it can only diffuse a short distance from the blood vessels. Hemoglobin, a protein present in red blood cells, transports the vast majority of oxygen (about 98.5%). Because of its strong affinity for oxygen, hemoglobin may bond to oxygen molecules firmly. Oxyhemoglobin is created when hemoglobin binds to oxygen. The oxygen is released and diffuses into the cells after being carried by oxygen hemoglobin to the tissues. This dual mechanism ensures efficient oxygen transport: oxygen dissolved in plasma provides rapid delivery to tissues, while oxygen bound to hemoglobin offers a reservoir that can be readily released where it's needed, particularly in regions with lower oxygen tension.

18. Correct option: (C) Both I and II are correct and II explains I

The answer is C, "Both I and II are correct and II explains I." The phloem, which is responsible for moving organic nutrients like sugars and amino acids in plants, moves in both directions, making statement I true. This implies that nutrients can travel both uphill and downward (from source to sink, such as leaves to roots) (from sink to source, like roots to developing fruits). Since the source-sink connection in plants is flexible, Statement II is also true. The distribution of nutrients between regions of production (source, often mature leaves) and regions of use or storage is described by this connection (sink, like growing fruits or roots). The diversity in this relationship is influenced by elements like the plant's growth stage, the environment, and nutrient requirements. The flexibility of the source-sink

relationship leads to the bidirectional movement in the phloem, which is an important way that statement II explains statement I. The dynamic nature of the source-sink connection makes it easier for nutrients to move in both directions when some plant parts are functioning as sources and others are acting as sinks.

19. Correct option: (B) Potassium influx and efflux

The most valid and recent description for stomatal movements is option B, "Potassium influx and efflux." Intake of carbon dioxide for photosynthesis and release of oxygen and water vapor are regulated by stomata, tiny pores on the surface of plant leaves and stems. Changes in the turgor pressure of guard cells, which surround the stomatal pore, are the main factor controlling how stomata open and close. Ions of potassium are crucial to this procedure. Guard cells enlarge as a result of water entering the cells by osmosis when potassium ions enter the guard cells and raise their internal osmotic potential. The stomatal pore opens as a result of this enlargement.

In contrast, the osmotic potential falls when potassium ions are pushed out of the guard cells, forcing water to leave the cells and causing them to contract, which closes the stomatal pore. Since the influx and efflux of potassium ions effectively control the turgor pressure and, in turn, the opening and closing of stomata, plants are able to manage gas exchange and water loss in response to external factors.

20. Correct option: (C) Water is first lost from the cytoplasm and then from the vacuole

The correct answer is C, "Water is first lost from the cytoplasm and then from the vacuole." When plant cells lose water in a hypertonic environment, a condition known as plasmolysis takes place, which causes the cell to shrink away from the cell wall. Due to the increased solute concentration outside the cell, water first leaves the cytoplasm during this process, which causes the turgor pressure to fall. The central vacuole, which often contains a significant amount of water and solutes, also releases water as it continues to travel out of the cytoplasm, causing it to contract. The contraction and separation of the cell from the cell wall brought on by this series of occurrences—water loss from the cytoplasm followed by that from the vacuole—leads to plasmolysis. When the cell is submerged in a hypotonic solution, the process can be reversed, enabling water to enter once more and returning the cell to its initial state.

21. Correct option: (B) Increase in number

The partial pressure of oxygen in the air is lower at high altitudes than it is at sea level. As a result, the body has less oxygen at its disposal. The body makes extra red blood cells to make up for this (RBCs). Since RBCs are in charge of delivering oxygen to the tissues, the body can receive more oxygen if there are more RBCs.

An erythropoietin-producing hormone is responsible for the increase in RBC count (EPO). The kidneys respond to low oxygen levels by producing EPO. The bone marrow is stimulated by EPO to create more RBCs.

It takes roughly a week for the RBC count to rise. The RBCs may also enlarge and assume an oval form as a result. This is due to the necessity for the RBCs to transport more oxygen to the tissues.

High altitude causes an increase in RBC count, which is a typical physiological reaction. It aids in maintaining a healthy amount of oxygen delivery to the tissues and aids the body in adjusting to the decreasing oxygen levels.

22. Correct option: (C) B

A person with antigen B in RBC and antibody A in the plasma belongs to blood group B.

The ABO blood type system is based on whether or not red blood cells have the two antigens A and B on their surface. Red blood cells from those with blood type A carry the A antigen,

those with blood group B carry the B antigen, those with blood group AB carry both the A and B antigens, and those with blood group O carry neither the A nor the B antigen.
Red blood cells from people with blood group B contain the B antigen, and their plasma contains antibodies to the A antigen. This is due to the fact that the body makes antibodies against antigens that are not found on its own cells. A blood group B person will therefore have antibodies against the A antigen.

23. Correct option: (A) Active transport is used for both loading at the source and unloading at the sink.

Active transport is necessary for the transfer of glucose in plants from source to sink. Active transport uses energy to move materials up and down a concentration gradient. At the source, active transport loads the glucose into the phloem. A protein known as a sucrose transporter is responsible for this. The glucose is moved against the gradient of concentration by the sucrose transporter using ATP energy.
Active transport at the sink causes the glucose to be released from the phloem. A protein known as a sucrose transporter also performs this function. The glucose is moved against the gradient of concentration by the sucrose transporter using ATP energy. Without the use of energy, passive transport transfers materials along a gradient of concentration. Since the concentration of glucose is higher at the sink than at the source, passive transport cannot be employed to carry glucose from the source to the sink.

24. Correct option: (B) Carbon dioxide

Plants use the carbon dioxide in the air as a raw material during photosynthesis to create glucose and oxygen. Through microscopic openings on the surface of leaves known as stomata, carbon dioxide enters the plant. Glucose is produced inside the leaf as a result of a series of chemical reactions that take place when carbon dioxide, water, and sunshine are combined. Additionally, the oxygen created during photosynthesis is discharged back into the environment, raising the atmospheric oxygen levels.
Oxygen is not a necessary component of photosynthesis. It comes from photosynthesis. The basic materials for photosynthesis do not include nitrogen. Additionally, hydrogen is not a component in photosynthesis. Despite not being directly utilized in photosynthesis, it is necessary for plants to synthesize chlorophyll.

25. Correct option: (B) Erythroblastosis

The correct answer is B, "Erythroblastosis." Erythroblastosis fetalis, also known as hemolytic illness of the newborn, is a condition associated with the Rh factor, sometimes referred to as the Rh antigen or D antigen. When a Rh-negative mother's blood and a Rh-positive fetus's blood are incompatible, this disease develops. The mother's immune system may create antibodies against Rh-positive blood cells if she is Rh-negative and has been exposed to the Rh antigen before (often during a prior pregnancy with a Rh-positive foetus). These antibodies have the potential to cause hemolysis—the death of red blood cells—in the foetus if they cross the placenta and enter the bloodstream. This can result in a number of major health issues, including anemia and jaundice.
A protein called hemoglobin transports oxygen through the blood. The process of producing red blood cells is called erythropoiesis. Thyroid hormone insufficiency is the root cause of myxedema. Erythroblastosis fetalis development is not directly impacted by it.

26. Correct option: (C) Bicuspid and tricuspid valve close quickly

The correct answer is C, "Bicuspid and tricuspid valve close quickly." The bicuspid (mitral) and tricuspid valves close during the cardiac cycle, causing the first heart sound (S1), sometimes known as the "lub" sound, to be audible. This happens at the start of the phase known as ventricular systole, during which the ventricles contract and pump blood from the heart. The beginning of ventricular contraction and the ejection of blood into the

pulmonary and systemic circulation is signaled by the closing of the bicuspid and tricuspid valves, which stop blood from flowing backward into the atria from the ventricles. The mechanical actions connected with the heartbeat begin with the first heart sound.

27. Correct option: (D) Digested food from the small intestine to the other parts of the body

The correct answer is D, "Digested food from the small intestine to the other parts of the body." Blood is essential for moving a variety of things throughout the body, including nutrients like molecules from digested food, oxygen, carbon dioxide, hormones, and waste products. Nutrients from the digested meal are absorbed into the bloodstream after digestion in the small intestine. Blood then carries these nutrients, such as glucose, amino acids, and fatty acids, to various tissues and organs where they are utilized for energy, growth, and other metabolic processes. This transportation of nutrients to different parts of the body is essential for the proper functioning and survival of cells and tissues.

28. Correct option: (A) They would be unable to move water and minerals to the top of the plant during the day

The correct answer is A, "They would be unable to move water and minerals to the top of the plant during the day." The stomata of CAM (Crassulacean Acid Metabolism) plants open at night, when the air is cooler and more humid, as a special adaptation to lessen water loss. They take in carbon dioxide at night and fix it into an organic acid that is then stored in vacuoles. These organic acids are then broken down to release carbon dioxide for photosynthesis throughout the day when the stomata are closed to prevent water loss. Even while this water-saving technique works, it reduces the amount of water and minerals that are available during the day for other crucial processes like nutrient transfer and growth. Because of this restriction, CAM plants often develop more slowly and are generally shorter than plants with other types of water-conserving mechanisms.

29. Correct option: (D) Turgor pressure

The correct answer is D, "Turgor pressure." Turgor pressure is the force that a plant cell's cell wall experiences as a result of the osmotic flow of water inside the cell. The cell swells as a result of water entering through osmosis, pushing the hard cell wall with the cell membrane in opposition. Turgor pressure is the name given to this force exerted on the cell wall. Turgor pressure gives plant cells structural support, preserving the rigidity and shape of the cell. Additionally, it is essential for many physiological activities, including controlling stomata's opening and shutting, assisting with nutrient transport, and promoting cell proliferation. Osmotic potential, or the propensity of water to penetrate a semipermeable membrane as a result of variations in solute concentration, has an impact on turgor pressure.

30. Correct option: (A) (i), (ii), (iii)

(i) A decrease in pressure on cell exerted by the wall: As water enters by osmosis and the pressure on the cell wall diminishes, the cell can perhaps enlarge and become more flexible.
(ii) An increase in water potential of the surroundings: Between places with higher and lower water potential, water flows. Water can enter a cell if there is a difference between the water potential of the inside and the surrounds.
(iii) The uptake of solutes by the cell: A decreased water potential occurs inside the cell when solutes (dissolved particles) from the environment are taken in by the cell. When the concentration of solutes is balanced, water will enter the cell.
(iv) An increase in the tension on the surrounding solution: The proper response does not include this choice. The transport of water by osmosis is primarily driven by changes in solute concentration (water potential), not tension in the surrounding solution.

6. Reproduction in Plants
Solutions

LEVEL-I

1. Correct option: (D) All the above

There are a number of characteristics that help seeds spread through the air. First of all, because they have wings, they are light and may be easily moved by wind currents. They may now go farther and find new places to germinate thanks to this. Second, the tiny size of these seeds contributes to their capacity to be carried by air. They are easily carried by even light winds due to their compact size, which lowers air resistance. Lastly, some of these seeds have hairy structures, such as tufts or bristles, which further enhance their ability to catch onto air currents and be carried away. Overall, these features collectively contribute to the successful dispersal of seeds through air.

2. Correct option: (C) Both A and B

Different characteristics of seeds that are disseminated through the air help them do so. They can easily be moved by wind currents because they have wings and are light. This enables them to travel farther and arrive at new locations for germination. Second, the size of these seeds is so little that it also helps them travel through the air. Even light breezes can lift them due to their small size, which lowers air resistance. Furthermore, some of these seeds include hairy features like tufts or bristles that improve their capacity to attach to air currents and be blown away. Overall, these characteristics help seeds disperse well through the air.

3. Correct option: (B) Sweet potato

A root that has undergone structural modifications to carry out particular tasks other than anchoring the plant and absorbing water and nutrients is referred to as a modified root. The sweet potato is the right response in the case of the provided possibilities. The sweet potato is a tuber, which is a modified root. To store starch and other nutrients for the plant, it gets bloated and expanded. As a storage organ, the sweet potato differs from a normal root in that it enables the plant to endure adverse conditions and supply energy for growth and reproduction.

The potato is basically a modified stem, more particularly, a tuber, which is an underground stem. The onion is a modified stem, more precisely, a bulbous subterranean stem. Garlic is also a modified stem known as a bulb, much like the onion.

4. Correct option: (A) Spores

By means of spores, bread mould reproduces. Small reproductive cells called spores are created by the mould. These spores are dispersed into the air and are readily transferable to other surfaces, such as additional slices of bread. The spores can germinate and develop into new mould colonies when the right circumstances are present, such as when there is sufficient moisture and food available. This method of reproduction allows bread mould to quickly and efficiently spread and colonize new areas, making it a successful and resilient organism.

5. Correct option: (D) All the above

All of the given options, A) Caltropis, B) Maple, and C) Milkweed, are dispersed by air. Plants frequently disperse their seeds by means of the air to reach new areas. Small seeds produced by Caltropis, also known as Gomphrena globosa, have feathery characteristics that allow them to be carried by the wind over a variety of distances. The characteristic "helicopter" seeds of maple trees rotate as they fall, allowing them to escape the parent tree by catching the wind. In contrast, milkweed produces seeds with silky hairs that

promote wind dispersal. All three of these plants have developed systems to benefit from air currents in order to spread their seeds widely and increase the likelihood that the seeds will find favourable habitats for germination and growth.

6. Correct option: (D) All the above

Animals are responsible for dispersing seeds with hooks, thorns, and hairy portions. As animals move through their surroundings, these adaptations allow seeds to cling to their fur, feathers, or bodies. Burdock and cocklebur seeds, for example, have hooks that allow them to cling to an animal's fur or feathers and travel to new areas. Thorny seeds, such as those in roses or blackberries, can cling to an animal's skin or fur and travel from one place to another. Similarly, seeds from plants with hairy portions, such as dandelions or cattails, can get tangled in an animal's fur or feathers and travel to other locations. Animals are therefore in charge of disseminating all of these different kinds of seeds.

7. Correct option: (C) Both A and B

Both birds and insects act as natural pollinators. For some kinds of plants, especially those with enormous, vividly colored blooms that yield a lot of nectar, birds are known to be efficient pollinators. They frequently have lengthy tongues and beaks that enable them to reach deep inside the blossom to get at the nectar. On the other hand, insects are the most prevalent and effective pollinators. Flowers draw them in because of their vivid hues and alluring aromas. They unintentionally carry pollen from the male reproductive organs (stamens) to the female reproductive organs (pistils) as they go from flower to bloom, aiding in fertilization. Some insects, such as bees, butterflies, and beetles, are particularly important pollinators due to their ability to carry large amounts of pollen on their bodies. Overall, both birds and insects play crucial roles in the natural process of pollination, ensuring the reproduction and survival of many plant species.

8. Correct option: (D) Bryophyllum

Since bryophyllum is a special plant that reproduces through buds on its leaves, it is the right response. It is possible for a plant to create new progeny by a technique known as vegetative propagation that does not require seeds or blooms. Along the edges of the leaves, the buds, also known as plantlets, form before falling off, whereupon they take root and develop into new plants. Since it enables quick and effective multiplication, this type of reproduction is beneficial for Bryophyllum because it ensures the survival and expansion of the species.

Rhizomes are used in the rhizomes of sugarcane, a long, perennial grass. The seeds of the biennial turnip are used to spread the plant. Its tubers are used to spread the herbaceous perennial plant known as the potato.

9. Correct option: (B) Balsam

Impatiens spp., also known as balsam plants, are noted for their explosive seed dissemination technique. Balsam ripe seed pods quickly break open when touched or otherwise disturbed because of the strain that the walls of the pods have built up from drying. The result is a forced ejection of the seeds from the pod, which frequently disperse far from the parent plant. By being projected away from the parent plant, these seeds have a better chance of locating places where they may grow and gerinate. This type of seed dispersal aids the balsam plant in colonizing new areas.

The other options (A) Coconut, (C) Mango, and (D) Tamarind do not disperse seeds by bursting of fruits; they have different modes of seed dispersal.

10. Correct option: (B) Entomophily

The term "entomophily" describes flower pollination that is aided by insects. As they gather nectar, insects like bees, butterflies, moths, and beetles are essential in moving pollen from one blossom to another. Bright colors, alluring fragrances, and nectar cues

are common characteristics of flowers suited for entomophily to entice insects. Numerous coevolutionary adaptations between plants and their insect pollinators have resulted from this type of ubiquitous pollination method in the plant kingdom.
Pollination by creatures other than insects, such as bats, birds, and rodents, is known as zoophily. Pollination by snails and slugs is known as malacophily. Ornithophily is the practice of using birds to pollinate plants.

11. Correct option: (D) Ovule

The ovule in flowering plants transforms into a seed during fertilization. The female gametophyte and egg cell are located in the ovule, a structure in the ovary of the flower. When a pollen tube transports a pollen grain to the ovule, the male and female gametes can unite to produce fertilization. A zygote is created as a result of this fusion, and it grows into an embryo inside the ovule. The ovule undergoes modifications as the embryo grows, ultimately becoming a seed. The embryo, a food source for the growing plant (in certain cases endosperm), and a protective seed coat are all parts of the seed.

12. Correct option: (D) All the above

Adapted for wind dispersal, seeds with hairy structures use their hairs or tiny fibers to catch the wind and move away from the parent plant. In the case of cotton, the hairs help spread seeds and the fibers are well-known for their usage in textiles. Another plant with hairy seed structures that can be carried by air currents is called madar (Calotropis). Dandelion seeds, on the other hand, contain feathery hairs that form a "parachute," enabling the seeds to be lifted by even the smallest wind. The existence of these hairy appendages in cotton, madar, and dandelion seeds demonstrates their wind dispersal adaptations, increasing the likelihood that the seeds would land in favorable conditions for germination and growth.

13. Correct option: (D) All the above

The correct answer is D) All the above. This is so because each of the pairs is matched to a particular component of the structure of a flower. A) The term "calyx" describes the outermost whorl of a flower made up of sepals, which are typically green and serve to shield the budding bud. B) The corolla, or second whorl, is made up of petals that are frequently colorful and appealing, assisting in pollination. C) The third whorl, which contains stamens, the male reproductive organs that create pollen, is where an androecium is located. As a result, each pair closely matches a distinct part of a flower's anatomy, including its defense, appeal, and reproductive functions.

14. Correct option: (D) All the above

The answer is D) All the above. A flower is said to be complete when it has all four major components: sepals, petals, stamens, and pistils. Hibiscus, Datura, and Ipomea are a few examples of plants with full blooms. The flower's outermost structure, called sepals, is typically green and has a protective function. The flower's vibrant and enticing petals are what frequently draws pollinators. While pistils are the female reproductive organs that house the ovary and receive pollen for fertilization, stamens are the male reproductive organs that create pollen. All three of the aforementioned plants are regarded as full flowers because they include all four of these components.

15. Correct option: (D) All the above.

The female reproductive organs of a flower receive pollen from the male reproductive organs through pollinating agents, which allows fertilization and the development of seeds. Animals, air, and water are a few examples of these agents. Pollen grains are spread onto plant female reproductive organs through airborne pollination, which is caused by wind currents. Pollen is transported across water to reach the female reproductive organs in aquatic plants, which pollinate less frequently. The most common

and effective mode of pollination is by animals, with bees, butterflies, birds, and bats playing a critical part in the movement of pollen between flowers. Therefore, the correct answer is D) All the above, as all three pollinating agents - air, water, and animals - contribute to the pollination process in different ways.

16. Correct option: (B) Madar

Adapted for wind dispersal, seeds with hairy structures use their hairs or tiny fibers to catch the wind and move away from the parent plant. In the case of cotton, the hairs help spread seeds and the fibers are well-known for their usage in textiles. Another plant with hairy seed structures that can be carried by air currents is called madar (Calotropis). Dandelion seeds, on the other hand, contain feathery hairs that form a "parachute," enabling the seeds to be lifted by even the smallest wind. The existence of these hairy appendages in cotton, madar, and dandelion seeds demonstrates their wind dispersal adaptations, increasing the likelihood that the seeds would land in favorable conditions for germination and growth.

17. Correct option: (A) Fleming

The answer is A) Fleming, referring to Sir Alexander Fleming. In 1928, he is credited with discovering penicillin. Fleming discovered a mould called Penicillium notatum generated a chemical that killed a variety of bacteria when he was working at St. Mary's Hospital in London. As a result of this unintentional discovery, antibiotics were created, changing medicine by effectively treating bacterial illnesses. The discovery of penicillin as a potent and well-known antibiotic was made possible by Fleming's observation of the mold's antibacterial characteristics.

18. Correct option: (B) Ovary

The correct answer is B) Ovary. After successful fertilization, the ovary, which is a component of the flower's reproductive structure, turns into a fruit. The female reproductive cells are housed in one or more of its ovules. The fertilization process takes place inside the ovary after a flower has been pollinated, often by the transfer of pollen from the stamen to the stigma. The ovary undergoes a transformation into a fruit as a result of this fertilization, which can serve a variety of purposes, including preserving and distributing the seeds produced by the fertilized ovules. By causing the growth of fruits, the ovary thus plays a critical part in the reproductive cycle of flowering plants.

19. Correct option: (B) Budding

The correct answer is B) Budding. Budding is the process through which yeast cells divide. A little protrusion or bud develops on the parent yeast cell's surface during budding. This bud becomes bigger over time while still being connected to its parent cell. The bud eventually splits off from the parent cell and develops into a distinct yeast cell. By using this process, yeast cells can create progeny that are genetically identical to themselves. Asexual reproduction known as budding is typical of single-celled organisms like yeast.

20. Correct option: (C) Thalamus

The correct answer is C) Thalamus. The inflated base of the flower, often referred to as the thalamus or receptacle, is where all of the floral organs are whorl-mounted. It gives the flower's numerous parts, such as the sepals, petals, stamens, and carpels, support and structure (which include the ovary). The reproductive and defensive organs of the flower are positioned on the thalamus, which facilitates their coordinated growth and operation. It is a crucial component of the flower's anatomy, assisting in the appropriate positioning and display of the floral parts to maximize their capacity for reproduction.

21. Correct Matching is: a-R, b-P, c-Q, d-T, e-S

a) Thalamus - R) Seat of a flower:

The bulging base of a flower, sometimes referred to as the receptacle or thalamus, is where all the floral organs are attached. It acts as a base or "seat" for the arrangement of the other sections, offering support and structure for the flower's constituent parts.

b) Calyx - P) Protects the parts of a flower:

The flower's outermost whorl, the calyx, is made up of sepals. It frequently offers support while also enclosing and guarding the developing inner flower elements. The fragile reproductive components within the flower can be protected by sepals.

c) Corolla - Q) Attracts the insects:

The second whorl of the flower, which is made up of petals, is called the corolla. The flower's petals, which are frequently vividly colored and come in a variety of forms, draw insects and other pollinators. It is essential for pollinators to be attracted to the corolla in order to promote pollination.

d) Androecium - T) Filament with bulb-like pollen sac

The stamens, which are the male reproductive components of the flower, are referred to as the androecium. A filament (stalk) and an anther (pollen sac), which is where pollen is made, make up each stamen. The pollen-containing anther, which is supported by the filament, occasionally has a bulb-like form.

e) Gynoecium - S) Pistil

The flower's female reproductive system, which may include one or more carpels, is represented by the gynoecium. The ovary, style, and stigma are located inside the carpel. The term "pistil" is frequently used synonymously with "carpel," and it refers to the female reproductive system, which includes the stigma, which receives pollen, the style, which supports the stigma, and the ovary, which houses the ovules.

22. Correct Matching is: a-T, b-S, c-Q, d-R, e-P

a) Complete flower - T) A flower that has all the four whorls

Sepals, petals, stamens, and pistils are the four major whorls of floral organs that make up a full flower. These are the fundamental elements of a flower that work together to support its reproduction process.

b) Incomplete flower - S) A flower in which any of the four whorls is missing

One or more of the four primary whorls are absent in a type of flower known as an incomplete flower. This indicates that one or more of the flower's structural components—sepals, petals, stamens, or pistils—is absent.

c) Unisexual flower - Q) A flower that has either stamens or pistil

A flower that is missing one or more of its four primary whorls is said to be incomplete. This denotes a structural flaw in the flower, such as the absence of sepals, petals, stamens, or pistils.

d) Bisexual flower - R) A flower that has both stamens and pistil

A bisexual flower, often known as a perfect flower, has both female (pistil) and male (stamen) reproductive organs. This enables the flower to carry out both male and female reproductive processes, enabling cross-pollination and self-pollination.

e) Anther - P) Pollen sac

The area of the stamen where pollen is produced called the anther. Pollen grains are held in structures resembling sacs called pollen sacs. The male gametes required for fertilization are contained in these grains.

23. Correct Matching is: a-T, b-S, c-Q, d-R, e-P

a) Pollination - T) Transfer of pollen grains from anther to stigma

Pollination is the process of moving pollen grains from a flower's male reproductive structure, or anther, to its female reproductive structure, or another flower, called a stigma. The success of fertilization depends on this transmission.

b) Fertilization - S) Fusion of male and female gametes

The process through which the male and female gametes (sperm and egg cells) join forces to create a zygote is called fertilization. The plant's subsequent generation is starting with this zygote.

c) Zygote - Q) Formed after the fertilization process

The cell that results from the union of the male and female gametes during fertilization is known as the zygote. It is the first cell of an embryo, which then develops into a new individual plant.

d) Vegetative reproduction - R) Reproduction by somatic parts of a plant

In asexual plant reproduction known as vegetative reproduction, non-reproductive plant elements like stems, roots, or leaves are used to create new individuals. Neither seeds nor spores are produced during this process.

e) Spores - P) Asexual reproduction of fungus

A few plants and fungi develop reproductive structures called spores. They are microscopic cells that, in the right circumstances, can grow into new individuals, enabling asexual reproduction in organisms like fungus.

24. Correct Matching is: a-R, b-P, c-S, d-Q

	Group A	**Group B**
a)	Pollen grain	R) Anther
b)	Ovule	P) Ovary
c)	Reproduction through eyes	S) Potato
d)	Reproduction through leaves	Q) Bryophyllum

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LEVEL - II

1. Correct option: (B) Formation of seed without syngamy and meiosis.

The term "apomixis" describes the generation of seeds without the meiosis and syngamy processes. A zygote is created through the union of gametes, which is known as syngamy. On the other hand, meiosis is a type of cell division where the number of chromosomes is cut in half. In the event of apomixis, neither of these two critical processes is involved in the formation of the seeds. Instead, without fertilization, the seeds are created by the direct growth of the egg cell. As there has been no genetic recombination or variation introduced, this indicates that the offspring generated through apomixis are genetically identical to the parent plant. As a result, option B is the right response because it explains apomixis in an exact manner.

2. Correct option: (C) Papaveraceae

The Papaveraceae family of plants have the censar mechanism of seed dispersion. When touched or agitated, the fruit capsule's unique dispersal mechanism causes the seeds to eject, frequently as a result of the fruit walls' drying up or withering. This process in Papaveraceae plants—including poppies—projects seed out of the fruit capsule quickly and frequently violently, facilitating their dispersal across a larger region. By lowering competition with parent plants and assisting in seed distribution, this adaptation encourages the colonization of new areas.

3. Correct option: (D) Chara

Chara is a monoecious plant. On the same individual plant, monoecious plants have both male and female reproductive organs. Green algae known as Chara are typically found in freshwater habitats. The male reproductive structures (antheridia) and female reproductive structures (oogonia) are found on the same thallus in monoecious Chara species (plant body). Due to the proximity of both types of reproductive organs on the same plant, this configuration permits the possibility of self-fertilization.

4. Correct option: (A) Carrion flies

True, carrion insects are mostly responsible for pollination in Rafflesia. One of the largest individual flowers in the world are produced by the parasitic flowering plant genus Rafflesia. These blooms give off an unpleasant smell that is similar to that of rotting flesh, luring carrion flies. The flies are drawn to the flower because they believe it to be a food source, and while they are searching for food, they unintentionally spread pollen between the male and female flowers of various plants, aiding in pollination. Rafflesia's potent aroma and unusual pollination method are adaptations to its particular ecological niche.

5. Correct option: (B) 8

The fruit fly, Drosophila melanogaster, has eight chromosomes in each meiocyte (a cell undergoing meiosis). Fruit flies contain two sets of chromosomes since they are diploid organisms. However, the number of chromosomes is decreased by half during meiosis, the cell division process that results in the creation of gametes (sperm and egg cells). When it comes to fruit flies, the reduction division creates four haploid cells with eight chromosomes apiece. This haploid number of chromosomes guarantees that the subsequent diploid zygote will have the typical number of chromosomes for the fruit fly species, which is 16, when fertilization takes place and the sperm and egg unite.

6. Correct option: (A) Great synchrony between sexes

Externally fertilized organisms, in which the union of the gametes takes place outside the body, demonstrate excellent sex-to-sex synchronicity. Both males and females discharge their gametes (sperm and eggs) into the environment during external fertilization, frequently in close temporal and geographical proximity. In aquatic creatures, the gametes must meet and fuse in the water before they scatter or are eaten, therefore this

synchronization is essential for proper fertilization. Coordination of timing and position during external fertilization is necessary to increase the likelihood that gametes will come into contact during successful reproduction. Internal fertilization, which is common among terrestrial animals, allows for greater gamete protection from external variables and reduces the requirement for synchronized release in the short term.

7. Correct option: (B) Multicarpellary syncarpous

The papaver has a multicarpellary syncarpous pistil, which is the plant's female reproductive system. As a result, the pistil is made up of many carpels that have been united into a single structure. The separate carpels of papaver combine to create a syncarpous pistil. The stigma, which is where pollen lands, the style, which is a stalk connecting the stigma to the ovary, and the ovary normally make up each carpel (containing ovules that will develop into seeds after fertilization). In many flowering plants, the fusing of carpels into a syncarpous pistil is a common structural adaptation. In papaver, it makes reproductive activities like pollination and seed development easier.

8. Correct option: (D) Continue the species forever

A crucial biological process that allows living things to perpetuate their species is reproduction. By reproducing, organisms create children who inherit their genetic code, preserving genetic diversity and allowing for environmental adaption. While an organism's ability to develop, use its energy, and maintain its various organs is necessary for its survival and well-being, reproduction performs the special function of transferring genetic features to succeeding generations. Because of their ability to adapt to changing environmental conditions and evolve over time, animals are able to have long lives on Earth.

9. Correct option: (C) 2 polar nuclei + 1 male gamete nucleus

The fusing of two polar nuclei and one male gamete nucleus results in the formation of the main endosperm nucleus (PEN). When flowering plants undergo twofold fertilization, this procedure takes place. Two male gametes are released when a pollen tube enters the ovule; one combines with the egg cell to make the zygote, and the other unites with the two polar nuclei in the embryo sac's center cell to form the PEN. The endosperm, a nutrient-rich tissue that feeds the growing embryo, is created from the primary endosperm nucleus.

10. Correct option: (D) Tapetum nourishes the developing pollen

A specialized layer of cells called the tapetum is present in the anther of a flower. Its primary purpose is to nourish and nurture the growing pollen grains. It secretes a variety of compounds, including proteins, lipids, and enzymes, which are crucial for the pollen's healthy growth and development. The pollen grains would not be able to grow and perform effectively for pollination without the tapetum because they would not be provided with the essential nutrients. Therefore, by guaranteeing the successful production of pollen grains, the tapetum plays a critical part in flowering plants' reproductive process.

Exine refers to the pollen's tough outer covering. The tissue from which microspores are produced is known as sporogenous tissue. The endothecium is the innermost layer of the anther wall, and it does not produce the microspores.

11. Correct option: (A) allogamy

In the real genetic meaning, cross-pollination is the act of moving pollen from one plant's reproductive organs, like stamens, to another plant's reproductive organs, like pistils, within the same species. This could happen because of the wind, insects, birds, or other forces. The term "allogamy," denoted by option A, especially alludes to cross-pollination, in which pollen from the blooms of one plant fertilises the ovules of another plant. As a

result of this genetic exchange, a species' genetic diversity is encouraged, which can improve adaptation and evolutionary success.
Autogamy (B) is self-pollination, where pollen from the same plant reaches its own stigma. Cleistogamy (C) is the production of closed, self-fertilizing flowers. Geitonogamy (D) is a subset of allogamy, involving pollination between flowers of different individuals within the same population. Among these options, allogamy (A) best captures the essence of true cross-pollination, fostering genetic variability.

12. Correct option: (C) Apogamy

In the fern life cycle, the prothallus—a tiny, heart-shaped structure—can develop into a new plant without the requirement for fertilization. This process is known as apogamy. This process involves the direct formation of a sporophyte plant from a gametophyte (the prothallus) without the fusion of gametes (sperm and egg cells). This avoids the typical meiosis and fertilization stages of sexual reproduction. Due to the fact that asexual reproduction does not entail the recombination of genetic material from two separate sources, it is possible to produce sporophytes that are genetically identical to their parent plants. In some fern species, this may support clonal reproduction and genetic stability.
Apospory is the formation of a gametophyte (n) directly from a cell of the sporophyte (2n) without meiosis. Parthenocarpy is the development of fruits without fertilization. Organogenesis is the process of forming organs.

13. Correct option: (A) 1 diploid and 1 triploid cell

One diploid cell and one triploid cell make up a mature ovule following double fertilization, which happens in angiosperms. The procedure entails the merging of two sperm cells with two distinct nuclei inside the ovule's embryo sac. The embryo is created when one sperm cell combines with an egg cell to create a diploid zygote. A triploid cell known as the major endosperm nucleus is created when the other sperm cell merges with two polar nuclei. The endosperm, a nutrient-rich tissue that sustains the growing embryo, will develop from this triploid nucleus. Due to this special property of twofold fertilization, the embryo and endosperm within the seed develop in a coordinated manner, which helps flowering plants to carry out their reproductive strategy.

14. Correct option: (B) Sucker

Plants that reproduce vegetatively by a process known as "suckering" include elm (Ulmus), dandelion (Taraxacum), and different rose species. Suckering is the process through which new shoots or stems develop from the parent plant's roots. These "sucker" shoots, which are also known as lateral buds, develop from the plant's subterranean parts, usually the roots or rootstock. These suckers produce new plants that are genetically similar to the parent plant as they mature into independent shoots. This kind of vegetative reproduction aids in the growth of plant populations and the formation of clonal colonies, which are collections of plants that are virtually identical clones of one another and descended from the same parent.

15. Correct option: (B) small, producing a large number of dry pollen grains.

Flowers that are pollinated by the wind often have a tiny size and produce a lot of dry pollen grains. These flowers lack the eye-catching hues and features that draw pollinators from animals. The efficient generation and spread of pollen, which is frequently tiny, light, and non-sticky, is instead given top priority. The huge amount of pollen grains thrown into the air makes up for the absence of nectar and aesthetic features. The tremendous pollen production and the pollen grains' light weight boost the likelihood of successful wind pollination because wind is these plants' major method of moving pollen from one flower to another. Wind-pollinated plants can improve their reproductive efficiency thanks to this adaptation without the need for animal pollination.

16. Correct option: (C) Morphology, vigour and disease resistance

Morphology, vigour, and disease resistance are the traits that carry over from generation to generation in vegetative propagation by tuber. This means that each new generation of tubers grown using this technique will maintain the same physical characteristics, pace of growth, and disease resistance. Farmers and gardeners benefit from this since they can rely on these characteristics to be present in the progeny, assuring a reliable and appealing harvest or plant. Farmers may guarantee a high-quality output and lower the chance of crop loss due to illnesses by choosing tubers with the necessary morphology, vigour, and disease resistance.

17. Correct option: (D) Pollen and endosperm

The process known as "Xenia" occurs when the endosperm, a nutritive component that surrounds the embryo inside a seed, exhibits traits that are influenced by the maternal parent. Metaxenia, on the other hand, describes how the paternal parent affects the properties of both the endosperm and the embryo. These phrases explicitly refer to the contributions made by the mother (endosperm) and father (pollen) in the formation of seeds. Endosperm feeds the developing embryo while pollen carries the genetic material that joins with the egg cell to create the embryo. Xenia and metaxenia effects are seen in the traits expressed by the endosperm and sometimes the embryo, which can be influenced by the genetic makeup of the pollen donor, thus highlighting the contribution of both pollen and endosperm to the characteristics of the seed.

18. Correct option: (C) The flowers pollinated by the flies and beetles secrete foul odor to attract them

The statement that is not true is C) "The flowers pollinated by the flies and beetles secrete foul odor to attract them." In actuality, flowers that are pollinated by beetles and flies usually have strong, enticing scents or even imitate the aroma of decomposing stuff. Because they deposit their eggs in conditions that resemble rotting materials, flies and beetles are frequently drawn to those smells. By producing scents that are similar to these ones, flowers have evolved to take advantage of this behaviour and draw these pollinators. This adaptation emphasizes the variety of tactics used by plants to draw various pollinators, each one catered to the distinct traits and behaviors of the pollinating insects.

19. Correct option: (B) Sporopollenin

Due in large part to the fact that sporopollenin makes up their exine, pollen grains have the amazing capacity to withstand high temperature changes and desiccation. The pollen grain's outer layer of protection is made of the strong and chemically resistant biopolymer known as sporopollenin. By providing protection from extremes in temperature, UV radiation, and moisture loss, this special chemical protects pollen from environmental challenges. In order to successfully pollinate and fertilize flowers, pollen grains must be able to preserve their viability and integrity along the travel from the anther to the stigma of the flower. The inclusion of sporopollenin in the exine is an essential trait that helps plants reproduce and survive.

20. Correct option: (C) It can propagate vegetatively at a phenomenal rate

The main factor that makes water hyacinth one of the most invasive weeds is its exceptional capacity for vegetative propagation at an incredibly quick rate. The Eichhornia crassipes, sometimes known as the water hyacinth, is distinguished by its aggressive growth and reproductive strategy, which involves the development of multiple daughter plants via vegetative mechanisms like stolons and runners. These daughter plants can quickly encircle bodies of water, generating dense mats that obstruct sunlight, water flow, and oxygen exchange, upsetting aquatic ecosystems. Water hyacinth outcompete native plants and swiftly colonize and take over aquatic settings. This results in serious

ecological and economic issues, including decreased biodiversity, altered water quality, and obstructions to navigation and leisure activities.

21. Correct option: (D) Monocystis

An instance of sexual reproduction known as isogamy occurs when the gametes are morphologically identical, or the same size and shape. Many unicellular creatures, like the parasitic flatworm Monocystis, exhibit isogamy. The Coccidia class of parasitic protozoa includes the genus Monocystis. It is distinguished by having an isogamous reproduction mode in which a zygote is produced by the union of two isogametes that are structurally similar in size and appearance. This is distinct from anisogamy, which occurs in many other organisms and involves the fusion of gametes of different sizes. As a part of its life cycle, Monocystis reproduces isogamously in the reproductive organs of earthworms. Malaria is brought on by the multicellular parasite Plasmodium. A flatworm called a planaria reproduces through fission. Hydra is a freshwater polyp that reproduces by budding.

22. Correct option: (C) Perennial plants

It is true that perennial plants do not go through clean-cut vegetative, reproductive, and senescent periods. Long lifespans and the absence of clearly defined phases, which are present in annual and biennial plants, are characteristics of perennial plants. Annual plants evolve from vegetative growth through reproduction and senescence during the span of one year. Contrarily, biennial plants have a two-year life cycle, with the first year being devoted to vegetative growth and the second year being characterised by reproduction and eventual senescence. Perennial plants, on the other hand, lack such distinct periods. Over a longer period of time, they continue to develop, reproduce, and age in an unpredictable way, frequently alternating between vegetative and reproductive growth. This flexibility in growth and reproductive patterns is a characteristic feature of perennial plants.

23. Correct option: (B) Pre-fertilization, fertilization, post fertilization

B) Pre-fertilization, fertilisation, and post-fertilization is the correct response. The three primary stages of sexual reproduction are as follows. Pre-fertilization, the initial phase, sees the production and development of gametes (sperm and egg). Processes like gametogenesis, in which gametes are created by meiosis, are included in this stage. The sperm and egg combine to produce a zygote during the second stage, which is called fertilization. Because it mixes the genetic makeup of both parents, this period is critical. Post-fertilization involves the zygote's growth and development into an embryo, and it is the last step. Cell division, differentiation, and the development of diverse tissues and organs are all part of this period. The continuance of the species through sexual reproduction depends on these three stages overall.

24. Correct option: (A) Node

Nodes are essential for producing new plants during vegetative propagation. Nodes are certain locations along a stem from which buds, branches, or leaves grow. Nodes have the power to form roots and produce new plants when they come into touch with moist soil or water. Meristematic tissues, which are in charge of cell division and proliferation within nodes, support this process. A new plant begins to grow once the node establishes contact with an appropriate medium, going through physiological changes that allow for a type of asexual reproduction where genetically identical children are created from a parent plant.

The other options are incorrect. A shoot is a young, growing plant. An axial shoot is a shoot that grows from the main stem of a plant. A seed is a reproductive structure that contains an embryo plant.

25. Correct option: (C) Cellulose and pectin

The correct answer is C) Cellulose and pectin. The inner layer of the pollen grain's wall is called the pollen intine, and it is mainly made of cellulose and pectin. The pollen grain and other plant components, including cells, are structurally supported by cellulose, a complex carbohydrate. Pectin is another polysaccharide that is involved in cell adhesion and serves a role in preserving the integrity of the cell wall. The pollen grain is protected and is able to maintain its structure while it is subjected to a variety of environmental stressors during pollination and fertilization thanks to the strength and elasticity that cellulose and pectin together provide to the pollen intine. Pollen grains may successfully carry male gametes to female reproductive organs in plants thanks to their composition.

The pollen grain's exine is its outer coating. It is formed of sporopollenin, a very resilient substance that aids in shielding the pollen grain from the outside elements. The cell membranes of both plants and animals contain a particular form of steroid called ergsterol. Pollen grains don't contain any of it. Although they are absent from pollen grains, phospholipids and cholesterol are also present in cell membranes.

26. Correct option: (A) Vascular cambium is absent

Grafting is the process of uniting plant tissues from two separate plants so that they develop as a single organism. Grafting is possible in dicot plants because they include vascular cambium, a layer of meristematic tissue that promotes lateral growth and the development of secondary vascular tissues. These tissues enable the integration of the circulatory networks of the two plants, preserving the flow of nutrients and water. However, vascular cambium is missing in monocot plants like grasses and lilies, and their vascular bundles are organized in a dispersed way as opposed to creating the distinct rings found in dicots. It is difficult to successfully graft monocots due to the lack of vascular cambium and the different vascular arrangement because there is insufficient tissue to form a vascular connection between the scion (upper part of the graft) and the rootstock (lower part of the graft), which ultimately reduces graft compatibility and success.

The other options are incorrect. Meristematic tissue is present in both monocots and dicots. Collateral open vascular bundles are found in both monocots and dicots. Radial vascular bundles are only found in dicots.

27. Correct option: (D) Synergids, egg cell, micropylar

A flowering plant's ovule contains a specific structure called the egg apparatus at one end of the embryo sac. One egg cell and two synergids make up this organism. Supportive cells called synergids help the pollen tube go toward the embryo sac during fertilization. The zygote, or first cell of the new person, is created when a sperm cell fertilizes an egg cell, the female gamete. The embryo sac's micropylar end, which serves as the entryway for the pollen tube to enter during pollination, is not far from the egg apparatus.

This arrangement facilitates the fertilization process by positioning the egg cell and synergids in close proximity to the site of pollen tube entry, increasing the chances of successful fertilization.

28. Correct option: (B) a – ii; b – i; c – iii; d – iv

a) Eyes of - ii) Potato:

The little depressions or buds that may be seen on a potato's surface are referred to as "eyes" in this context. These "eyes" are actually latent buds that, given the right conditions, can sprout and develop into new potato plants. Potatoes are a good example of vegetative multiplication through these bud structures because each eye has the capacity to produce a new plant.

b) Rhizome of - i) Ginger:

A rhizome is a form of modified stem that develops horizontally underground, and ginger is a rhizome. It functions as the plant's storage organ and is capable of producing new shoots and roots. A piece of ginger rhizome can sprout new shoots and grow into a new ginger plant when it is planted.

c) Bulbil of - iii) Agave:

On the plant's flower stalks, agave plants generate bulbils, which are tiny aerial structures. When these bulbils fall to the ground and take root, they might develop into new plants. Plants can create progeny without the use of seeds thanks to asexual reproduction known as bulbils.

d) Leaf buds of - iv) Bryophyllum:

Succulent bryophyllum reproduces vegetatively by means of its leaf buds. Small plantlets or buds develop around the borders of its leaves. These buds have the ability to root and develop into new plants when they fall to the ground or come into contact with soil. Bryophyllum and related plants are particularly prone to this type of proliferation.

29. Correct option: (C) a – iii; b – iv; c – i; d – ii

a) Fission - iii) Algae, fungi, Monera:

Asexual reproduction via fission occurs when an organism divides into two or more parts, each of which develops into a new individual. Algae, fungus, and Monera (bacteria) are examples of single-celled organisms that frequently go through this process, in which the parent cell divides to produce genetically identical daughter cells.

b) Budding - ii) Oedogonium, Chlamydomonas:

Asexual reproduction known as budding occurs when a new organism arises as a bud from the parent organism. This process is used by several algae, including Oedogonium and Chlamydomonas, where a tiny bud grows and ultimately separates to become a separate individual.

c) Fragmentation - i) Ulothrix, saprolegnia:

The splitting of an organism's body into multiple parts, each of which has the potential to develop into a new organism, is known as fragmentation. Water moulds like Saprolegnia and algae like Ulothrix can reproduce via fragmenting, in which parts of the parent plant separate and grow into new individuals.

d) Sporulation - iv) Diktyoma, fucus, Monera:

Sporulation is the process by which specialized reproductive cells with the capacity to develop into new organisms, known as spores, are formed. This process is seen in a variety of organisms, including certain bacteria (Monera) and algae (Dickyota and Fucus), where spores are created and spread to create new individuals.

30. Correct option: (B) (I), (iv)

Asexual reproduction techniques like stem cutting and tissue culture both produce offspring that are genetically related to the parent plant. In stem cutting, a section of the stem of a plant is removed, planted, and allowed to grow roots and become a new plant. Since the new plant is effectively a clone of the parent plant, this approach produces children that are genetically identical to the parent plant. Similar to this, tissue culture entails cultivating a little portion of plant tissue—such as a cell or a cluster of cells—in a sterile environment with nutrients and hormones in order to transform it into a new plant. Due to the fact that the new plants are created from the same original tissue sample, tissue culture also produces plants that are genetically identical.

As seeds are created through sexual reproduction including the union of male and female gametes, and mutations can result in alterations to a plant's genetic make-up, on the other hand, seed production (ii) and mutation (iii) bring genetic variety.

7. Reproduction in Animals

Solutions

LEVEL-I

1. Correct option: (C) Leopard

Animals that produce living offspring after the embryo has internally grown within the mother's body are said to be viviparous. The leopard in this instance is a viviparous animal. In the course of gestation, the foetus grows inside the mother's uterus in female leopards. The mother gives birth to live cubs once the foetus is fully formed. When compared to species that lay eggs or have less developed young, species like leopards benefit from this reproductive strategy because their young require a higher amount of protection and nurturing.

2. Correct option: (D) Cat

Mammals, including domestic cats, have epidermal hair, commonly referred to as fur or coat, which is a distinguishing characteristic. Cats have a thick layer of epidermal hair on their bodies, which the animals use for communication, camouflage, insulation, and other purposes. By trapping air near the skin, the hair aids cats in controlling their body temperature. In their native environments, it helps them blend in with their surroundings so they can stalk prey or avoid predators. They also receive sensory information about their surroundings through the tactile nature of cat fur, which improves their ability to navigate and interact with their environment.

3. Correct option: (D) All the above

Numerous creatures, including Hydra, algae, and fungi, have been found to reproduce asexually, which is a manner of reproduction devoid of gamete fusion. Simple aquatic organisms like the hydra reproduce asexually by budding. Photosynthetic organisms known as algae frequently reproduce asexually through fragmentation or spore production. Molds and yeasts are examples of fungi, which also use spore generation, budding, and fragmentation as asexual reproduction strategies. These species use asexual reproduction to rapidly produce young that are genetically identical or similar, allowing them to quickly adapt to their environment, fill new habitats, and continue to exist in different ecosystems.

4. Correct option: (B) Amoeba

Asexual reproduction known as binary fission occurs when a single parent cell divides into two identical daughter cells. A single-celled organism called an amoeba reproduces by binary fission. The amoeba's cytoplasm and nucleus divide during this phase, creating two daughter cells that are genetically identical to one another. Some unicellular creatures, like amoeba and bacteria, frequently reproduce in this way in order to proliferate and colonise their surroundings.

5. Correct option: (D) Stamen

The stamen, a flower's male reproductive organ, is absent from female flowers. The filament, which supports the anther, and the anther, which produces pollen grains, are the two primary components of the stamen. The pistil, which has three main components—the ovary, the style, and the stigma—is the reproductive structure in female flowers. Ovules are found in the ovary, where they undergo fertilisation to become seeds. The stigma is the receptive surface where pollen settles and germinates, leading to fertilisation, and the style is a tube-like structure that connects the stigma to the ovary.

6. Correct option: (B) Testes

The testes are one of the most important parts of the male reproductive system, which also comprises other organs. Sperm and male sex hormones like testosterone are produced by the testes. The epididymis, which matures and stores sperm, the vas deferens (sperm ducts),

which transport sperm from the testes to the urethra, and accessory glands that generate fluids that combine with sperm to make semen are further parts of the male reproductive system. The testes are unique to the male reproductive anatomy, in contrast to the other alternatives (Fallopian tube, uterus, and ovary), which are components of the female reproductive system.

7. Correct option: (C) Uterus

The uterus, a crucial reproductive organ in all female mammals, including humans, is where the embryo grows. The zygote (fertilised egg) passes through numerous cell divisions after fertilisation to become a blastocyst. The blastocyst is subsequently implanted into the uterine lining, where it receives nourishment and assistance for growth. During pregnancy, the uterus offers a favourable environment for the embryo to transform into a foetus. This process results in the development of diverse embryonic structures as well as the growth of organs and systems, which ultimately results in the birth of a child that is fully developed.

8. Correct option: (A) Frog

Fertilization normally occurs externally in frogs. Male and female frogs both discharge sperm and eggs into the water during the breeding season. This external fertilisation takes place in watery habitats like ponds, lakes, or streams when the female frog's eggs and the male frog's sperm come into touch. Tadpoles are produced from the fertilised eggs, and these creatures eventually change into adult frogs. This process of fertilisation, which is an illustration of external fertilisation as opposed to internal fertilisation, in which fertilisation takes place inside the female's body, is ideally adapted to the frog's aquatic existence.

9. Correct option: (B) Tadpole

Tadpoles are the name for a frog's larval stage. Aquatic animals known as tadpoles are the offspring of frog eggs. As they grow into adult frogs, they often have a long tail, external gills for breathing underwater, and go through a transformation process. Tadpoles eventually lose their gills during metamorphosis, and their tails shorten and their legs begin to develop. They eventually change into froglets, which then grow into adult frogs that can survive on land. Tadpoles are an important stage in the life cycle of the frog, allowing them to colonise aquatic habitats before switching to a terrestrial way of life.

10. Correct option: (A) Male flower

The male reproductive organs of blooming plants are called pollen grains. They are created by the stamen and other male flower components. Anthers and filament make up the stamen. The pollen grain, which contains the male gametes, is created by the anther (sperm cells). When pollen grains are discharged from the anthers, they are frequently carried to female flowers by wind, insects, or other pollinators where they can fertilise the ovules and start the seed-forming process.

11. Correct option: (C) In Vitro Fertilisation

In vitro fertilisation is a technique used to create "test tube babies," commonly referred to as "IVF babies." An egg (ova) is fertilised by a sperm during IVF outside of the body, typically in a lab dish or culture medium. After fertilisation, the resultant embryos are given a few days to mature before being placed into the uterus in preparation for possible implantation and pregnancy. When natural conception is difficult or not possible, IVF is a reproductive technology used to treat infertility and help people or couples become pregnant.

12. Correct option: (C) 270–280 days

Human gestation, sometimes referred to as pregnancy, usually lasts between 270 and 280 days, or around 9 months. From the first day of the woman's last menstrual period until the baby is born, this period is measured. The precise duration of pregnancy can vary slightly from woman to woman and can be impacted by a number of variables, including

the mother's and the baby's health, genetics, and environmental influences. However, the typical length is between 270 and 280 days.

13. Correct option: (D) Camel

The other choices, fish, frog, and lizard, are all vertebrate animals that fall within the classifications of Reptilia (for lizards) and Amphibia (for frogs) (for lizards). The physiological and anatomical properties of these animals are comparable. The camel, on the other hand, is a mammal and a member of the Mammalia phylum. All of the animals on the list are vertebrates, but the camel stands out from the other three due to its warm-bloodedness, the fact that it has mammary glands for producing milk, and its adaptations to desert settings. The camel differs from the other animals in that it is not aquatic, amphibious, or reptilian in nature.

14. Correct option: (D) Ostrich

The ostrich is an example of an oviparous animal, which are those that deposit eggs. African ostriches are enormous, flightless birds. They lay their eggs in ground nests, which are either incubated by the female ostrich or, in certain situations, also by the male. Prior to being laid, the eggs are internally fertilised, and they normally hatch after a predetermined incubation period. In order to secure the survival of their young in a variety of settings and conditions, oviparous animals, like the ostrich, have evolved this reproductive strategy. This technique frequently entails protective nests and systems to preserve the ideal environment for embryo development.

15. Correct option: (D) Reproduction

The fundamental life process that is essential to the survival of a species is reproduction. Through the development of children, it entails the creation of new individuals. In order to sustain genetic diversity and the survival of the species, reproduction makes sure that genetic information is transferred from one generation to the next. There are many different ways for different species to reproduce, including asexual reproduction, in which children are created from a single parent, and sexual reproduction, in which male and female gametes are fused to create offspring. A species' ability to reproduce and maintain its population over time would be compromised without it, which would eventually cause it to decrease and go extinct.

16. Correct option: (B) Ian Wilmut

An important historical milestone was reached when the first instance of cloning was carried out by pioneering British embryologist Ian Wilmut. He was an important contributor to the group at the Roslin Institute in Scotland that cloned a sheep dubbed "Dolly" successfully in 1996. Dolly was a significant accomplishment since she was the first mammal to be cloned utilising the somatic cell nuclear transfer method from an adult somatic cell. In addition to showing that it was possible to clone animals, Wilmut's discovery also provided new opportunities for genetic study, biotechnology, and the knowledge of cellular reprogramming. The study of biology and genetics advanced significantly as a result of this ground-breaking achievement.

17. Correct option: (C) Metamorphosis

A larval stage of an organism goes through considerable structural and physiological changes during the process of metamorphosis to become an adult form. This metamorphosis involves a number of developmental phases, many of which have unique external characteristics and ecological functions. The majority of insects, amphibians, and some aquatic species exhibit this behaviour. Examples of metamorphosis include the development of a tadpole into a frog or a caterpillar into a butterfly. In order to adjust to the new adult lifestyle and surroundings, it often entails the reorganisation and reabsorption of tissues as well as the restructuring of bodily parts.

18. Correct option: (A) Asexual

Asexual reproduction is defined as reproduction that occurs without the release of male and female gametes. Asexual reproduction produces children without the fusing of gametes, which are specialised reproductive cells, from two distinct parents. Since there is no genetic material mixing as occurs during sexual reproduction, this technique produces children that are genetically identical or highly similar. Binary fission, budding, and fragmentation are asexual reproduction techniques examples where the offspring receive the genetic features straight from a single parent.

19. Correct option: (D) Hermaphrodite and Bisexual animal

Hermaphrodites are animals with both male and female reproductive systems. As a result, they can reproduce with either males or females of their species since they have the capacity to generate both eggs and sperm. When referring to an animal that can reproduce in both sexes, the terms "bisexual animal" and "hermaphrodite" are frequently used interchangeably. As a result, D) Hermaphrodite and Bisexual Animal is the right response to the question because it includes both names and depicts creatures with both male and female reproductive systems.

20. Correct option: (B) Budding

In yeast cells, asexual reproduction occurs by budding. A tiny bud or protrusion on the parent cell emerges during budding, and as it expands and separates from the parent cell, it develops into a new organism. Under ideal circumstances, this mechanism enables yeast cells to multiply quickly and successfully. Yeast cells benefit from budding because it enables them to rapidly increase their population and adapt to changing conditions. Additionally, because the offspring receives a copy of the parent's genetic material, budding ensures genetic resemblance between the parent and offspring. Overall, yeast cells primarily reproduce asexually by budding, which aids in their survival and growth.

LEVEL – II

1. Correct option: (B) Reproduction

The biological process of reproduction is how living things create new members of the same species. It is a basic aspect of life that guarantees the survival of species over generations. It is possible for creatures to reproduce asexually and sexually, and this allows them to pass on their genetic makeup and characteristics to the following generation. The ability to make new people of the same sort is not immediately related to respiration, on the other hand, which is a metabolic activity involved in energy production and the exchange of gases.

2. Correct option: (A) 2

Asexual reproduction and sexual reproduction are the two basic types of reproduction. Asexual reproduction refers to the generation of genetically identical or extremely similar children from a single parent organism. Contrarily, sexual reproduction entails the union of gametes (sperm and eggs) from two distinct parents, resulting in genetic variation in the progeny. These two methods of reproduction are important tactics used by different creatures to maintain their species.

3. Correct option: (A) Asexual

One parent usually participates in asexual reproduction, which is a sort of reproduction. Asexual reproduction is the process of producing offspring from a single parent organism without the joining of gametes or the use of genetic material from another person. Since there is no recombination of genetic material as occurs during sexual reproduction, this technique produces children who are genetically identical or highly similar. Numerous

organisms have asexual reproduction, which is advantageous in circumstances that favour rapid population increase and colonisation.

4. Correct option: (B) Asexual reproduction

Amoebas are single-celled organisms that frequently reproduce asexually through a process known as binary fission. In binary fission, the amoeba's cell nucleus splits in half, then the remaining components of the cell divide as well. Due to this, two daughter cells are created that are genetically identical to one another and grow and develop into separate amoebas. Under the right circumstances, asexual reproduction enables amoebas to multiply quickly and colonise their surroundings.

5. Correct option: (C) Budding

The Cnidaria phylum member Hydra is a basic aquatic organism that reproduces asexually via a process known as budding. A new person, known as a bud, develops as a protrusion on the parent organism during this process. The bud gradually matures and separates from the parent to become a distinct and autonomous organism. Hydra frequently reproduce asexually by budding, which enables them to effectively increase their number under favourable circumstances.

6. Correct option: (D) Hydra

When an organism reproduces asexually, it creates offspring that are genetically identical to it. Simple creatures like bacteria and plants frequently reproduce in this way. Asexual reproduction is possible for the small aquatic organism Hydra, which is a member of the phylum Cnidaria. This process is known as budding. A new individual develops as an outgrowth or bud on the parent organism during budding. This bud develops, eventually separating to become a separate entity. Asexual reproduction through budding allows hydra to efficiently produce genetically identical offspring, promoting population growth in suitable environments.

7. Correct option: (B) Viviparous

The animals that directly reproduce young ones are called viviparous. Viviparity is a form of reproduction where the embryos grow inside the mother's body and are fed by the mother herself. The growing embryos in viviparous creatures, such as most mammals and certain reptiles, are given nutrition and oxygen through a placental link or other specialised structures, allowing them to grow and develop before giving birth as living offspring. Opposing this mode of reproduction is oviparity, in which animals lay their eggs outside of their bodies, and the embryos grow outside the mother's body. Therefore, the correct answer is B) Viviparous.

8. Correct option: (D) Sperm

Sperm makes up the male gamete. In many animal species, including humans, males generate sperm, which are specialised reproductive cells. Sperm cells are created with fertilisation in mind. They often have a tail (flagellum), which allows them to travel through fluid environments like the female reproductive canal to reach the egg. Sperm are essential for sexual reproduction because their nuclei contain genetic information. The formation of a new person with a special blend of genetic features from both parents begins with the union of a sperm and a female gamete (egg), which creates a zygote. Therefore, sperm are necessary for a species to continue through sexual reproduction.

9. Correct option: (A) Sponges

Internal budding, commonly referred to as gemmulation, is an asexual reproduction process that is seen in sponges (phylum Porifera). Gemmules, which are tiny bud-like structures, develop within the parent sponge's body throughout this phase. The cells that make up these gemmules are grouped together and covered in a layer of defence. The parent sponge can discharge these gemmules into the nearby water when the environment becomes

adverse, such as during periods of hard weather. The gemmules serve as a kind of survival capsule, shielding the sponge cells from harmful elements. When conditions are better, the gemmules can separate and grow into new sponge individuals, enabling sponges to endure and spread over a variety of aquatic settings. Internal budding in sponges is a remarkable adaptation that aids in their ability to reproduce asexually and withstand challenging ecological circumstances.

10. Correct option: (C) Fragmentation

Asexual reproduction known as fragmentation allows for the growth of a whole new organism from specific parts or cells of the original organism. The parent organism splits into various pieces throughout this process, and each piece has the ability to regenerate and grow into a brand-new, autonomous individual. Each fragment finally develops into a complete organism after the missing body parts are grown again by the pieces. Many different animals, including flatworms and some plant species, exhibit fragmentation, which enables them to procreate without the necessity for mating or the development of specific reproductive cells. The parent organism can produce several offspring using its own body parts thanks to this reproductive strategy, which increases population size.

11. Correct option: (B) Fragmentation.

Flatworms, also known as planarians, are capable of reproducing through a process called fragmentation. A flatworm can be divided into two or more parts via this technique of reproduction, and each portion can grow back into a whole creature. Each piece develops into a distinct individual and grows any missing body parts, such as the head and tail. Flatworms can produce new individuals from dismembered or separated body parts thanks to this type of asexual reproduction. While the animal kingdom also employs alternative asexual reproduction strategies, such as binary fission and budding, flatworms stand out for their use of fragmentation.

12. Correct option: (B) Sexual reproduction

An example of sexual reproduction occurs when male and female gametes from two distinct parents are fused. Genetic material is transferred between the parents during sexual reproduction, resulting in a genetically diverse progeny. This procedure typically calls for the development of specialised reproductive cells, or gametes, like sperm and eggs, which have half as many chromosomes as usual. These gametes join together during fertilisation to form a new person that combines genetic characteristics from both parents. Contrarily, asexual reproduction entails the generation of children without the requirement for gamete fusion, resulting in offspring that are genetically identical to their parents. B) Sexual reproduction is the right response, so.

13. Correct option: (C) Zygote

A sperm or egg (ova) cell is an example of a specialised reproductive cell known as a gamete. Gametes are necessary for sexual reproduction and are produced by both male and female members of a species. A zygote is the structure formed when a male gamete (sperm) combines with a female gamete (egg). The first cell to form during fertilisation is called a zygote, and it combines genetic material from both parents. The embryo develops into a multicellular organism after undergoing fast cell divisions and development from the single-celled zygote. Thus, the development of sexual reproduction begins with the union of two gametes to produce a zygote.

14. Correct option: (B) Bisexual organism

An organism that has both male and female reproductive systems and is able to produce both male and female gametes (sperm and eggs) within a single individual is referred to as a hermaphrodite or a bisexual organism. With the use of one's own sperm and eggs, an individual may self-fertilize via this reproductive technique. Many animal species,

particularly invertebrates like some fish, worms, and snails, are found to be hermaphrodites. The ability to reproduce even without a mate is ensured by this technique, however genetic variety may be lessened compared to cross-fertilization between different individuals.
Unisexual organisms, on the other hand, produce only one type of gamete. Male organisms produce sperm, and female organisms produce eggs. Some examples of unisexual organisms include humans, dogs, and cats.

15. Correct option: (C) Testes

As a component of the male reproductive system, the testes are where sperm, also known as spermatozoa, are created. The testes are two oval-shaped organs that hang outside of the male body in the scrotum. Seminiferous tubules, a large number of small tubules within the testes, are where spermatogenesis occurs. Sperms are created during the process known as spermatogenesis from unique cells known as spermatogonia. These cells go through a number of divisions and changes before becoming fully developed sperm cells. The mature sperm are released into the epididymis, a coiled tube attached to the testes, where they continue to develop and are kept until ejaculation, once they have fully developed. In the male reproductive system, the testes serve as the major location for the generation of sperm.

16. Correct option: (C) Sperm ducts

Sperm are transported from the testes to the penis by sperm ducts, commonly known as the vas deferens or ductus deferens. The male reproductive system includes these ducts, which act as a route for sperm to follow after ejaculation. The testes produce sperm, which mature as they pass through the testes' epididymis, a coil-like structure. In order to get to the urethra near the base of the penis, the sperm must first travel through the sperm ducts, which run from the epididymis through the pelvic cavity. The sperm are launched through the sperm ducts during ejaculation and combine with secretions from other accessory glands to form semen, which is then ejected from the penis. Thus, the sperm ducts play a crucial role in the transport of sperm for fertilization.

17. Correct option: (B) Four

A specialised cell used in sexual reproduction is a sperm. The head, the midpiece, the tail (sometimes referred to as the flagellum), and the neck are its four primary components. The genetic material is housed in the compact nucleus of the cranium. During fertilisation, the acrosome, a structure at the tip of the head, provides enzymes required for the sperm to pierce the egg's outer layers. There are many mitochondria in the midpiece, which supply the sperm with the energy it needs to migrate. The sperm can move through fluid environments thanks to the tail, or flagellum, which is a long, whip-like structure. The midpiece and the head are connected by the neck. These four components work together to facilitate the sperm's journey toward the egg for the potential process of fertilization.

18. Correct option: (B) Female

The correct answer is B) Female. The flagellum, or tail, of a sperm is a whip-like feature that helps the sperm move forward in a fluid environment. The sperm's tail is principally in charge of enabling travel into the female reproductive canal in the context of human reproduction. The sperm must pass through the female reproductive system after being ejaculated during sexual activity in order to get to the fallopian tubes, where fertilisation may eventually take place. The sperm can travel via the cervical mucus, the uterus, and finally the fallopian tube where they may come into contact with an egg for fertilisation thanks to the movement of the tail. As a result, the tail's main job is to facilitate sperm passage inside the female reproductive system.

19. Correct option: (B) Penetration

A sperm cell's head contains a unique component called the acrosome. It is essential to the fertilisation process. When the sperm makes contact with the zona pellucida, the outer covering of the egg, enzymes from the acrosome, which are present, are released. These enzymes aid in removing the egg's defensive barriers, allowing the sperm to enter and unite with the egg cell. This penetration is a crucial stage in fertilisation because it enables the sperm's genetic material to unite with the egg's genetic material to create a zygote, which will later develop into an embryo. As a result, the acrosome's main purpose is to make it easier for the egg to penetrate.

20. Correct option: (A) Ova

The female reproductive system's egg-producing organs are the ovaries. A different name for the eggs is ova. One ovary is situated on either side of the uterus in the pelvic. The female gametes or eggs, known as ova, are produced by the ovaries, which are a component of the female reproductive system. These eggs, which contain the genetic material of the female, are necessary for sexual reproduction. Each of the countless ovarian follicles found in ovaries is home to an immature egg. Typically, one egg is chosen and matures within a follicle during the menstrual cycle before being released through the process of ovulation. Following the release of the egg and the subsequent fusing of a sperm and an egg, fertilisation can result in the creation of a foetus.

21. Correct option: (B) Ovulation

A developed egg, or ovum, is released from the ovary into the fallopian tube during ovulation, a biological process. The pelvis contains two tiny, almond-shaped structures called ovaries. They generate hormones and eggs. The ovarian follicle that contains the egg ruptures during ovulation, releasing the egg into the pelvic cavity. The fallopian tube, which is lined with small hair-like structures known as cilia and aids in capturing the egg there, then directs it toward the uterus. The sperm cell must fertilise the egg for it to embed in the uterine lining and begin a pregnancy. Ovulation often occurs 14 days or so before a woman's period begins.

22. Correct option: (C) Inside the female body

Male gametes (sperm) are placed inside the female body during internal fertilisation, where they fertilise the female gametes (eggs). Mammals, birds, and some reptiles frequently undergo this sort of fertilisation. In mammals, the male ejaculates sperm after inserting his penis into the female's vagina. The sperm next proceed to the ovaries, where they can fertilise the eggs, via the uterus and fallopian tubes. Compared to external fertilisation, internal fertilisation provides a number of benefits. The fact that the eggs are shielded from predators and the elements is one benefit. The sperm are more likely to fertilise the eggs, which is an additional benefit.

23. Correct option: (A) Fertilization

The creation of a zygote occurs when a male gamete (sperm) combines with a female gamete (egg or ovum) during the process of fertilisation. This is the first step in a person's growth. A sperm cell enters the interior of the egg during fertilisation after breaking through its outer layers of protection. Once within, the sperm and egg nuclei combine to form a single cell, fusing their genetic material. The two parents' genetic material combines to form a unique set of DNA that determines the offspring's genetic features and characteristics. The crucial process of fertilisation takes place inside the female reproductive system, frequently in the fallopian tubes in humans.

24. Correct option: (B) Fallopian tube

Human fallopian tubes are where fertilisation occurs. The slender, tube-like fallopian tube, often referred to as the oviduct, joins the ovaries with the uterus. An egg is discharged from

one of the ovaries during ovulation, and it travels through the fallopian tube. Fertilization can take place if both sperm are in the fallopian tube at the same moment. To get to the fallopian tube, the sperm must pass through the cervix and enter the uterus. The fertilised egg, also known as a zygote, travels via the fallopian tube after fertilisation in order to reach the uterus, where it implants and grows into a foetus. Therefore, human fertilisation takes place in the fallopian tube.

25. Correct option: (A) Fish

In fish, external fertilisation is seen. The male fish releases his sperm to externally fertilise the eggs while the female fish discharges her eggs into the water throughout this procedure. This sort of fertilisation takes place outside the female fish's body, typically in bodies of water like rivers, lakes, or oceans. Fish benefit from external fertilisation because it makes it possible to fertilise many eggs at once, increasing the likelihood of fertile offspring. Additionally, it gives sperm and eggs a chance to freely mix in the water, enhancing the genetic diversity of the offspring. However, external fertilisation also subjects the sperm and eggs to a variety of environmental elements, leaving them open to predators and other dangers. Overall, external fertilisation is a unique reproductive strategy observed specifically in fish

26. Correct option: (C) Implantation

The procedure through which the embryo fuses to the uterine wall is known as implantation. It is an important stage in the progression of a pregnancy. The endometrium, the uterus' lining, is where the embryo first penetrates. The embryo's trophoblast cells then infiltrate the endometrium and establish a blood vessel between it and the mother. This process, which enables the development of the placenta, a crucial organ created later in gestation, allows the embryo to obtain needed nutrients and oxygen from the mother's bloodstream, is crucial for the progression of pregnancy. In most cases, implanting happens 6 to 10 days after conception. After fertilisation, it can happen as soon as 4 days later or as late as 12 days later.

27. Correct option: (B) Foetus

The term "Fetus" refers to the developmental stage at which an embryo's individual bodily components may be recognised. The embryo has grown and differentiated significantly during this stage, and its main organs and structures have formed. Rapid cell division and the building of the framework for many organs and systems characterise the development in the stage before this one, known as the embryo stage. The embryo's fundamental structures have formed during the foetal stage, which is a later stage of development.

28. Correct option: (B) Uterus

The uterus is not a part of the male reproductive organ. It is a female reproductive system that is essential to pregnancy. During pregnancy, the uterus serves as the foetus' home and source of nutrition. The reproductive system in males consists of the penis, testes, and seminal vesicles. Urine and semen are transported through the penis during sexual activity. The seminal vesicles help to create semen, whereas the testes produce sperm and male sex hormones. The only organ available that is not a part of the male reproductive system is the uterus.

29. Correct option: (D) Both B and C

Animals that give birth to live offspring are said to be viviparous. Animals that lay eggs include hens and silkworms, which are both oviparous. Young that can survive without their mother emerge from the eggs.

Eggs laid by hens develop into chicks. Within hours of hatching, the chicks are independent. Eggs laid by silkworms develop into caterpillars. The caterpillars undergo several moults before pupating to become moths. The moths have the ability to fly and lay eggs.

Humans are viviparous creatures. The fertilised egg grows into a foetus after implanting in the mother's uterus. Through the placenta, the mother feeds the foetus. After roughly nine months of gestation, the baby is delivered.

30. Correct option: (D) Hen

Hens do not undergo metamorphosis, a process that results in a significant shift in body composition. Hens grow and mature gradually without going through the specific metamorphic changes that these other organisms do, but animals like silkworms, fish, and butterflies go through metamorphic transformations during their life cycles.

Silkworms and butterflies emerge from their eggs as caterpillars, and as they develop, they undergo a series of moults. While the butterfly's last moult results in the adult butterfly, which has wings, the silkworm's final moult results in the adult silkworm, which is a moth. Fish grow through a series of moults after emerging from their eggs as larvae. The adult fish, which has scales and fins, is created after the last moult.

8. Our Forest

Solutions

LEVEL-I

1. Correct option: (B) Sunderlal Bahuguna

The Chipko movement was started by Sunderlal Bahuguna in the Garhwal forests. The movement was aimed at stopping the felling of trees for commercial purposes. The movement was non-violent and the villagers hugged trees to stop them from being felled.

2. Correct option: (D) 20 %

The percentage of forest area of our county is 21.71%. The total forest cover in India is 80.9 million hectares (x 100 square kilometres) which makes 21.71 percent of the total geographical area in 2021. According to a recent study, India's forest/tree cover has increased by more than 1,500 sq.

3. Correct option: (C) Both A and B

The Chencho's hunt wild animals like boar and deer, but with the increasing interest in wildlife conservation, they are content to hunt small animals like lizards, rabbits and wild birds.

4. Correct option: (A) Anantha Giri Forest

Konda Mallipudivana Samrakshana Samithi is located at the foothills of Anantgagiri forest.

5. Correct option: (D) All of the above

Dandakaranya forest holds importance due to its mention in Ramayana. The location housed various demons and deadly creatures. It is also believed that Hindu deities lived there. Panchavati forest is present in Nashik city, on the banks of the Godavari River. Nallama forest is located in Andhra Pradesh and Telangana. The forest is part of Eastern Ghats.

6. Correct option: (B) Conservation

Nature conservation is the moral philosophy and conservation movement focused on protecting species from extinction, maintaining and restoring habitats, enhancing ecosystem services, and protecting biological diversity.

7. Correct option: (A) Fauna

Total number of animal species present in a region are called fauna. Wildlife refers to animals that cannot be domesticated by human beings and known as 'fauna'. India is also home to a variety of wildlife. Tigers are found throughout India - from the Himalayas to Kanyakumari. The Royal Bengal Tiger is our national animal.

8. Correct option: (C) Soil erosion

Soil erosion is the denudation or wearing away of the upper layer of soil. It is a form of soil degradation. This natural process is caused by the dynamic activity of erosive agents, that is, water, ice, snow, air, plants, and animals.

9. Correct option: (D) Forest

The plants release oxygen through the process of photosynthesis. The plants help to provide oxygen to animals for respiration. They also maintain the balance of oxygen and carbon dioxide in the atmosphere. That is why forests are called green lungs.

10. Correct option: (B) Rosewood

Rosewood is a species that can be used in projects to restore degraded lands, which can be applied to thousands of hectares of degraded lands across the region impacted by extractive activities. Moreover, the trees planted in this project can serve as a seed bank for others who may wish to reforest and conduct ecological restoration that includes rosewood, either for strict conservation or for managed, non-destructive leaf harvest, which could generate additional income for local communities.

11. Correct option: (A) Forest of cold areas
Alpine forests are mostly found in mountainous areas. Alpine forests can be found across the Himalayas, at elevations ranging from 2,900 to 3,500 m or indeed up to 3800 m above sea level, based on location and plant variety.
12. Correct option: (D) All of the above
overpopulation, urbanisation and Industrialisation are the causes for deforestation.
13. Correct option: (D) All the above
Limiting the use of firewood, using recycled paper and using biofuels for cooking all will result in preventing deforestation
14. Correct option: (A) Biodiversity
Biodiversity is the variety of plant and animal life in a particular habitat.
15. Correct option: (D) Resources
Materials present in large quantities for future needs are called Resources. A resource is a physical material that humans need and value such as land, air, and water. Resources are characterized as renewable or non-renewable; a renewable resource can replenish itself at the rate it is used, while a non-renewable resource has a limited supply.
16. Correct option: (A) 25Km
17. Correct option: (D) all the above
Elephants, rabbits and monkey etc are found in the forestry of Adilabad district.
18. Correct option: (D) all the above
Writing pads, cot and pencil are made up from wood.
19. Correct option: (D) all the above
We all depend on the forest for groundwater, rainfall and medicinal plants.
20. Correct option: (D) kerosene
Kerosene can be produced from distillation of crude oil (straight-run kerosene) or from the cracking of heavier petroleum streams (cracked kerosene).

LEVEL-II

1. Correct option: (B) Crown
The branchy part of a tree above the stem is known as the crown
2. Correct option: (C) Both A and B
A Forest is a purifier of water and air because it helps in bringing rain and helps in maintaining ground water and provides fresh water. Moreover, forests are responsible for maintaining a balance between oxygen and carbon dioxide gas in the atmosphere and provide fresh air.
3. Correct option: (C) Food chain
The interdependence of animals for food is called a food chain. For example, one kind of interdependent relationship would be the bees and butterflies that help pollinate many kinds of flowers or the bats and birds that carry pollen to other areas so that new plants can grow. Even a predator-prey relationship is a form of interdependence.
4. Correct option: (D) honey
Honey is produced from bees (Yes, insects are considered animals) and therefore is not a vegan product. According to PETA, honey is the energy source for bees and they would starve without it, especially during winter and poor weather conditions. Conventional beekeeping works to harvest the largest amount of honey.
5. Correct option: (B) humus
Decomposers feed on dead plants and animal tissues. They convert them into humus. This activity also helps in increasing the soil fertility.

6. Correct option: (C) Porcupine
Porcupines are large rodents with coats of sharp spines, or quills, that protect them against predation. Sal, teak and kachnar are trees.
7. Correct option: (C) Decomposers
Nature has its own recycling system: a group of organisms called decomposers. Decomposers feed on dead things: dead plant materials such as leaf litter and wood, animal carcasses, and faces.
8. Correct option: (D) All of these
Understoreys are formed due to different types of crowns, different size of crown, different height of trees. The lower layer of the forest formed due to different types of crowns in the forest is known as the understory. The constituents of the understorey are herbs and shrubs. The plants which are usually short and have green, tender stems are known as herbs.
9. Correct option: (B) Canopy
Tall trees which look like roof over other plants in the forest is called canopy
10. Correct option: (C) Porcupine
Porcupines are large rodents with coats of sharp spines, or quills, that protect them against predation. Sal, teak and kachnar are trees.
11. Correct option: (A) Bird, rabbit, elephant
Primary consumers are a group of organisms that are placed in the second trophic level. They feed on producers i.e grass and green plants. They are therefore also known as herbivores.
12. Correct option: (A) Cinchona
Cinchona bark contains quinine, which is a medicine used to treat malaria.
13. Correct option: (C) Furniture industry
Industries which are totally dependent upon the products of forests are - Paper Industry, Match Industries, Plywood Industries and Fibre Board Industries.
14. Correct option: (B) Decomposing
Several beetles and grubs were feeding on the heap and a bunch of seedlings was sprouting. “These seedlings are of the herbs and shrubs the animals also disperse the seeds of certain plants and help the forest to grow and regenerate. The decaying animal dung also provides nutrients to the seedlings to grow”, said Prof Ahmad.
15. Correct option: (C) Recycling of paper bags
Recycling of cardboard cartons and paper bags will lead to a decrease in the level of tree cutting and hence conserve forests.
16. Correct option: (B) Forest
Forest is a natural absorber of rain water. It allows rain water to seep through. The roots of the trees absorb the water and prevent it from flowing away. The trees present in the forest prevents the rain from directly hitting the ground which may result in floods. It rather hits the ground slowly. Hence, trees in the forest slow down the movement of running water and thus help in preventing floods.
17. Correct option: (C) Humus
Decomposers or saprotrophs recycle dead plants and animals into chemical nutrients like carbon and nitrogen that are released back into the soil, air, and water. Fungi like mushrooms, mildew, Mold, and toadstools can't make their own food.
18. Correct option: (B) Thermocol
Thermocol is not prepared from the wood obtained from the forest. Paper, matchsticks, and plywood are prepared from the wood.
19. Correct option: (B) Global Warming
If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.

20. Correct option: (A) Grass → insects → frog → snake → eagle
Grass is a primary producer which is consumed by insects, insects are further eaten by amphibians like frogs which are consumed by snakes, and snakes are consumed by eagles.
21. Correct option: (C) Creating flood conditions
Forest is a natural absorber of rain water. It allows rain water to seep through. The roots of the trees absorb the water and prevent it from flowing away. The trees present in the forest prevents the rain from directly hitting the ground which may result in floods. It rather hits the ground slowly. Hence, trees in the forest slow down the movement of running water and thus help in preventing floods.
22. Correct option: (B) plants and animals in a forest are not dependent on one another.
Animals are dependent on plants for shelter and food, while plants are dependent on animals for pollination and seed dispersal. The plants filter water and make it clean, and animals require clean water for drinking purposes.
23. Correct option: (D) All the above are correct.
Absence of forest will result in no food and shelter for animals and If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.
24. Correct option: (A) Amount of CO2 in air will increase
If forests disappear, the amount of carbon dioxide in air will increase, resulting in the increase of earth's temperature.
25. Correct option: (B) Dodo
The dodo (*Raphus cucullatus*) is an extinct flightless bird that was endemic to the island of Mauritius, east of Madagascar in the Indian ocean. It became extinct in 14th century from Mauritius because of large-scale hunting. Dodo was helpful in pollinating and propagating seed of this tree species. Subfossil remains show the dodo was about 1 metre (3 ft 3 in) tall and may have weighed 10.6–21.1 kg (23–47 lb) in the wild.
26. Correct option: (A) Semal and hornbill
Semal (Silk Cotton) tree is a type of native cotton tree with large red flowers. It grows rapidly to attain height up to 30 m and a girth of 3-4 m and the largest hornbill species in India, Great Hornbills are also called Great Indian or Great Pied Hornbills.
27. Correct option: (D) None of the above.
The plants are called producers, the animals are called consumers and Saprotrophs are commonly known as decomposers are correct.
28. Correct option: (D) None of the above.
The branchy part of a tree above the stem is known as the crown of the tree, the layer of vegetation in a forest which is just below the canopy is called under story and the uppermost branches and leaves of tall trees which act like a roof over the forest ground is called canopy.
29. Correct option: (B)
a – iv) The forest canopy is a structurally complex and ecologically important subsystem of the forest.
b – iii) Forest is a renewable resource. But, each year, the earth loses about 36 million acres of forest.
c – ii) Plants gives out oxygen.
d – i) Small trees are Under storeys trees are shade tolerant species, about 15 to 50 feet in height, that grow under the crowns of larger trees.
30. Correct option: (D)
a – iii) Herbs are grasses.
b – i) Watermelon is a creeper.
c – v) Pea plants are climber.
d – ii) Hibiscus is a shrub.
e – iv) Banyan is a tree.

9. Cell-The Basic Unit of Life

Solutions

LEVEL-I

1. Correct option: (A) Oak tree

Cork has dead cells which resembles the cells found in Matchstick. Onion peel cells and Cheek cells are living cells.

2. Correct option: (B) Robert Brown

Robert Brown discovered nucleus in 1831.

3. Correct option: (D) Nucleus

Robert Brown observed nucleus in a plant cell while studying the process of fertilization and claimed that it is an integral part of any cell, in the year 1831.

4. Correct option: (A) Orchid

Robert Brown coined the term 'nucleus' in 1831. He first observed the nucleus in the epidermis of the orchids as an opaque spot.

5. Correct option: (A) Cytoplasm

Cytoplasm is the jelly-like substance present between the cell membrane and the nucleus. It is mainly composed of water, salts, and proteins. All the membrane bound cell organelles like endoplasmic reticulum, golgi complex, etc are present in the cytoplasm.

6. Correct option: (D) All of the above

Unicellular organisms include all prokaryotes including Bacteria, and Eukaryotes including Chlamydomonas sp, and Amoeba sp.

7. Correct option: (C) Amoeba sp.

Pseudopodia or false feet are cytoplasmic extensions present in unicellular protists, and functions in locomotion and capturing of prey.

8. Correct option: (A) Protection

The dermal tissue covers the outer surface of the stem and usually functions to waterproof, protect, and control gas exchange.

9. Correct option: (D) All of these

Any process that is commonly exhibited by living organisms on a daily basis for carrying out daily metabolic processes can be termed as a life process.

10. Correct option: (C) Janssen

A Dutch father-son team named Hans and Zacharias Janssen invented the first so-called compound microscope in the late 16th century when they discovered that, if they put a lens at the top and bottom of a tube and looked through it, objects on the other end became magnified.

11. Correct option: (D) Cell

The cell is the basic structural and functional unit of life. Cells are independent, single-celled organisms that take in nutrients, excrete wastes, detect and respond to their environment, move, breathe, grow, and reproduce.

12. Correct option: (C) Microscope

Microscope, instrument that produces enlarged images of small objects, allowing the observer an exceedingly close view of minute structures at a scale convenient for examination and analysis.

13. Correct option: (D) Sclerenchyma cells

Mature sclerenchyma cells are usually dead cells that have heavily thickened secondary walls containing lignin. The cells are rigid and non-stretchable and are usually found in nongrowing regions of plant bodies, such as the bark or mature stems.

14. Correct option: (A) The number of cells present
The size of an organism primarily depends upon the number of cells in its body. When the number of cells increases then the size of the organism also increases.
15. Correct option: (A) Cell wall
The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress.
16. Correct option: (D) Both A and B
The functions of pseudopodia include locomotion and ingestion: Pseudopodia are critical in sensing targets which can then be engulfed; the engulfing pseudopodia are called phagocytosis pseudopodia.
17. Correct option: (A) 1-A; 2-D; 3-B; 4-C
Robert Hooke in the year 1665 discovered cell in cork tissue. Robert brown discovered nucleus in orchids in the year 1831. Antonie van Leuwenhoek made the first microscope and visualized live cells. A plant cell is a living cell due to a proper functioning metabolism.
18. Correct option: (D) 2 only
Robert Brown was the one who discovered nucleus while studying orchid leaves and coined the term 'nucleus' in 1831.
19. Correct option: (D) Both A and B
The shape of the human red blood cell is known to be a biconcave disk.
20. Correct option: (D) 90-100 cm
The nerve cell is the structural and functional unit of the nervous system. They are generally long and branched cells and can be 90 - 100 cm long.

LEVEL II

1. Correct option: (B) Cytoplasm
The cytoplasm is a thick, clear, jelly-like substance present inside the cell membrane. Most of the chemical reactions within a cell take place in this cytoplasm. The cell organelles such as endoplasmic reticulum, vacuoles, mitochondria, ribosomes, are suspended in this cytoplasm.
2. Correct option: (D) Cell wall, Chloroplast, Mitochondria, Cell membrane, Nucleus
The following are the components of animal cells: Nucleus, Ribosomes, Endoplasmic reticulum, Lysosomes, Golgi bodies, Cell membrane, Cell wall, nuclear membrane, Mitochondria.
3. Correct option: (C) c and d
As red blood cells contain no nucleus, protein biosynthesis is currently assumed to be absent in these cells. Since its an animal cell it doesn't have a plastid.
4. Correct option: (C) b and d
A nerve cell or neuron is so long about 100 cm so that it can carry messages over long distances from brain to other parts of the body. It has numerous projections to make contact with other neurons.
5. Correct option: (B) Euglena, Chlorella, Escherichia
Euglena is a unicellular protist, Chlorella is an unicellular algae growing in fresh water. Escherichia coli is a gram-negative bacterium hence an unicellular organism.
6. Correct option: (C) A-ii, B-iv, C-I, D-iii
A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.
7. Correct option: (A) Mitochondria - synthesize the respiratory enzymes
Mitochondrion is the site for aerobic respiration in cells. It contains respiratory enzymes necessary for ATP synthesis. A nucleus is the membrane-enclosed organelle within a cell that contains the chromosomes. Cytoplasm is the gelatinous liquid that fills the inside of a

cell. It is composed of water, salts, and various organic molecules. A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.

8. Correct option: (B) X – Plastid, Y – Chloroplast, Z – Chromoplast

The chloroplast is the photosynthetic plastid, and it is named for its green colour. Based on pigmentation, the non-photosynthetic plastids can be broadly divided into leucoplasts, the 'white' or colourless plastids, and chromoplasts, the coloured plastids notable for their accumulation of carotenoids.

9. Correct option: (A) a and b only

Nucleus is the largest cell organelle and controls the activities of the cell. The living material of a cell that is made up of both nucleus and cytoplasm is called protoplasm. Chloroplasts are only one, albeit the most prominent, member of a larger family of plant organelles called plastids. All plastids contain the same genome as chloroplasts, but they differ in both structure and function.

10. Correct option: (D) iii and iv only

Plant cells have a cell wall, chloroplasts and other specialized plastids, and a large central vacuole, whereas animal cells do not.

11. Correct option: (A) These are composed of tiny spherical structures surrounded by a double membrane

Lysosomes are solitary membrane-bound organelles that contain digestive enzymes that break down food scraps, bacteria, macromolecules, worn-out organelles, and viruses that have been eaten. Lysosomes do not have a double membrane. They are also called suicide bags of cell as they can digest their own cell.

12. Correct option: (A) i, iii and iv only

Anything inside the plasma membrane is called a protoplasm. This term was firstly introduced by J. E. Purkinje. Rudolf Virchow gave the statement *'Omnis cellula-e-cellula'*. It means "All cells arise from pre-existing cells. The Cell Theory was first given by M.J Schleiden and Theodor Schwann in 1839.

13. Correct option: (B) The egg cell has half as many chromosomes as the skin cell

Egg and sperm cells have just 23 chromosomes each. That's half as many chromosomes as regular cells. Through the process of fertilization, egg and sperm join to make a cell with 46 chromosomes (23 pairs), called a zygote.

14. Correct option: (D) A pancreatic cell engaged in the production of digestive enzymes

Endoplasmic reticulum helps in the transport of substances within the cell. Rough endoplasmic reticulum (RER) bears ribosomes on the cytoplasmic surface and synthesises secretory and membranous proteins. Then the modified proteins move into the transitional area where they are further processed and packed into secretory vesicles. Thus, RER is rich in cells which are actively engaged in protein synthesis and secretory activity, e.g., pancreatic cells engaged in the production of digestive enzymes, etc.

15. Correct option: (B) X could be Amoeba whereas Y could be Mycoplasma

An amoeba is an eukaryotic unicellular cell, that have a well-organized nucleus which is separated from the cytoplasm by the nuclear membrane. Mycoplasmas are the smallest bacterial cells that do not have a true nucleus and their nuclear material is in direct contact with cytoplasm, also they do not have a cell wall.

16. Correct option: (D) Leucoplasts help in the storage of Food

Plastid is the organelle that found only in plant cells. The three types of plastids are green plastid, nongreen chromoplast, and leucoplasts. The leucoplasts are the non-colored plastid which is used for the storage of the food. The oil, starch, and fats are generally stored in these plastids.

17. Correct option: (C) Only D

Chloroplast are involved in synthesis of food for plant through photosynthesis. Nucleus is a central organelle responsible for controlling all metabolic reactions and contains genetic material of the cell. The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress. Mitochondria is involved in production of energy in the form of ATP.

18. Correct option: (D) Only II

A vacuole is a membrane-bound cell organelle. In animal cells, vacuoles are generally small and help sequester waste products. In plant cells, vacuoles help maintain water balance. Sometimes a single vacuole can take up most of the interior space of the plant cell.

19. Correct option: (C) A is true but R is false

The endoplasmic reticulum which lacks ribosomes is called smooth endoplasmic reticulum and is involved in lipid biosynthesis.

20. Correct option: (A) Golgi complex

Golgi complex receives the substances synthesized and release by the ER, condenses, modifies, packs and releases them in the form of secretory vesicles.

21. Correct option: (C) A – Incoming transport vesicle, B – Lumen, C - Newly form vesicle, D – Cisternae, E –Secretory vesicle

Golgi apparatus also called Golgi complex and found universally in both plant and animal cell. It is comprised of a series of membrane-covered sacs called cisternae that look something like a stack. The cis face of a Golgi stack is also known as forming a face which has a convex face. The trans face is known as the maturing face which has a concave face. From the forming face, several vesicles are pinched off forming.

22. Correct option: (C) Both the statements are correct

Antonie van Leeuwenhoek was the first to see living bodies under a Microscope in the year 1674. Robert Brown discovered the nucleus in the Orchid leaf in the year 1831.

23. Correct option: (A) D, C, B, A

Take a scaly leaf of onion and pull a transparent peel. Place the peel onto a drop of water on a glass slide. Add a drop of methylene blue and place a coverslip. Observe the slide under a microscope.

24. Correct option: (D) A – v, B – iii, C – i, D – ii, E – iv

Onion cell is plant cell that have a cell wall surrounding its membrane. Chromosomes are strands of DNA interwound together and forms a distinct mass during cell division. Methylene blue is a bluish coloured stain. Microscope is an instrument that magnifies and aids in visualization of tiny creatures. Ribosomes are protein manufacturing organelles that maybe freely suspended in cytoplasm or maybe attached on the membrane of rough endoplasmic membrane.

25. Correct option: (D) i, ii, and iii

In 1972, Singer and Nicolson proposed the fluid mosaic model of the plasma membrane. The fluid mosaic model describes the plasma membrane structure as a mosaic of components -including phospholipids, cholesterol, proteins, and carbohydrates. The proportions of proteins, lipids, and carbohydrates in the plasma membrane vary with the cell type.

26. Correct option: (A) Both A and R are true, and R is the correct explanation of A

Chloroplasts are the green plastids. They contain a green coloured pigment called chlorophyll. Chlorophyll is the key component in the process of photosynthesis which is the preparation of food for the plant. Therefore, chloroplast is known as the kitchen of the cell.

27. Correct option: (A) Vacuoles

Vacuoles are membrane-bound organelles found in the cytoplasm of eukaryotic cells. They can vary in size, but they are typically large, bulged, and transparent structures that play various essential roles in the cell. One of their primary functions is to store water and maintain cell turgidity. Vacuoles can also store waste products, excretory substances, and even toxic materials in some cases.

28. Correct option: (D) a, b, and c

Chromosomes are present inside nucleus and are seen only during cell division and carries genes on it.

29. Correct option: (D) Cell Wall

Cell wall is a protective outermost layer, which is present surrounding the cell membrane. It is present in plant cells. It supports the contents of cell and protects from pathogen attack and wear and tear. It prevents water loss from plant cells. It gives shape to the plant. It is made up of cellulose and lignin.

30. Correct option: ** A-Thylakoid, B- Granum, C- Outer Membrane, D- Intermembranous space, E- Inner Membrane

Chloroplasts are found in all higher plants. It is oval or biconvex, found within the mesophyll of the plant cell. The size of the chloroplast usually varies between 4-6 μm in diameter and 1-3 μm in thickness. They are double-membrane organelle with the presence of outer, inner and intermembrane space. There are two distinct regions present inside a chloroplast known as the grana and stroma. Grana are made up of stacks of disc-shaped structures known as thylakoids or lamellae. The grana of the chloroplast consists of chlorophyll pigments and are the functional units of chloroplasts.